# Critical Invariant Polygons and the Farey–Ito Boundary of Stochastic Spectra

Boundary Contacts, First Returns, and the Karpelevič Theorem in Ito's Formulation

Brecht Verbeken[1,2] and Vincent Ginis[1,2,3]

[1]Department of Business Technology and Operations, Data Analytics Laboratory, Vrije Universiteit Brussel (VUB), Pleinlaan 2, 1050 Brussels, Belgium
[2]imec-SMIT, Vrije Universiteit Brussel, Pleinlaan 9, 1050 Brussels, Belgium
[3]School of Engineering and Applied Sciences, Harvard University, Cambridge, Massachusetts 02138, USA

brecht.verbeken@vub.be vincent.ginis@vub.be

**Abstract**

Let $T$ be an orientation-preserving real-linear contraction of an oriented plane with nonreal eigenvalues, and let its polygonal complexity be the least number of vertices of a nondegenerate convex polygon $P$ satisfying $TP \subseteq P$. We first develop an intrinsic contact theory for radially critical maps. For polygonal complexity $N \geq 4$, contact on every side and at every image vertex, a half-open side-contact assignment, exact vertex replacements, a finite cyclic reduction, and the unimodular lattice sail lead to a first-return normal form. A projective argument then proves that, when some cyclic-shift orbit contributes more than one relative-interior contact, the first return reaches the next such contact in cyclic order. In an adapted complex coordinate, the resulting recurrence relations yield a finite product equation with varying parameters and an exact equation for chosen real arguments, with the arithmetic encoded by Farey neighbors. Together with a direct treatment of orders at most three, we then apply this structure to determine, for every $n$, the union $\Theta_n$ of the spectra of all real row-stochastic $n \times n$ matrices. Jensen's inequality for a strictly convex log-sine function shows that equality can occur only when the factor parameters agree and gives the sharp radial equation. Sparse stochastic matrices attain every equality point, while scalar Farey refinement proves nesting from order $n-1$ to order $n$. Together with filling each radial segment by convex combinations with the identity matrix and the unit-circle analysis, this establishes the full boundary description of the Karpelevič region in Ito's formulation from critical invariant polygons.

**Keywords.** critical invariant polygons; elliptic contractions; vertex replacement; projective geometry; Farey sequences; stochastic matrices; eigenvalue regions; Karpelevič theorem in Ito's formulation.

## Contents

# Part I: Critical Invariant Polygons and Farey Return Normal Forms

## 1 The intrinsic theorem

Let $V$ be a real plane. For every real-linear map $A : V \to V$, define its polygonal complexity by

$$\nu_{\mathrm{poly}}(A) = \min\Big\{|\mathrm{Ext}\, P| : P \subset V \text{ is a nondegenerate compact convex polygon and } AP \subseteq P\Big\}, \tag{1.1}$$

with the value $\infty$ when the class is empty. An invertible map $T : V \to V$ is *elliptic* if

$$(\operatorname{tr} T)^2 < 4 \det T,$$

and it is an *elliptic contraction* when, in addition, its spectral radius belongs to $(0, 1)$. Defining (1.1) for all linear maps is essential: the radial comparison maps $tT$ in definition 1.1 need not remain contractions.

**Definition 1.1** (Radial polygonal criticality)**.** For $N \geq 3$, an elliptic contraction $T$ is $N$-*critical* if

$$\nu_{\text{poly}}(T) = N, \qquad \nu_{\text{poly}}(tT) > N \quad \text{for every } t > 1. \tag{1.2}$$

The second inequality allows the value $\infty$.

Both conditions are invariant under real-linear conjugacy. They are therefore properties of the planar dynamics, not of a chosen Euclidean metric, complex coordinate, polygon, or vertex numbering.

We write $\operatorname{Ext} K$ for the set of extreme points of a compact convex set $K$, $\partial K$ for its boundary, $\operatorname{relint} F$ for relative interior, and $\operatorname{aff} S$ for affine hull.

**Definition 1.2** (Polygon and vertex-list convention)**.** Whenever a planar convex polygon $P$ is presented as an $N$-gon or with a displayed cyclic vertex list $x_0, \dots, x_{N-1}$, we require that $P$ have nonempty interior and that the displayed list contain every point of $\operatorname{Ext} P$ exactly once, in cyclic order, and no other points. Thus adjacent sides are not collinear, every displayed side is a maximal boundary segment, and the interior of the normal cone at every vertex is nonempty. A supporting line $L$ of $P$ *exposes* a vertex $v$ when $P \cap L = \{v\}$.

For a polygon $P$ with an oriented boundary and the vertex-list convention of definition 1.2, let $\mathcal{E}(P)$ be its cyclic set of sides. For $e \in \mathcal{E}(P)$ write $\operatorname{tail}(e)$ and $\operatorname{head}(e)$ for its tail and head, let $\operatorname{succ}(e)$ be the next side, and put

$$e^{\triangleright} = (\operatorname{tail}(e), \operatorname{head}(e)].$$

A *half-open contact bijection* for $TP \subseteq P$ is a cyclic-order-preserving bijection

$$\chi : \operatorname{Ext} P \longrightarrow \mathcal{E}(P)$$

such that $Tv \in \chi(v)^{\triangleright}$ for every vertex $v$. The sides whose assigned image points lie in their relative interiors form the set $I$, and the induced cyclic shift of the side labels is denoted by $\sigma$:

$$I = \left\{ e \in \mathcal{E}(P) : T\chi^{-1}(e) \in \operatorname{relint} e \right\}, \qquad \sigma(e) = \chi(\operatorname{head}(e)). \tag{1.3}$$

Both maps $e \mapsto \operatorname{head}(e)$ and $\chi$ preserve cyclic order. Their composition $\sigma$ is therefore an automorphism of the finite cyclic order; after identifying $\mathcal{E}(P)$ with $\mathbb{Z}/N\mathbb{Z}$, it is a translation. When $e \in I$ and $\operatorname{succ}(e) \notin I$, the permitted vertex replacement at $e$ replaces $\operatorname{head}(e)$ by the image point $T\chi^{-1}(e) \in \operatorname{relint} e$. Whenever this replacement produces a polygon $P'$ with the complete cyclic vertex list prescribed in definition 1.2, put $v' = T\chi^{-1}(e)$ and let

$$b_e : \mathcal{E}(P) \longrightarrow \mathcal{E}(P')$$

be the *label-preserving bijection between the old and new side sets* defined by

$$b_e(f) = \begin{cases} [\operatorname{tail}(e), v'], & f = e, \\ [v', \operatorname{head}(\operatorname{succ}(e))], & f = \operatorname{succ}(e), \\ f, & f \notin \{e, \operatorname{succ}(e)\}, \end{cases}$$

where a side in the last line is identified with the geometrically unchanged boundary segment of $P'$. This bijection preserves cyclic order. If $\operatorname{succ}'$ is the successor map on $\mathcal{E}(P')$, then $b_e \circ \operatorname{succ} = \operatorname{succ}' \circ b_e$.

**Theorem 1.3** (Contact and first-return structure of an $N$-critical invariant polygon)**.** *Let $N \geq 4$ and let $T$ be an $N$-critical elliptic contraction. Then there exist an invariant $N$-gon $P$ and a choice of one of the two orientations of $V$ such that, in that orientation, a half-open contact bijection has the following properties.*

(i) Contact on every side and at every image vertex. *Every polygon $Q$ with nonempty interior, at most $N$ vertices, and $TQ \subseteq Q$ has exactly $N$ vertices; every side of $Q$ intersects $TQ$, and every vertex of $TQ$ lies on $\partial Q$.*

(ii) Equivariant vertex-replacement update. *Every permitted vertex replacement at $e$ produces another invariant $N$-gon $P'$. Let $b = b_e$ be the label-preserving bijection between the old and new side sets, and let $I'$ be the set of sides with relative-interior contact and $\sigma'$ the induced cyclic permutation of the primed side labels. Then*

$$\begin{aligned} I' &= b\big((I \setminus \{e\}) \cup \{\sigma(e)\}\big), \\ \sigma' &= b \circ \sigma \circ b^{-1}. \end{aligned} \tag{1.4}$$

(iii) Interval reduction. *Among the finitely many sets of sides with relative-interior contact realized by finite permitted vertex-replacement sequences, there is a representative for which $I$ is a nonempty cyclic interval. If $\delta$ is the number of $\sigma$-orbits and $\varphi = |I|$, then $\varphi \geq \delta$.*

(iv) First-return cases. *If $\sigma$ is the identity, every contact lies in the relative interior of its assigned side. Assume $\sigma$ is not the identity. If $\varphi = \delta$, then $I$ contains exactly one side from each $\sigma$-orbit (equivalently, it is a complete orbit transversal), and every base has return time $N/\delta$. If $\varphi > \delta$, the first-return map of $\sigma$ to $I$ is the cyclic successor on $I$. There exist integers $q > 0$ and $h \geq 0$ such that every first-return orbit segment has length $q$ or $q + h$, and these orbit segments form a disjoint partition of the $N$ vertex labels.*

*All assertions are invariant under invertible real-linear conjugacy.*

The orientation asserted in the theorem is the one in which the right-half-open side assignment holds; it selects an intermediate eigenvalue for the first-return analysis. The later theorem on the finite product equation may select the conjugate eigenvalue after reflecting the completed chain of recurrence relations.

The restriction $N \geq 4$ is used in the non-exhausting-boundary-chain and factor-argument-bound steps; no assertion about the $N = 3$ case is made here.

These first-return cases are the substantive conclusion. Before the projective argument is used, the first-return map is an arbitrary positive cyclic shift of $I$. The theorem says that, when some orbit contributes more than one relative-interior contact, the first return reaches the cyclic successor.

For a nonzero complex number $z$ that is not positive real, write

$$\arg_+(z) \in (0, 2\pi)$$

for its positive lifted argument. The orientation used to construct the right-half-open contact assignment is an intermediate choice. The finite-product theorem is allowed to use either that orientation or the opposite orientation, and it makes this choice only after the return case is known. The exact reflection of the final chain of recurrence relations is proved in lemmas 8.2 and 8.3; no vertex-replacement or cyclic-successor argument is transported across that reflection.

**Theorem 1.4** (Consecutive Farey fractions and a finite product equation)**.** *Let $N \geq 4$ and let $T$ be an $N$-critical elliptic contraction. There is an adapted complex coordinate in one of the two orientations of $V$, hence one of the two conjugate eigenvalues $\mu$ of $T$, such that*

$$Tz = \mu z, \qquad 0 < |\mu| < 1.$$

*Let $\vartheta = \arg_+(\mu)$ and put $y = \vartheta/(2\pi)$. There are consecutive reduced fractions*

$$\frac{p}{q} < y < \frac{r}{s}$$

*in the Farey sequence $F_N$ whose ordered denominators satisfy $q < s$. Put*

$$d = \lfloor N/q \rfloor, \qquad e = s - dq \in \mathbb{Z}.$$

*Then there are parameters*

$$0 \le \beta_j < 1, \qquad \alpha_j = 1 - \beta_j > 0 \qquad (1 \le j \le d) \tag{1.5}$$

*for which $\mu$ satisfies the following polynomial equation:*

$$\mu^s \prod_{j=1}^{d} (\mu^q - \beta_j) = \mu^{dq} \prod_{j=1}^{d} \alpha_j. \tag{1.6}$$

*Since $\mu \neq 0$, this is equivalent to the Laurent equation*

$$\mu^e \prod_{j=1}^{d} (\mu^q - \beta_j) = \prod_{j=1}^{d} \alpha_j. \tag{1.7}$$

*The integer closing exponent $e$ is not asserted to be nonnegative. In the reflected one-contact-per-orbit case it is negative; in that case the polynomial equation (1.6), rather than the Laurent equation (1.7), is the primary formulation.*

*Put*

$$A = q\vartheta - 2\pi p, \qquad M = \arg_+(\mu^q - 1).$$

*Then $0 < A < M < \pi$, and for each $j$ there is a unique $u_j \in [A, M)$ such that*

$$e^{iu_j} = \frac{\mu^q - \beta_j}{|\mu^q - \beta_j|}. \tag{1.8}$$

*These chosen real lifts satisfy the equation*

$$e\vartheta + \sum_{j=1}^{d} u_j = 2\pi(r - dp). \tag{1.9}$$

*The factors with $\beta_j = 0$ are introduced only to extend the finite list of recurrence relations to its closing index; they are not claimed to be additional relative-interior contacts.*

The stochastic statement is deferred to section 9. Its proof will consist only of identifying $N$-criticality through the invariant-polytope criterion and applying theorem 1.4.

### 1.1 Relation with the stochastic eigenvalue literature

The final application belongs to the classical study of eigenvalue regions of stochastic matrices initiated by Dmitriev–Dynkin and Karpelevič; see [1, 4]. Farey descriptions and explicit matrix realizations appear in several later forms [2, 3, 5, 6]. The result proved here is organized in the opposite direction: the parent statement is intrinsic to a critical planar linear map, while stochastic matrices enter only through the invariant-polytope criterion in section 9.

### 1.2 Logical architecture

The proof is divided into five modules with one-way dependencies:

$$\boxed{\text{criticality}} \Longrightarrow \boxed{\text{contact on every side and at every image vertex}}$$

$$\Longrightarrow \boxed{\text{half-open contacts and vertex replacements}} \Longrightarrow \boxed{\text{first returns of a cyclic permutation}}$$

$$\Longrightarrow \boxed{\begin{matrix}\text{adjacent first return when an orbit}\\ \text{contains multiple contacts, for } N\geq 4\end{matrix}}.$$

The finite product equation is obtained by multiplying the recurrence relations along the final closed chain and cancelling its nonzero endpoint vertices. The finite-rotation module is proved through consecutive lattice-sail vectors; it does not use a parity-indexed Euclidean remainder chain. The projective module contains no modular arithmetic. Detailed determinant and endpoint audits occur only inside the vertex-replacement theorem and the boundary-chain deformation where they are logically necessary.

For reference, the following table records the input and conclusion of each module:

| module | input | exported conclusion |
|---|---|---|
| contact on every side and at every image vertex | $N$-criticality | every invariant polygon with at most $N$ vertices has all sides and all image vertices in the stated boundary contact |
| half-open contact assignment | the preceding contact theorem | a cyclic half-open contact bijection and an exact, geometrically realized vertex-replacement update |
| first returns of a cyclic permutation | one reduced cyclic interval | a bijective two-height tower partition with unimodular return data |
| projective adjacent-return proof | $N \geq 4$, tower partition, exact boundary-edge classification, and validity after each vertex replacement | a globally defined invariant deformation with one escaped image vertex; hence, when an orbit contains multiple contacts, the first return is the adjacent successor |
| finite product equation | a closed chain of recurrence relations along adjacent returns | the Farey interval, varying-parameter product equation, and equation for chosen real arguments |

No later module uses the internal indices of an earlier proof; it uses only the theorem displayed in the last column. This is the stability mechanism of the argument.

### 1.3 Foundational tools

The proof uses only finite-dimensional linear algebra, elementary planar convexity, and the index formula for a rank-two sublattice of $\mathbb{Z}^2$. For completeness, the precise forms of Perron–Frobenius, strict separation, polarity, Hausdorff continuity of polygonal area, and the lattice parallelogram count used below are proved in appendix A. Thus every nonstandard dependency of the main argument is available inside the paper. We use in their standard forms Cayley–Hamilton and Jordan normal form, elementary compactness and subsequence arguments, dominated convergence, Smith normal form for rank-two integer lattices, and elementary real projective geometry.

## 2 Elliptic coordinates and convex preliminaries

**Proposition 2.1** (Adapted complex structures)**.** *Let $T$ be elliptic. Then* $\det T > 0$*. Put*

$$\rho = \sqrt{\det T}, \qquad \cos\theta = \frac{\operatorname{tr} T}{2\rho}, \qquad 0 < \theta < \pi,$$

*and define*

$$J_+ = \frac{\rho^{-1}T - \cos\theta\, I}{\sin\theta}, \qquad J_- = -J_+. \tag{2.1}$$

*Then $J_+^2 = J_-^2 = -I$, and the two complex structures induce the two opposite orientations of $V$. There is an inner product for which both are isometries and*

$$T = \rho(\cos\theta\, I + \sin\theta\, J_+) = \rho(\cos\theta\, I - \sin\theta\, J_-). \tag{2.2}$$

*Thus $T$ is multiplication by $\rho e^{i\theta}$ in the $J_+$ complex coordinate and by $\rho e^{-i\theta}$ in the $J_-$ coordinate. In the adapted inner product,*

$$T^* = \rho(\cos\theta\, I - \sin\theta\, J_+),$$

*and $T^*$ is conjugate to $T$ by an orientation-reversing reflection.*

*Proof.* The characteristic polynomial of $\rho^{-1}T$ is

$$X^2 - 2\cos\theta\, X + 1.$$

Cayley–Hamilton gives

$$(\rho^{-1}T - \cos\theta I)^2 = -\sin^2\theta I,$$

so $J_+^2 = -I$, and the same is true of $J_- = -J_+$. A complex structure on a real plane determines an orientation by declaring $(v, Jv)$ positive; replacing $J$ by $-J$ reverses that orientation. Hence $J_+$ and $J_-$ induce precisely the two orientations of $V$.

Starting from any inner product $g_0$, set

$$g(x, y) = g_0(x, y) + g_0(J_+x, J_+y).$$

Then $g(J_+x, J_+y) = g(x, y)$, so $J_+$, and therefore also $J_-$, is orthogonal. Formula (2.2) follows directly from (2.1); its two complex interpretations follow from the rule that multiplication by $a + ib$ acts as $aI + bJ$. Since $J_+^* = -J_+$, the formula for $T^*$ follows. In an orthonormal $J_+$-adapted basis, reflection in the first coordinate satisfies $SJ_+S^{-1} = -J_+$, and consequently $STS^{-1} = T^*$. □

**Proposition 2.2** (Real-linear invariance of polygonal complexity). *For every invertible real-linear map $A$,*

$$\nu_{\text{poly}}(ATA^{-1}) = \nu_{\text{poly}}(T).$$

*Consequently $N$-criticality is invariant under conjugacy, and $\nu_{\text{poly}}(tT^*) = \nu_{\text{poly}}(tT)$ for every $t > 0$ in the adapted inner product.*

*Proof.* The correspondence $P \mapsto AP$ preserves convexity, nondegeneracy, the number of extreme points, and inclusion. Moreover

$$TP \subseteq P \iff (ATA^{-1})(AP) \subseteq AP.$$

The first assertion follows. The second uses the reflection conjugacy in proposition 2.1. □

**Proposition 2.3** (Real-linear covariance of faces and boundary incidences). *Let $A : V \to V$ be invertible, put*

$$\widetilde{T} = ATA^{-1}, \qquad \widetilde{P} = AP,$$

*and suppose that $P$ is presented with the complete cyclic vertex list of definition 1.2. Then the following statements hold.*

(i) *A maps* $\operatorname{Ext} P$ *bijectively onto* $\operatorname{Ext} \widetilde{P}$*, maps every side of* $P$ *onto a side of* $\widetilde{P}$*, and preserves relative interiors and proper inclusions.*

(ii) *Every nonempty proper face of a polygon is exposed by a nonzero linear functional. For every such face* $F$ *of* $P$*, the image* $AF$ *is the corresponding face of* $\widetilde{P}$*. In particular, side contacts, vertex contacts, vertex-exposing contacts, and the assertions "every side is touched" and "every image vertex lies on the boundary" are transported exactly by* $A$*.*

(iii) *If* $TP \subseteq P$*, then* $\widetilde{T}\widetilde{P} \subseteq \widetilde{P}$*. If a polygon* $P'$ *is obtained from* $P$ *by clipping with a closed half-plane, replacing a vertex by a contact point, or performing a finite sequence of such operations, then* $AP'$ *is obtained from* $AP$ *by the corresponding transformed operations. All vertex counts, incidence equalities, complete-vertex-list conditions, and invariance statements are equivalent before and after the transformation.*

(iv) *Give* $\widetilde{P}$ *the boundary orientation transported by* $A$*. If* $\chi : \operatorname{Ext} P \to \mathcal{E}(P)$ *is a half-open contact bijection, then*

$$\widetilde{\chi}(Av) = A\chi(v)$$

*is a half-open contact bijection for* $\widetilde{T}\widetilde{P} \subseteq \widetilde{P}$*. If* $\operatorname{succ}_P$ *and* $\operatorname{succ}_{\widetilde{P}}$ *denote the respective successor maps and* $A_{\mathcal{E}} : \mathcal{E}(P) \to \mathcal{E}(\widetilde{P})$ *is the induced side bijection, then*

$$\widetilde{I} = A_{\mathcal{E}}(I), \qquad \operatorname{succ}_{\widetilde{P}} = A_{\mathcal{E}} \circ \operatorname{succ}_P \circ A_{\mathcal{E}}^{-1}, \qquad \widetilde{\sigma} = A_{\mathcal{E}} \sigma A_{\mathcal{E}}^{-1}. \tag{2.3}$$

*If* $A$ *reverses the originally displayed ambient orientation, then the transported orientation on the target is the opposite displayed orientation. Re-expressing the target in the original displayed orientation reverses cyclic order and may change which half-open side an endpoint is assigned to; this orientation and endpoint-assignment issue is recorded in lemma 2.4.*

*Proof.* An invertible real-linear map is a homeomorphism that maps line segments to line segments and preserves convex combinations. A point $x \in P$ is extreme precisely when an equality $x = tu + (1-t)v$ with $u, v \in P$ and $0 < t < 1$ forces $u = v = x$; applying $A$ and $A^{-1}$ proves the first assertion about extreme points. Maximal boundary segments, relative interiors, and proper inclusions are therefore preserved as well.

Every nonempty proper face of a polygon is exposed by a nonzero linear functional. If $\ell$ supports $P$ on such a face $F = \{x \in P : \ell(x) = \max_P \ell\}$, then $\widetilde{\ell} = \ell \circ A^{-1}$ supports $AP$ on

$$AF = \{y \in AP : \widetilde{\ell}(y) = \max_{AP} \widetilde{\ell}\}.$$

Thus all face and vertex-exposure assertions are equivalent. The identity

$$\widetilde{T}(AP) = A(TP)$$

proves the invariance and image-contact claims. Finally, $A(P \cap H) = AP \cap AH$ for every half-plane $H$, and replacing a point $x$ by a point $c$ in a displayed convex hull commutes with applying $A$. Iteration proves the statement for finite sequences of these clipping and vertex-replacement operations.

With the transported orientation, $A$ preserves tail, head, cyclic successor, and every right-half-open side:

$$A\big((\operatorname{tail}(e), \operatorname{head}(e)]\big) = (A\operatorname{tail}(e), A\operatorname{head}(e)].$$

Thus $\widetilde{\chi}$ satisfies the required half-open-side membership condition, and the successor maps, set $I$, and cyclic shift $\sigma$ satisfy (2.3). The condition for the permitted vertex replacement is expressed only through membership in $I$ and the successor map, so it is preserved. The final orientation statement follows because the sign of every oriented area is multiplied by $\det A$. ∎

**Consequences used in later topics.** Any later construction that uses only the side set $\mathcal{E}$, its successor map, $I$, and $\sigma$, and is unchanged by relabelling is transported by $A_{\mathcal{E}}$. Each such construction is defined and checked when it is introduced.

Whenever an adapted complex coordinate is fixed, we identify $V$ with $\mathbb{C}$ and use

$$\langle z, w\rangle = \mathrm{Re}(\overline{z}w), \qquad \det(z, w) = \mathrm{Im}(\overline{z}w). \tag{2.4}$$

The determinant is positive on positively oriented ordered bases. For a nonempty compact convex set $K$, its support function is

$$h_K(u) = \max_{z\in K}\langle u, z\rangle. \tag{2.5}$$

Until a consistent assignment to half-open sides is selected, we use the $J_+$ coordinate and write

$$Tz = \lambda z, \qquad \lambda = \rho e^{i\theta}, \qquad 0 < \rho < 1, \qquad 0 < \theta < \pi.$$

The opposite orientation labels the same real map by $\overline{\lambda} = \rho e^{-i\theta}$, whose positive lifted argument is $2\pi - \theta$. No geometric property of the real-linear dynamics depends on this choice.

**Lemma 2.4** (Coordinate reversal, orientation, and endpoint assignment)**.** *Let $C : \mathbb{C} \to \mathbb{C}$ be complex conjugation. For every $\lambda \in \mathbb{C} \setminus \mathbb{R}$ and every compact convex polygon $P$,*

$$C(\lambda P) = \overline{\lambda}\, C(P), \qquad \lambda P \subseteq P \iff \overline{\lambda}\, C(P) \subseteq C(P). \tag{2.6}$$

*Conjugation preserves the number of extreme points, complete cyclic vertex lists, relative interiors, supporting incidences, side-intersection properties, image-vertex-on-boundary properties, and Hausdorff-continuous deformation families.*

*If $x_i$ is a positively indexed boundary list for $P$ and*

$$\widetilde{x}_i = \overline{x}_{-i},$$

*then $\widetilde{x}_i$ is a positively indexed boundary list for $C(P)$. However, conjugation reverses the half-open convention: the image of the right-half-open side $(x_{j-1}, x_j]$ is the left-half-open side*

$$[\widetilde{x}_{-j}, \widetilde{x}_{1-j}). \tag{2.7}$$

*Consequently a contact assignment to right-half-open sides is not, in general, carried to another such assignment by conjugation when endpoint contacts are present. Thus the endpoint convention must be checked again after conjugation.*

*Proof.* The identity

$$C(\lambda z) = \overline{\lambda}\, C(z)$$

gives (2.6). Since $C$ is an invertible real-affine map, it preserves convex combinations, extreme points, faces, relative interiors, and inclusions. A line supports $P$ if and only if its conjugate supports $C(P)$, with the conjugate contact face; this proves all incidence, side-intersection, and image-vertex-on-boundary assertions. Applying $C$ pointwise to a deformation family preserves Hausdorff convergence because $C$ is an isometry in the adapted metric.

Complex conjugation reverses the cyclic order on the boundary. Hence $\widetilde{x}_i = \overline{x}_{-i}$ restores positive order. The side of $C(P)$ from $\widetilde{x}_{-j} = \overline{x}_j$ to $\widetilde{x}_{1-j} = \overline{x}_{j-1}$ is the conjugate of the unoriented segment $[x_{j-1}, x_j]$. The old half-open set excludes $\overline{x}_{j-1} = \widetilde{x}_{1-j}$ and includes $\overline{x}_j = \widetilde{x}_{-j}$, which is exactly (2.7). Thus conjugation swaps which endpoint is included, proving the final assertion. ∎

**Roadmap for later topics.** Later arguments must reconstruct the half-open-side assignment after conjugation rather than assume that it is unchanged.

**Lemma 2.5** (Interior origin for a nonreal contraction)**.** *Let $P$ be a non-singleton compact convex hull of finitely many points, possibly a segment, and let $\lambda = \rho e^{i\theta}$ with $0 < \rho < 1$ and $\theta \not\equiv 0, \pi \pmod{2\pi}$. If $\lambda P \subseteq P$, then $P$ has nonempty interior and $0 \in \operatorname{int} P$.*

*Proof.* For any $z \in P$, invariance gives $\lambda^k z \in P$, while $\lambda^k z \to 0$; since $P$ is closed, $0 \in P$. If $P$ were a segment, its affine line would pass through 0 and would have to contain the segment of positive length $\lambda P$; hence multiplication by $e^{i\theta}$ would preserve a real line, forcing $\theta \equiv 0$ or $\pi$, a contradiction. Thus $P$ has nonempty interior.

Suppose $0 \in \partial P$. Choose a nonzero real linear functional $\ell$ such that $\ell \geq 0$ on $P$ and choose $z \in P$ with $\ell(z) > 0$. Invariance would give

$$\lambda^k z = \rho^k e^{ik\theta} z \in P, \qquad \rho^k \ell(e^{ik\theta} z) \geq 0.$$

Because $\rho^k > 0$, it follows that $\ell(e^{ik\theta} z) \geq 0$ for every $k \geq 0$. If $e^{i\theta}$ has finite order $m$, then $m \geq 3$ and $\sum_{k=0}^{m-1} e^{ik\theta} z = 0$; summing the displayed nonnegative quantities forces them all to vanish, contrary to $\ell(z) > 0$. If $e^{i\theta}$ has infinite order, its powers are dense in the unit circle, so some $e^{ik\theta} z$ lies in the open half-plane where $\ell < 0$, again a contradiction. Hence 0 is interior. □

**Lemma 2.6** (Oriented order on a convex boundary)**.** *Let $K \subset \mathbb{R}^2$ be a compact convex set with nonempty interior, with $\partial K$ given its positive orientation. For three distinct noncollinear points $a, b, c \in \partial K$,*

$$\det(b - a, c - a) > 0 \tag{2.8}$$

*if and only if $a, b, c$ occur in that positive cyclic order on $\partial K$. Consequently, if all vertices of a polygon $Q$ lie on $\partial K$, then the positive cyclic order of the vertices of $Q$ is exactly the order induced from $\partial K$.*

*Proof.* Choose $o \in \operatorname{int} \operatorname{conv}\{a, b, c\}$. Since the triangle $\operatorname{conv}\{a, b, c\}$ is contained in $K$, one has $o \in \operatorname{int} K$. Every ray from $o$ meets $\partial K$ in exactly one point, so the radial map

$$\pi_o : \partial K \longrightarrow \mathbb{S}^1, \qquad \pi_o(z) = \frac{z - o}{|z - o|},$$

is a homeomorphism. With the positive boundary orientation it has degree $+1$: every ray from $o$ meets the boundary exactly once, and traversing the positively oriented boundary once makes the direction $z - o$ traverse the unit circle once counterclockwise. Hence $\pi_o$ preserves cyclic order. Because $o$ lies in the interior of the triangle $abc$, the three rays from $o$ to $a, b, c$ occur counterclockwise in the order $a, b, c$ exactly when the oriented area of that triangle is positive, which is (2.8).

No three vertices of $Q$ are collinear: a middle point of three collinear vertices would be a convex combination of the other two and therefore would not be extreme. For the final assertion, apply the first part both to $K$ and to $Q$. For every triple of vertices of $Q$, the sign of the same determinant characterizes its positive cyclic order on $\partial K$ and on $\partial Q$. Thus the two ternary cyclic-order relations coincide, and therefore so do the full cyclic orders. □

**Lemma 2.7** (Triple-sign criterion for a complete convex vertex list)**.** *Let $z_0, \ldots, z_{m-1}$ be a cyclic list of points, $m \geq 3$. The list is exactly the complete cyclic vertex list of a convex $m$-gon, in the displayed orientation and in the sense of definition 1.2, if and only if there is a sign $\varepsilon \in \{1, -1\}$ such that*

$$\varepsilon \det(z_j - z_i, z_k - z_i) > 0 \tag{2.9}$$

*for every triple of distinct indices $i, j, k$ occurring in that cyclic order. The displayed orientation is positive precisely when $\varepsilon = 1$. Here and below, indices attached to a cyclic vertex, side, or normal list are read modulo the length of that list. More explicitly, three distinct classes $i, j, k \in \mathbb{Z}/m\mathbb{Z}$ occur in that cyclic order when they have integer representatives $\widetilde{i}, \widetilde{j}, \widetilde{k}$ with*

$$\widetilde{i} < \widetilde{j} < \widetilde{k} < \widetilde{i} + m.$$

*Proof.* If the list is the complete cyclic vertex list of a polygon, apply lemma 2.6 in the displayed orientation; reversing orientation changes all determinant signs simultaneously.

Conversely, first take $\varepsilon = 1$. The inequalities force the points to be distinct and no three of them to be collinear. Fix $i$. For every $k \notin \{i, i+1\}$, the indices $i, i+1, k$ occur in positive cyclic order, so

$$\det(z_{i+1} - z_i, z_k - z_i) > 0.$$

Thus every other point lies strictly in the left half-plane of the directed line from $z_i$ to $z_{i+1}$. Hence $[z_i, z_{i+1}]$ is an exposed edge of $\operatorname{conv}\{z_0, \ldots, z_{m-1}\}$. This holds for every $i$, so all displayed points are extreme, the displayed consecutive segments are exactly the boundary edges, and the resulting polygon has the displayed list as its complete vertex list and is positively oriented. The case $\varepsilon = -1$ is identical with left replaced by right, or follows by reversing the displayed orientation. □

**Lemma 2.8** (Simultaneous preservation of convexity and side conditions under perturbation)**.** *Let $I_0 \subset \mathbb{R}$ be an open interval containing $0$, and let*

$$z_0(\tau), \ldots, z_{N-1}(\tau) \qquad (\tau \in I_0)$$

*be continuous maps from $I_0$ to $V$ such that $z_0(0), \ldots, z_{N-1}(0)$ are the complete, positively ordered vertex list of an $N$-gon. Let $\mathcal{A}$ and $\mathcal{B}$ be finite sets. For every $a \in \mathcal{A}$, fix a side index $k(a)$ and a continuous map $w_a : I_0 \to V$ such that, throughout $I_0$,*

$$w_a(\tau) \in \operatorname{aff}\big(z_{k(a)-1}(\tau), z_{k(a)}(\tau)\big), \qquad w_a(0) \in \operatorname{relint}[z_{k(a)-1}(0), z_{k(a)}(0)].$$

*For every $b \in \mathcal{B}$, fix a side index $k(b)$ and a continuous map $u_b : I_0 \to V$ such that*

$$\det\big(z_{k(b)}(0) - z_{k(b)-1}(0), u_b(0) - z_{k(b)-1}(0)\big) > 0. \tag{2.10}$$

*Then there is an open interval $I \subseteq I_0$ containing $0$ on which all of the following hold simultaneously:*

(i) Convexity is preserved: *the list $z_0(\tau), \ldots, z_{N-1}(\tau)$ consists of distinct extreme points in the same positive cyclic order and therefore defines an $N$-gon $P_\tau$ with that complete vertex list;*

(ii) Relative-interior membership is preserved: *$w_a(\tau) \in \operatorname{relint}[z_{k(a)-1}(\tau), z_{k(a)}(\tau)]$ for every $a \in \mathcal{A}$;*

(iii) The prescribed strict determinant inequalities are preserved: *the strict inequality obtained from (2.10) by replacing $0$ with $\tau$ holds for every $b \in \mathcal{B}$.*

*For each such $P_\tau$, its interior is the intersection of the $N$ open left half-planes determined by its positively oriented sides. In particular, a continuous test point satisfying all $N$ strict side inequalities belongs to $\operatorname{int} P_\tau$.*

*At the unperturbed polygon, a point in the relative interior of one side satisfies the strict side inequality for every other side. Equivalently, if $w \in \operatorname{relint}[z_{c-1}(0), z_c(0)]$, then*

$$\det\big(z_k(0) - z_{k-1}(0), w - z_{k-1}(0)\big) > 0 \qquad (k \neq c). \tag{2.11}$$

*Proof.* For a positively oriented polygon with its complete cyclic vertex list, the directed line of side $[z_{k-1}(0), z_k(0)]$ supports the polygon, the polygon lies in its closed left half-plane, and the contact face is exactly that side. If $w \in \operatorname{relint}[z_{c-1}(0), z_c(0)]$ and $k \neq c$, then $w$ does not belong to the contact face of side $k$: distinct maximal sides of a polygon have disjoint relative interiors and adjacent sides meet only at their common endpoint. The supporting inequality is therefore strict, proving (2.11). The same support description shows that the polygon is the intersection of the closed left half-planes and its interior is the intersection of their open counterparts.

For every cyclically ordered triple of distinct labels, the corresponding oriented determinant is positive at $\tau = 0$. There are finitely many such determinants, so continuity gives a common neighborhood on which they all remain positive. The triple-sign criterion lemma 2.7 proves that convexity is preserved, item (i).

Fix $a \in \mathcal{A}$. Choose a real linear functional $f_a$ satisfying

$$f_a\big(z_{k(a)}(0) - z_{k(a)-1}(0)\big) \neq 0.$$

Continuity gives an interval about 0 on which the corresponding denominator remains nonzero. Intersecting these finitely many intervals with the interval already chosen gives one common interval, and the collinearity hypothesis gives the unique scalar

$$\alpha_a(\tau) = \frac{f_a\big(w_a(\tau) - z_{k(a)-1}(\tau)\big)}{f_a\big(z_{k(a)}(\tau) - z_{k(a)-1}(\tau)\big)}$$

with

$$w_a(\tau) = (1 - \alpha_a(\tau))z_{k(a)-1}(\tau) + \alpha_a(\tau)z_{k(a)}(\tau).$$

This scalar is continuous and belongs to $(0, 1)$ at 0; finiteness of $\mathcal{A}$ shows that relative-interior membership is preserved, item (ii), on one common interval. Finally, the determinants in item (iii) are finitely many continuous functions that are positive at 0. Intersecting their positivity intervals with the interval already chosen preserves all their signs and proves item (iii). The interior assertion follows from the open half-plane description proved in the first paragraph. □

**Lemma 2.9** (Test for whether a supporting line exposes a vertex)**.** *Let $P$ be a polygon with complete cyclic vertex list $x_i$, and let $L$ be a supporting line of $P$ through $x_i$. Then $L$ exposes $x_i$ if and only if neither adjacent vertex $x_{i-1}$ nor $x_{i+1}$ lies on $L$. Equivalently, once support is known, exposure is characterized by the two nonzero determinants*

$$\det(v, x_{i-1} - x_i) \neq 0, \qquad \det(v, x_{i+1} - x_i) \neq 0,$$

*where $v$ is any nonzero direction vector of $L$.*

*Proof.* The contact set $P \cap L$ is a face of the polygon. If it is more than the single point $x_i$, it contains a nondegenerate boundary segment having $x_i$ as one endpoint. In a polygon the maximal nondegenerate faces are exactly the displayed sides, so this segment lies in either $[x_{i-1}, x_i]$ or $[x_i, x_{i+1}]$, and the corresponding adjacent vertex is on $L$. Conversely, if an adjacent vertex lies on $L$, then the contact face contains the side joining it to $x_i$, so $L$ does not expose $x_i$. The determinant formulation is the line-membership test for the two adjacent vertices. □

**Lemma 2.10** (Angular monotonicity along a polygonal boundary)**.** *Let $P$ be a polygon with $0 \in \operatorname{int} P$, and list its vertices $x_i$ in positive cyclic order. There are lifted arguments $\Theta_i$ with $\Theta_{i+N} = \Theta_i + 2\pi$ such that*

$$0 < \Theta_i - \Theta_{i-1} < \pi \qquad (i \in \mathbb{Z}). \tag{2.12}$$

*Moreover the polar argument is strictly increasing along every positively oriented side $[x_{i-1}, x_i]$.*

*Proof.* The interior of a positively oriented polygon lies strictly to the left of every oriented side. Applying this to the interior point 0 gives

$$\det(x_{i-1}, x_i) > 0.$$

Lift the polar argument continuously along the positively oriented boundary. By the radial-map argument in lemma 2.6, a positively oriented convex boundary has winding number $+1$ about every interior point. One circuit therefore increases this lift by $2\pi$. The displayed determinant sign places each consecutive angular increment in $(0, \pi)$, giving the lifted sequence and (2.12). For $z(t) = (1-t)x_{i-1} + tx_i$, $0 \le t \le 1$, the segment avoids 0 and

$$\frac{d}{dt} \operatorname{Arg} z(t) = \frac{\det(z(t), x_i - x_{i-1})}{|z(t)|^2} = \frac{\det(x_{i-1}, x_i)}{|z(t)|^2} > 0.$$

This proves the sidewise assertion. □

## 3 Contact on every side and at every image vertex

Let $P$ be an invariant polygon containing 0 in its interior, with its complete cyclic vertex list understood as in definition 1.2. Write its outward unit normals in cyclic order as

$$u_i = e^{i\phi_i}, \qquad \phi_1 < \cdots < \phi_m < \phi_1 + 2\pi,$$

and put $r_i = \mathbb{R}_{\ge 0} u_i$. With indices read cyclically, the full normal fan $\Phi$ of $P$ consists of the zero cone $\{0\}$, the edge-normal rays $r_i$, and the two-dimensional vertex-normal cones $\operatorname{cone}(u_i, u_{i+1})$. For the coefficient matrix below, we use the cyclically ordered unit ray generators $u_i$. Let $h_i = h_P(u_i) > 0$ be the support numbers. For each $i$, express $e^{-i\theta}u_i$ in the cone generated by the two adjacent fan rays that contain it:

$$e^{-i\theta}u_i = a_i u_j + b_i u_{j+1}, \qquad a_i, b_i \ge 0. \tag{3.1}$$

The resulting full coefficient vector is unique and defines a nonnegative matrix $B_\Phi(\theta)$. If the direction lies on a fan ray, either adjacent cone gives that same vector: the coefficient of the unit generator of that ray is 1 and the other displayed coefficient is 0.

**Proposition 3.1** (Support-function criterion in a fixed normal fan)**.** *With $h = (h_i)$,*

$$\lambda P \subseteq P \quad \Longleftrightarrow \quad \rho B_\Phi(\theta) h \le h. \tag{3.2}$$

*Moreover, for the complex vector $u = (u_i)^T \in \mathbb{C}^m$,*

$$B_\Phi(\theta) u = e^{-i\theta} u. \tag{3.3}$$

*The support vectors for which this fixed fan remains the complete normal fan form an open subset of $\mathbb{R}^m$.*

*Proof.* If $v = au_j + bu_{j+1}$ belongs to the cone generated by two adjacent normal rays, with $a, b \ge 0$, the two adjacent support inequalities are maximized at their common polygon vertex. For every $z \in P$ one therefore has

$$\langle v, z \rangle \le a h_P(u_j) + b h_P(u_{j+1}),$$

with equality at that common vertex. Hence $h_P(v) = a h_P(u_j) + b h_P(u_{j+1})$. Applying this to $v = e^{-i\theta} u_i$ gives

$$h_P(e^{-i\theta} u_i) = a_i h_j + b_i h_{j+1} = (B_\Phi h)_i.$$

The support function of $\lambda P$ in direction $u_i$ is $\rho h_P(e^{-i\theta}u_i)$, proving the equivalence. The second identity is the vector decomposition itself.

For openness, the two boundary lines with normals $u_i, u_{i+1}$ intersect at a vertex depending linearly on $(h_i, h_{i+1})$. At $P$, each such vertex satisfies all nonincident side inequalities strictly, and all cyclic turn determinants have the convex nonzero sign. These are finitely many strict inequalities in continuous functions of $h$ and therefore persist under a sufficiently small perturbation. □

**Theorem 3.2** (Contact on every side and at every image vertex)**.** *Let $T$ be $N$-critical and let $R$ be a nondegenerate polygon with at most $N$ vertices such that $TR \subseteq R$. Then $R$ has exactly $N$ vertices. Every side of $R$ intersects $TR$, and every vertex of $TR$ lies on $\partial R$.*

*Proof.* By $\nu_{\text{poly}}(T) = N$, the polygon cannot have fewer than $N$ vertices. Hence it has exactly $N$. It has nonempty interior by the nondegeneracy assumption, and lemma 2.5 gives $0 \in \operatorname{int} R$; list its extreme points cyclically as prescribed in definition 1.2.

Write $R$ in its fixed normal fan and put $B = B_\Phi(\theta)$. From (3.2) and $h > 0$, the weighted-norm bound in lemma A.1 gives $\operatorname{spr}(\rho B) \leq 1$. Suppose the inequality were strict. Then

$$g = (I - \rho B)^{-1}\mathbf{1} = \sum_{k\geq 0} (\rho B)^k \mathbf{1} > 0$$

satisfies $g - \rho Bg = \mathbf{1}$. By the final assertion of proposition 3.1, for sufficiently small $s > 0$ the support vector $h_s = (1-s)h + sg$ defines a polygon with the same complete normal fan, and

$$h_s - \rho Bh_s = (1-s)(h - \rho Bh) + s\mathbf{1} > 0.$$

Put $d = h_s - \rho Bh_s > 0$. Every row of $B$ expresses a nonzero unit direction as a nonnegative combination, so $Bh_s > 0$ coordinatewise. Choose

$$0 < \eta < \min_i \frac{d_i}{\rho(Bh_s)_i}.$$

Then

$$h_s - (1+\eta)\rho Bh_s = d - \eta\rho Bh_s > 0,$$

so the same $N$-gon is invariant for $(1+\eta)T$. This contradicts radial criticality. Therefore

$$\operatorname{spr}(\rho B) = 1. \tag{3.4}$$

The nonnegative Perron theorem in lemma A.1 supplies a nonzero $w \geq 0$ such that

$$B^T w = \rho^{-1} w. \tag{3.5}$$

Since $h - \rho Bh \geq 0$,

$$w^T(h - \rho Bh) = w^T h - \rho(B^T w)^T h = 0.$$

Every summand is nonnegative, so

$$w_i(h_i - \rho(Bh)_i) = 0 \qquad (1 \leq i \leq N). \tag{3.6}$$

*Planar positive-cone fact.* If nonzero vectors $v_1, \ldots, v_r \in \mathbb{R}^2$ and positive coefficients $c_1, \ldots, c_r$ satisfy

$$\sum_{j=1}^r c_j v_j = 0,$$

then either all the $v_j$ lie in one line through 0, or their positive cone is all of $\mathbb{R}^2$.

*Proof of the fact.* After normalizing the coefficients, 0 belongs to $\operatorname{conv}\{v_1, \dots, v_r\}$. If the vectors are not collinear but 0 were on the boundary of this convex hull, a supporting functional at 0, chosen nonnegative on the hull, would have nonnegative values on all $v_j$. Their positive weighted sum is zero, so every one of those values would have to vanish, forcing all $v_j$ into the supporting line, a contradiction. Thus 0 is interior to their convex hull. A small positive multiple of every vector in $\mathbb{R}^2$ then belongs to that convex hull, proving that the positive cone is the whole plane.

We prove that every coordinate of $w$ is positive. Let $S = \{i : w_i > 0\}$. Pairing (3.5) with (3.3) gives

$$w^T u = 0,$$

because $\rho^{-1} \neq e^{-i\theta}$. Because every coefficient $w_i$ with $i \in S$ is positive, the planar fact gives the dichotomy: either the normals indexed by $S$ are contained in one line, or their positive cone is all of $\mathbb{R}^2$. We exclude the collinear alternative. If $S$ is the full index set and the normals were contained in $\{v, -v\}$, then the intersection of the side half-planes would be a strip, a half-plane, or the whole plane, never the bounded polygon $R$. Thus collinearity is impossible when $S = \{1, \dots, N\}$. Suppose instead that $S$ is proper. If its normals were contained in $\{v, -v\}$, then for every $k \notin S$ the equality

$$0 = (B^T w)_k = \sum_{i \in S} B_{ik} w_i$$

would force $B_{ik} = 0$ for every $i \in S$, because all summands are nonnegative and every $w_i$ with $i \in S$ is positive. Each row indexed by $S$ would then express $(Bu)_i$ as a real multiple of $v$, whereas $(Bu)_i = e^{-i\theta} u_i$ is not collinear with $v$. This contradiction excludes the remaining case. The normals indexed by $S$ are therefore not collinear, and the positive relation $w^T u = 0$ now implies that they positively span the plane.

Consider the intersection of the half-planes whose indices lie in $S$:

$$R_S = \bigcap_{i \in S} \{z : \langle u_i, z \rangle \leq h_i\}.$$

Positive spanning makes its recession cone trivial, so $R_S$ is bounded. It is an intersection of finitely many closed half-planes and is therefore closed and compact. Because every $h_i$ is positive, a sufficiently small disk about 0 satisfies all these inequalities strictly; hence $0 \in \operatorname{int} R_S$, so $R_S$ is nondegenerate. After redundant inequalities are deleted, each remaining boundary line carries one maximal side. Thus $R_S$ has at most $|S|$ sides and, being a nondegenerate planar polygon, at most $|S|$ extreme points. For $i \in S$ and $k \notin S$, the equality $0 = (B^T w)_k = \sum_{j \in S} B_{jk} w_j$ and positivity of every $w_j$ with $j \in S$ give $B_{ik} = 0$. Hence row $i$ of $B$ uses only columns in $S$. Thus, for $z \in R_S$,

$$\langle u_i, \lambda z \rangle \leq \rho \sum_{j \in S} B_{ij} h_j = \rho (Bh)_i = h_i,$$

where the last equality is complementarity. Hence $\lambda R_S \subseteq R_S$. If $S$ were proper, this would contradict $\nu_{\mathrm{poly}}(T) = N$. Consequently $w_i > 0$ for every $i$, and (3.6) gives

$$h = \rho B h. \tag{3.7}$$

For each side, equality of support values means that the side intersects $TR$. Thus $TR$ intersects every side of $R$.

It remains to prove that every vertex of $TR$ lies on $\partial R$. The polar polygon

$$R^\circ = \{y : \langle y, z \rangle \leq 1 \text{ for all } z \in R\}$$

is an $N$-gon; its vertices correspond to the maximal sides of $R$ and its maximal sides correspond to the vertices of $R$. Polarity and the real adjoint give

$$TR \subseteq R \quad \Longrightarrow \quad T^* R^\circ \subseteq R^\circ.$$

By proposition 2.2, $T^*$ is also $N$-critical. The side argument just proved therefore applies to $R^\circ$: every side of $R^\circ$ intersects $T^* R^\circ$.

Fix a vertex $x$ of $R$ and its dual side

$$F_x = \{y \in R^\circ : \langle y, x \rangle = 1\}.$$

Choose $y \in R^\circ$ with $T^* y \in F_x$. Then

$$1 = \langle T^* y, x \rangle = \langle y, Tx \rangle.$$

In particular, $y \neq 0$. Since $y \in R^\circ$ and $Tx \in R$, equality for the nonzero supporting functional $y$ is possible only on the boundary. Thus $Tx \in \partial R$. Invertibility of $T$ identifies the vertices of $R$ with the vertices of $TR$, completing the proof. □

*Remark* 3.3. The theorem can be applied again after a polygon has been modified: once the modified polygon is independently shown to remain an invariant $N$-gon with its complete cyclic vertex list, every vertex of its image must again lie on its boundary.

# 4 Half-open side-contact selection

Write the complete vertex list of a polygon $P$ in positive cyclic order as $x_i$ ($i \in \mathbb{Z}/N\mathbb{Z}$), and put

$$E_i = [x_{i-1}, x_i], \qquad E_i^+ = (x_{i-1}, x_i], \qquad E_i^- = [x_{i-1}, x_i).$$

All side, vertex, and image-vertex indices in this section are read modulo $N$. For $a, b \in \partial P$, write

$$[a, b]_{\partial P}, \qquad (a, b)_{\partial P}, \qquad (a, b]_{\partial P}, \qquad [a, b)_{\partial P}$$

for the closed, open, and half-open boundary arcs obtained by travelling from $a$ to $b$ in the positive orientation. We shall repeatedly use the following elementary consequence of the fact that the image polygon intersects every side.

**Lemma 4.1** (A side intersecting a polygon contains one of its vertices)**.** *Let $Q \subseteq P$ be polygons. If a side $E$ of $P$ intersects $Q$, then $E$ contains a vertex of $Q$.*

*Proof.* Let $\ell$ be a supporting functional of $P$ whose maximum face is $E$. If $E \cap Q \neq \varnothing$, then the maximum of $\ell$ on $Q$ equals its maximum on $P$. A linear functional on a polygon attains its maximum at a vertex, so a vertex of $Q$ belongs to $E$. □

## 4.1 Half-open boundary assignments

The following definition fixes the endpoint convention used in every cut and vertex-replacement argument. It is not an additional assumption; it is the right-half-open side convention written explicitly.

**Definition 4.2** (Assignment to half-open sides)**.** For a positively oriented polygon with its complete cyclic vertex list, side index $i$ refers to the half-open side

$$E_i^+ = (x_{i-1}, x_i].$$

Thus a polygon vertex $x_i$ belongs to the incoming half-open side $E_i^+$, not to the outgoing half-open side $E_{i+1}^+$. More generally, assigning a boundary point $z \in \partial P$ to side index $i$ means precisely that

$$z \in E_i^+.$$

In particular, assigning an image vertex $\xi_i$ to side index $i$ means $\xi_i \in E_i^+$. Either $\xi_i \in \operatorname{relint} E_i$, in which case $x_{i-1}, \xi_i, x_i$ occur in that positive boundary order, or $\xi_i = x_i$. If two consecutive image vertices $y_j, y_{j+1}$ both lie on $\partial P$, the intervening boundary arc follows the same endpoint convention: the positive half-open arc $(y_j, y_{j+1}]_{\partial P}$ includes its right endpoint and excludes its left endpoint.

**Lemma 4.3** (Half-open sides partition the boundary)**.** *Let $P$ be a positively oriented $N$-gon with complete vertex list $x_i$ and oriented sides $E_i = [x_{i-1}, x_i]$. For*

$$D_i(z) = \det(x_i - x_{i-1}, z - x_{i-1}),$$

*the following statements hold.*

(i) *The half-open sides $E_i^+ = (x_{i-1}, x_i]$ are pairwise disjoint and form a partition of $\partial P$.*

(ii) *If $z \in E_i^+$, then the vanishing-index set*

$$Z(z) = \{r : D_r(z) = 0\}$$

*is*

$$Z(z) = \{i\} \quad \textit{for } z \in \operatorname{relint} E_i, \qquad Z(x_i) = \{i, i+1\}.$$

(iii) *If $z \in P$ and $D_i(z) = 0$, then $z \in E_i$.*

(iv) *If $z \in E_i^+$ and $z \in E_j^+$, then $i = j$.*

*Proof.* The positively oriented side inequalities give the half-plane and face representations

$$P = \bigcap_i \{z : D_i(z) \geq 0\}, \qquad P \cap \{z : D_i(z) = 0\} = E_i.$$

Since the displayed sides are the maximal nondegenerate boundary faces, the nondegenerate boundary faces are precisely the displayed sides and the only sides incident with $x_i$ are $E_i$ and $E_{i+1}$. Thus a boundary point lies either in one relative side interior or at one displayed vertex, giving the vanishing-index list in (ii) and the implication in (iii). The half-open convention assigns every relative side interior to its side and assigns the vertex $x_i$ only to the incoming side $E_i^+$, proving the partition and disjointness statements. □

**Lemma 4.4** (A strict convex combination on the boundary lies in one side)**.** *Let $R$ be a convex polygon with nonempty interior, let $A, B \in R$ with $A \neq B$, and let $0 < s < 1$. If*

$$(1-s)A + sB \in \partial R,$$

*then $A$ and $B$ lie on one common side of $R$, and the whole segment $[A, B]$ is contained in that side.*

*Proof.* Choose a nonconstant supporting affine functional $f \leq c$ for $R$ at $z = (1-s)A + sB$. Since

$$c = f(z) = (1-s)f(A) + sf(B), \qquad f(A), f(B) \leq c,$$

positivity of both coefficients forces $f(A) = f(B) = c$. Thus $A$ and $B$ belong to the exposed face $R \cap \{f = c\}$. This face is nondegenerate because $A \neq B$, so it is a side of the polygon and contains $[A, B]$. □

**Lemma 4.5** (Locating the common side from adjacent half-open memberships)**.** *Let $P$ be a positively oriented polygon. Suppose*

$$A \in E_{j-1}^{+}, \qquad B \in E_j^{+}, \qquad A \neq B,$$

*and suppose that a strict convex combination $(1-s)A + sB$, $0 < s < 1$, lies on $\partial P$. Then*

$$A = x_{j-1}, \qquad [A, B] \subseteq E_j.$$

*In particular there is a unique $t \in (0, 1]$ such that*

$$B = (1-t)x_{j-1} + tx_j,$$

*and every strict convex combination of $A$ and $B$ lies in* $\operatorname{relint} E_j$.

*Proof.* By lemma 4.4, the two endpoints $A, B$ lie on one common side of $P$. Hence $Z(A) \cap Z(B)$ is nonempty. The half-open boundary partition gives

$$Z(A) \subseteq \{j-1, j\}, \qquad Z(B) \subseteq \{j, j+1\},$$

and the common index can only be $j$. The index $j$ belongs to $Z(A)$ only when $A$ is the right endpoint of $E_{j-1}^{+}$, namely $A = x_{j-1}$. The common side is therefore $E_j$, so $B \in E_j$ as well. Since $B \in E_j^{+}$ and $B \neq A = x_{j-1}$, the displayed representation with $t \in (0, 1]$ is unique. Since $A \neq B$, every strict convex combination of $A$ and $B$ lies in $\operatorname{relint}[A, B] \subseteq \operatorname{relint} E_j$. $\square$

**Lemma 4.6** (Determinant test for side membership and containment)**.** *Let $P$ be a positively oriented $N$-gon with its complete displayed vertex list $x_i$ and half-open sides $E_i^{+}$. Let $\eta_j$ be $N$ points such that, for every $j$, either*

$$\eta_j = x_j,$$

*or*

$$\eta_j = (1-t_j)x_{j-1} + t_j x_j, \qquad 0 < t_j < 1.$$

*Put*

$$C_{r,k} = \det(x_r - x_{r-1}, x_k - x_{r-1}),$$

*so $C_{r,r-1} = C_{r,r} = 0$, and put*

$$D_{r,j} = \det(x_r - x_{r-1}, \eta_j - x_{r-1}).$$

*Then $\eta_j \in E_j^{+}$ for every $j$ and*

$$D_{r,j} \geq 0 \qquad (0 \leq r, j < N).$$

*More precisely, if $\eta_j = x_j$, then $D_{r,j} = C_{r,j}$; if $\eta_j = (1-t_j)x_{j-1} + t_j x_j$, then*

$$D_{r,j} = (1-t_j)C_{r,j-1} + t_j C_{r,j}.$$

*In the second case $D_{j,j} = 0$ and $D_{r,j} > 0$ for every $r \neq j$. Moreover, the points $\eta_j$ are pairwise distinct, and* $\operatorname{conv}\{\eta_j\} \subseteq P$.

*Proof.* The endpoint $x_j$ belongs to $E_j^{+}$ and to no other half-open side, while a strict convex combination of $x_{j-1}$ and $x_j$ belongs to $\operatorname{relint} E_j \subset E_j^{+}$. This proves the asserted side-membership statements. The determinant identities follow by bilinearity. Because the displayed vertices are exactly the extreme points of $P$ and every displayed side is a maximal boundary segment, a nonincident vertex $x_k$ lies strictly in the inward half-plane determined by $E_r$. Thus $C_{r,k} > 0$ for every nonincident vertex, while $C_{r,r-1} = C_{r,r} = 0$ for the two incident vertices. Hence all entries of the displayed matrix are nonnegative, with strict positivity in the stated nonincident cases. Because $\eta_j \in E_j^{+}$ and the half-open sides are pairwise disjoint, $\eta_j \neq \eta_k$ whenever $j \neq k$. The inequalities $D_{r,j} \geq 0$ are exactly the oriented side inequalities of $P$, so every $\eta_j$ belongs to $P$ and therefore $\operatorname{conv}\{\eta_j\} \subseteq P$. $\square$

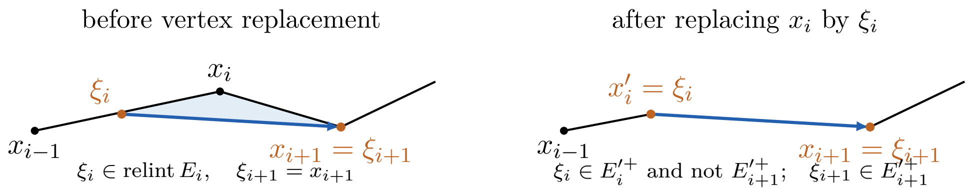


Figure 1: Half-open side membership for the unchanged image vertices in the elementary vertex-replacement pattern. The shaded region is the only part discarded from $P$; the blue edge is an actual edge of $Q = \lambda P$.

**Lemma 4.7** (Side membership after replacing one vertex)**.** *Let $P$ be a positively oriented $N$-gon with complete vertex list $x_r$ and sides $E_r = [x_{r-1}, x_r]$. Choose $\xi_i \in \operatorname{relint} E_i$ and define*

$$P' = \operatorname{conv}\Big((\operatorname{Ext} P \setminus \{x_i\}) \cup \{\xi_i\}\Big).$$

*Then $P'$ is an $N$-gon whose complete positive cyclic vertex list is obtained from that of $P$ by replacing $x_i$ by $x_i' = \xi_i$. In particular, the points $x_{i-1}, x_i' = \xi_i, x_{i+1}$ occur in that positive cyclic order. For the two new sides*

$$E_i' = [x_{i-1}, \xi_i], \qquad E_{i+1}' = [\xi_i, x_{i+1}],$$

*one has*

$$\xi_i \in E_i'^{+}, \qquad x_{i+1} \in E_{i+1}'^{+}.$$

*Proof.* There is a unique $t \in (0,1)$ such that

$$\xi_i = (1-t)x_{i-1} + t x_i.$$

Let $L = \operatorname{aff}(x_{i-1}, x_{i+1})$. Convexity and the completeness of the vertex list imply that $x_i$ lies in one open half-plane bounded by $L$, whereas every other vertex except $x_{i-1}, x_{i+1}$ lies in the opposite open half-plane. Since $\xi_i = (1-t)x_{i-1} + t x_i$ with $t > 0$, the point $\xi_i$ lies in the same open half-plane as $x_i$. It therefore does not belong to the convex hull of the unchanged vertices, so it is an extreme point of $P'$.

Each unchanged vertex remains extreme: a linear functional that strictly exposes it on $P$ still strictly exposes it among the displayed points defining $P'$ (its value at $\xi_i \in P$ is also strictly smaller). Thus all $N$ displayed points are extreme points of $P'$.

The supporting line of the old side $E_i$ meets these displayed points only at $x_{i-1}$ and $\xi_i$, so $[x_{i-1}, \xi_i]$ is an exposed side of $P'$. For every unchanged vertex $x_k \neq x_{i+1}$, the function $\xi \mapsto \det(x_{i+1} - \xi, x_k - \xi)$ is affine in $\xi$. Hence

$$\det(x_{i+1} - \xi_i, x_k - \xi_i) = (1-t)\det(x_{i+1} - x_{i-1}, x_k - x_{i-1}) + t\det(x_{i+1} - x_i, x_k - x_i) > 0.$$

The triple-sign criterion makes both terms nonnegative and at least one strictly positive (when $x_k = x_{i-1}$, the first term is zero and the second is positive). Hence all the other displayed points lie strictly in the left half-plane of the directed line from $\xi_i$ to $x_{i+1}$, so $[\xi_i, x_{i+1}]$ is also an exposed side of $P'$. Every old side not incident with $x_i$ remains exposed by its old supporting line. These sides join all $N$ extreme points in the displayed positive cyclic order. Consequently that list is exactly the complete positive cyclic vertex list of $P'$; in particular, $P'$ is an $N$-gon and the two sides incident with $\xi_i$ are $E_i' = [x_{i-1}, \xi_i]$ and $E_{i+1}' = [\xi_i, x_{i+1}]$.

The point $\xi_i$ is the right endpoint of $E_i'$ and the left endpoint of $E_{i+1}'$, so the right-half-open convention assigns it to $E_i'^{+}$. The point $x_{i+1}$ is the right endpoint of $E_{i+1}'$, so it belongs to $E_{i+1}'^{+}$. $\square$

## 4.2 Clipping along image edges

The cutting argument below uses only actual edges of $Q = \lambda P$. This makes the two endpoints distinct and specifies unambiguously which closed half-plane is retained, including the case in which the same line also supports $P$.

**Lemma 4.8** (Clipping along an image edge and bounding the number of vertices)**.** *Let $P$ be an $N$-gon, let $\lambda \neq 0$, and put $Q = \lambda P$. Assume $Q \subseteq P$ and that every vertex of $Q$ lies on $\partial P$. List the vertices of $Q$ in their positive cyclic order as $y_0, \dots, y_{N-1}$. Their order on $\partial P$ is the same cyclic order.*

*For the edge $[y_j, y_{j+1}]$ of $Q$, let $H_j$ be the closed half-plane bounded by its line and containing $Q$, and put*

$$P_j = P \cap H_j. \tag{4.1}$$

*For every $j$, define the closed and open positive boundary arcs*

$$A_j = [y_j, y_{j+1}]_{\partial P}, \qquad A_j^\circ = (y_j, y_{j+1})_{\partial P}.$$

*Also put*

$$k_j = \#(\operatorname{Ext} P \cap A_j),$$

*so an endpoint is counted exactly when it is already a vertex of $P$.*

*If $P_j \neq P$, the set removed by the cut is $P \setminus H_j = P \setminus P_j$ and has nonempty interior, while the part of the boundary that is removed is exactly*

$$\partial P \setminus P_j = A_j^\circ.$$

*In particular, $A_j^\circ$ contains no vertex of $Q$. In this case*

$$Q \subseteq P_j \subsetneq P, \qquad \lambda P_j \subseteq P_j, \qquad |\operatorname{Ext} P_j| \leq N + 2 - k_j. \tag{4.2}$$

*If $P_j = P$, the line through $[y_j, y_{j+1}]$ also supports $P$. In that case the positive half-open boundary arc $(y_j, y_{j+1}]_{\partial P}$ lies in one side of $P$ and contains at most one vertex of $P$.*

*Proof.* Because multiplication by $\lambda \neq 0$ is an invertible real-linear map with determinant $|\lambda|^2 > 0$, it preserves orientation, and $Q = \lambda P$ is a polygon whose image vertex list is complete. All of its vertices lie on $\partial P$, so lemma 2.6 shows that the positive cyclic order on $\partial Q$ is exactly the order induced from $\partial P$. Thus consecutive $Q$-vertices bound a unique positive boundary arc of $P$ containing no other $Q$-vertex.

The line of a $Q$-edge supports $Q$, so $Q \subseteq H_j$. Hence $Q \subseteq P_j \subseteq P$. Since $P_j \subseteq P$,

$$\lambda P_j \subseteq \lambda P = Q \subseteq P_j.$$

Suppose $P_j \neq P$, and write $L_j = \operatorname{aff}(y_j, y_{j+1})$. The full-dimensional polygon $Q$ has interior points in the open half-plane bounded by $L_j$ and contained in $H_j$, while $P_j \neq P$ supplies a point of $P$ in the opposite open half-plane. Thus $P$ has points on both sides of $L_j$. Since the interior of a full-dimensional convex set is dense in that set, the discarded open half-plane also contains an interior point of $P$. Therefore $P \setminus H_j$ has nonempty interior.

We claim that every point of $\operatorname{relint} I$, where $I = P \cap L_j$, lies in $\operatorname{int} P$. Suppose instead that $z \in \operatorname{relint} I \cap \partial P$. Choose a nonzero linear functional $\ell$ and a number $c$ such that

$$\ell \leq c \quad \text{on } P, \qquad \ell(z) = c.$$

With $d = y_{j+1} - y_j$, relative interior membership gives $z \pm \varepsilon d \in I$ for all sufficiently small $\varepsilon > 0$. Applying the two support inequalities at these points forces $\ell(d) = 0$. Hence the supporting line $\{\ell = c\}$ contains $z$ and has direction $d$, so it is exactly $L_j$. This would place all of $P$ in one closed half-plane bounded by $L_j$, contradicting the points of $P$ on both sides. The claim follows.

The points $y_j, y_{j+1}$ belong to $\partial P \cap I$, so the claim forces them to be the two endpoints of the compact segment $I$. If $z \in A_j^\circ$, then $y_j, z, y_{j+1}$ occur in positive cyclic order on $\partial P$. By lemma 2.6,

$$\det(z - y_j, y_{j+1} - y_j) > 0,$$

or equivalently $\det(y_{j+1} - y_j, z - y_j) < 0$. On the other hand, because $[y_j, y_{j+1}]$ is a positively oriented edge of $Q$, the polygon $Q$ lies in the closed half-plane where this last determinant is nonnegative; this is $H_j$. Thus $A_j^\circ$ lies in the discarded open half-plane.

For a point $z$ on the complementary open boundary arc $(y_{j+1}, y_j)_{\partial P}$, the points $y_{j+1}, z, y_j$ occur in positive cyclic order. The same determinant test, now with the endpoints reversed, gives

$$\det(y_{j+1} - y_j, z - y_j) > 0,$$

so this complementary arc lies in the interior of $H_j$. Moreover, the preceding description of $I = P \cap L_j$ gives $\partial P \cap L_j = \{y_j, y_{j+1}\}$. The endpoints themselves belong to $P_j$, and the two open arcs account for all other boundary points. Consequently

$$\partial P \setminus P_j = A_j^\circ.$$

This open arc contains no $Q$-vertex because $Q \subseteq H_j$.

Clipping removes every $P$-vertex on $A_j^\circ$. Each endpoint $y_j, y_{j+1}$ is retained, and it is a new candidate vertex precisely when it was not already a $P$-vertex. Thus the number of candidate vertices is

$$N - \#(\operatorname{Ext} P \cap A_j^\circ) + \#\{\text{endpoints not already } P\text{-vertices}\} = N + 2 - k_j.$$

Deleting any listed point that lies in the interior of a straight boundary segment can only lower the number of extreme vertices.

If $P_j = P$, the edge line supports $P$ as well. Its intersection with $P$ is a nondegenerate face and therefore one displayed side. The positive arc from $y_j$ to $y_{j+1}$ lies in that side, so after its left endpoint is excluded it contains at most one vertex of $P$. ☐

**Lemma 4.9** (Old-vertex bound on discarded boundary arcs)**.** *Let $T(z) = \lambda z$, where $0 < |\lambda| < 1$ and $\lambda \notin \mathbb{R}$, and assume that $T$ is $N$-critical. Among the $N$-gons satisfying $\lambda P \subseteq P$, impose the unit maximal-radius normalization*

$$\max_{z \in P} |z| = 1. \tag{4.3}$$

*The normalized class is nonempty and contains a member of least area; choose such a polygon $P$. Let $Q = \lambda P$ and use the notation of lemma 4.8. Then every proper image-edge cut, meaning $P_j \neq P$, satisfies*

$$k_j \leq 2, \tag{4.4}$$

*and at most one proper image-edge cut has $k_j = 2$.*

*Proof.* Because $\nu_{\mathrm{poly}}(T) = N$, an invariant polygon with exactly $N$ extreme vertices exists. Its maximal distance from the origin is positive, and rescaling it to make that maximum equal to 1 preserves both invariance and the number of extreme vertices. Thus the normalized class is nonempty. Write

$$\overline{\mathbb{D}} = \{z \in \mathbb{R}^2 : |z| \leq 1\}.$$

We next prove that the normalized class is compact in the Hausdorff metric. Take an arbitrary sequence in this class and display each member as the convex hull of its $N$ vertices, all of which lie in $\overline{\mathbb{D}}$. Compactness of $(\overline{\mathbb{D}})^N$ gives a subsequence for which the $N$-tuples converge. By lemma A.4, their convex hulls converge in the Hausdorff metric to a convex hull $P_\infty$ with at most $N$ extreme points; the same lemma preserves the maximal radius and the inclusion $\lambda P_\infty \subseteq P_\infty$.

The limit cannot be a singleton: if $P_\infty = \{v\}$, normalization gives $|v| = 1$, while invariance gives $\lambda v = v$, impossible because $0 < |\lambda| < 1$. It cannot be a nondegenerate segment either. Indeed, if $a \neq b$ are its endpoints, then $\lambda a, \lambda b \in P_\infty$, so the nonzero vectors $b - a$ and $\lambda(b-a)$ are parallel, which is impossible for nonreal $\lambda$. Thus $P_\infty$ has nonempty interior. If it had fewer than $N$ extreme points, it would be an invariant polygon with fewer than $N$ vertices, contrary to $\nu_{\text{poly}}(T) = N$. Hence it has exactly $N$ extreme points and belongs to the normalized class. We have shown that every sequence has a subsequence converging within the class, so the class is compact. Area is continuous by lemma A.4; therefore a least-area normalized representative exists.

By theorem 3.2, every vertex of $Q = \lambda P$ lies on $\partial P$. Thus all hypotheses of lemma 4.8 hold for this least-area polygon, and its clipping conclusions apply.

Every point of $P$ is a convex combination of its vertices, so convexity of the norm gives

$$\max_{z \in P} |z| = \max_{x \in \operatorname{Ext} P} |x|.$$

Choose a vertex $v$ with $|v| = 1$. Since $Q \subseteq \rho\overline{\mathbb{D}}$, where $\rho = |\lambda| < 1$, the point $v$ is not a vertex of $Q$. If a proper image-edge cut had $k_j \geq 3$, then $P_j$ would contain the full-dimensional polygon $Q$, would satisfy $\lambda P_j \subseteq P_j$, and would have at most $N-1$ vertices by (4.2). This contradicts $\nu_{\text{poly}}(T) = N$.

Suppose a proper image-edge cut has $k_j = 2$ and $v \notin A_j^\circ$. By the discarded-boundary equality in lemma 4.8, $v \in P_j$. Hence $P_j$ remains normalized. It contains the full-dimensional polygon $Q$, is invariant, and has at most $N$ vertices. Criticality rules out fewer than $N$ vertices, so $P_j$ has exactly $N$ extreme vertices and is another invariant $N$-gon. But $P_j \subsetneq P$ and both polygons have nonempty interior, so lemma A.5 gives $\operatorname{area}(P_j) < \operatorname{area}(P)$. This contradicts the choice of $P$.

The open positive arcs $A_j^\circ = (y_j, y_{j+1})_{\partial P}$ are pairwise disjoint and together cover $\partial P \setminus \operatorname{Ext} Q$. Since $v$ is not a $Q$-vertex, it belongs to exactly one of them. Therefore at most one proper image-edge cut with $k_j = 2$ can escape the preceding contradiction. □

**Lemma 4.10** (Finite cyclic endpoint count). *Let $(r_j)_{j\in\mathbb{Z}/N\mathbb{Z}}$ take values in $\{0,1,2\}$, satisfy $\sum_j r_j = N$, and contain at most one entry equal to 2. Let $(c_j)_{j\in\mathbb{Z}/N\mathbb{Z}}$ be a binary cyclic word. If an entry $r_s = 2$ occurs, assume*

$$c_s = 0, \qquad c_{s+1} = 1, \tag{4.5}$$

*and assume that every transition $c_j = 1$, $c_{j+1} = 0$ can occur only at an index $j$ for which $r_j = 0$. Then either $r_j = 1$ for every $j$, or there are unique indices $s, t$ such that*

$$r_s = 2, \qquad r_t = 0, \qquad r_j = 1 \quad (j \notin \{s,t\}),$$

*and in the latter case*

$$\ell_j := r_j + c_j - c_{j+1} = 1 \qquad (j \in \mathbb{Z}/N\mathbb{Z}). \tag{4.6}$$

*Proof.* If no entry is 2, then every $r_j \leq 1$ and the sum of the $N$ entries is $N$, so all entries equal 1. Suppose $r_s = 2$. Its surplus over 1 is one. Since no second entry can exceed 1, the equality $\sum_j r_j = N$ forces a unique zero $r_t = 0$ and makes every remaining entry equal to 1.

The transition $0 \to 1$ at $s$ in (4.5) forces at least one transition $1 \to 0$ in the cyclic binary word. By hypothesis it can occur only at the unique zero index $t$. In a cyclic binary word the number of $0 \to 1$ transitions equals the number of $1 \to 0$ transitions, because

$$\sum_j (c_{j+1} - c_j) = 0.$$

Consequently the transitions at $s$ and $t$ are the unique transitions of their respective types. Hence $c_j = c_{j+1}$ for $j \notin \{s, t\}$, while

$$(c_s, c_{s+1}) = (0, 1), \qquad (c_t, c_{t+1}) = (1, 0).$$

Substitution gives $\ell_s = 2 + 0 - 1 = 1$, $\ell_t = 0 + 1 - 0 = 1$, and $\ell_j = 1$ at every remaining index. $\square$

**Lemma 4.11** (Cyclic interlacing with endpoint bookkeeping)**.** *Let $P$ be the least-area representative of lemma 4.9. Then at least one of the following two alternatives holds:*
(i) *every arc $(y_j, y_{j+1}]_{\partial P}$ contains exactly one vertex of $P$;*
(ii) *every arc $[y_j, y_{j+1})_{\partial P}$ contains exactly one vertex of $P$.*
*In alternative* (ii)*, every right-half-open side of $P$ contains exactly one vertex of $Q$. In alternative* (i)*, the conjugate configuration*

$$\widetilde{P} = \overline{P}, \qquad \widetilde{Q} = \overline{Q} = \overline{\lambda}\,\widetilde{P}, \qquad \widetilde{x}_i = \overline{x}_{-i} \tag{4.7}$$

*has the same property when its vertices are read in positive cyclic order. Thus a right-half-open convention is obtained without ever reading a convex polygon in the wrong orientation.*

*Proof.* Put

$$c_j = \begin{cases} 1, & y_j \in \operatorname{Ext} P, \\ 0, & y_j \notin \operatorname{Ext} P, \end{cases} \qquad r_j = \#(\operatorname{Ext} P \cap (y_j, y_{j+1}]_{\partial P}).$$

The half-open arcs partition the $N$ vertices of $P$, so

$$\sum_{j=0}^{N-1} r_j = N. \tag{4.8}$$

The closed arc $A_j = [y_j, y_{j+1}]_{\partial P}$ is obtained from the right-half-open arc $(y_j, y_{j+1}]_{\partial P}$ by adding its left endpoint. Consequently, for every $j$—whether or not the image-edge line also supports $P$—one has

$$k_j = r_j + c_j. \tag{4.9}$$

When $P_j = P$, lemma 4.8 gives $r_j \le 1$. When $P_j \ne P$, lemma 4.9 and (4.9) give $r_j \le k_j \le 2$. Hence

$$0 \le r_j \le 2, \qquad \text{and at most one } r_j \text{ equals } 2. \tag{4.10}$$

Indeed, $r_j = 2$ is impossible when $P_j = P$. Thus any such index gives a proper image-edge cut; then (4.9) and $k_j \le 2$ force $c_j = 0$ and $k_j = 2$. By lemma 4.9, at most one proper image-edge cut has this value.

If no $r_j$ equals 2, the combination of (4.8) and (4.10) forces $r_j = 1$ for every $j$, which is alternative (i). Otherwise there are unique indices $s, t$ with

$$r_s = 2, \qquad r_t = 0, \qquad r_j = 1 \quad (j \ne s, t). \tag{4.11}$$

We determine whether the two exceptional endpoints are vertices of $P$.

First, $c_s = 0$ by (4.9). Also $c_{s+1} = 1$. Indeed, suppose that $y_{s+1}$ were not a vertex of $P$. Since $c_s = 0$, neither endpoint of the boundary arc from $y_s$ to $y_{s+1}$ would then be a vertex of $P$. Because $r_s = 2$, its open boundary arc would contain exactly two vertices of $P$ and no other vertex between them. They would therefore be consecutive vertices of $P$, while their joining side would contain no vertex of $Q$: the points $y_s$ and $y_{s+1}$ are consecutive vertices of $Q$. This contradicts lemma 4.1 and the fact that every side of $P$ intersects $Q$. Thus the binary cyclic word $(c_j)$ has a transition $0 \to 1$ at $s$.

Now consider a transition $c_j = 1$, $c_{j+1} = 0$ with $j \neq t$. Then $r_j = 1$, so (4.9) gives $k_j = 2$. The image-edge cut is proper. Indeed, if its line also supported $P$, the positive arc starts at the $P$-vertex $y_j$ and ends at a nonvertex $y_{j+1}$, so the arc must stop before the next $P$-vertex, so it would have $r_j = 0$, contrary to $r_j = 1$. It would therefore be a second proper image-edge cut with $k = 2$, which is impossible. Hence every transition $1 \to 0$ can occur only at the unique zero index $t$. The hypotheses of lemma 4.10 are now verified: (4.11) gives the unique two and zero, while the preceding paragraph gives the required $0 \to 1$ transition at $s$. Therefore the opposite half-open counts

$$\ell_j = \#(\operatorname{Ext} P \cap [y_j, y_{j+1})_{\partial P}) = r_j + c_j - c_{j+1}$$

all equal 1. This is alternative (ii).

It remains to pass from gaps between $Q$-vertices to sides between $P$-vertices. In alternative (ii), let $x_j$ be the unique $P$-vertex in $[y_j, y_{j+1})$. In the positive boundary order, $y_j$ lies after $x_{j-1}$ and no later than $x_j$; equivalently,

$$y_j \in (x_{j-1}, x_j]_{\partial P} = E_j^+.$$

No other $Q$-vertex lies there.

In alternative (i), let $x_j$ be the unique vertex in $(y_j, y_{j+1}]$. Then $y_{j+1} \in [x_j, x_{j+1})$. Apply complex conjugation and index the conjugate polygon positively by $\widetilde{x}_i = \overline{x}_{-i}$. The conjugate of $[x_j, x_{j+1})$ is, with its order reversed,

$$(\widetilde{x}_{-j-1}, \widetilde{x}_{-j}];$$

therefore it is a right-half-open side of $\widetilde{P}$. Conjugation is a bijection on vertices and takes $Q$ to $\widetilde{Q} = \overline{\lambda}\,\widetilde{P}$, so every such side contains exactly one vertex of $\widetilde{Q}$. This proves the final assertion with all endpoint inclusions explicit. $\square$

**Corollary 4.12** (Bijective global half-open assignment). *In either orientation supplied by lemma 4.11, the contact assignment is a genuine labeled bijection*

$$\{\textit{vertices of } Q\} \longleftrightarrow \{E_i^+ : i \in \mathbb{Z}/N\mathbb{Z}\}.$$

*Thus every image vertex belongs to exactly one half-open side, and every half-open side contains exactly one image vertex. After the possible conjugation in (4.7), the labels may be chosen so that the unique image assigned to $E_i^+$ is denoted by $\xi_i$.*

*Proof.* The half-open sides $E_i^+ = (x_{i-1}, x_i]$ are pairwise disjoint and their union is $\partial P$. In alternative (ii) of lemma 4.11, each interval $[y_j, y_{j+1})_{\partial P}$ contains exactly one vertex of $P$, and the proof there gives $y_j \in E_j^+$. Thus every side contains at least one image vertex. There are exactly $N$ sides and exactly $N$ image vertices, while the disjoint half-open partition prevents one image vertex from belonging to two sides; hence the assignment is a bijection. In alternative (i), the same argument applies after the conjugate reindexing (4.7). This proves that the labeling is bijective. $\square$

The next step has two logically distinct parts. The half-open assignment and cyclic order make the vertex-to-side assignment an orientation-preserving degree-one correspondence of two cyclic $N$-point sets, hence a shift modulo $N$. Passing that correspondence to the universal cover of the circle introduces a possible extra full turn at each side index. The interval estimate in the proof rules out those turns separately; summation then gives a strict rational bracket for the multiplier angle.

**Lemma 4.13** (Order-preserving half-open contact assignment). *After replacing $(P, \lambda)$ by the conjugate configuration (4.7) if required, there is an invariant $N$-gon representative, a positive cyclic indexing, and an integer $\kappa$ with $1 \le \kappa \le N$ such that*

$$\xi_i := \mu x_{i-\kappa} = \beta_i x_{i-1} + \alpha_i x_i, \qquad \alpha_i > 0, \quad \beta_i \ge 0, \quad \alpha_i + \beta_i = 1. \tag{4.12}$$

*Here $\mu$ is either $\lambda$ or $\overline{\lambda}$, indices are modulo $N$, and $\vartheta$ is the unique number in $(0, 2\pi)$ such that $\mu = |\mu| e^{i\vartheta}$. If $y = \vartheta/(2\pi)$, then*

$$\frac{\kappa - 1}{N} < y < \frac{\kappa}{N}. \tag{4.13}$$

*If $\kappa = N$, every contact lies in the relative interior of its assigned side.*

*Proof.* Choose the least-area normalized invariant $N$-gon from lemma 4.9. In alternative (ii) of lemma 4.11, put $\mu = \lambda$. In alternative (i), replace the configuration by (4.7) and put $\mu = \overline{\lambda}$. In either case the vertices are now in positive cyclic order and every $E_i^+$ contains exactly one vertex $\xi_i$ of $Q = \mu P$.

Multiplication by the nonzero complex number $\mu$ preserves cyclic order. Hence the order-preserving incidence bijection between source vertices and right-half-open sides is a single cyclic shift. Indeed, after identifying both cyclic $N$-sets with $\mathbb{Z}/N\mathbb{Z}$, a cyclic-order-preserving bijection sends each successor to the successor of its image and therefore has the form $j \mapsto j + \kappa$ modulo $N$. Choose the representative $\kappa \in \{1, \dots, N\}$ of this cyclic shift and write $\xi_i = \mu x_{i-\kappa}$. Because $\xi_i \in (x_{i-1}, x_i]$, it has the unique barycentric form (4.12) with $\alpha_i > 0$ and $\beta_i \ge 0$.

Choose lifted polar angles $\Theta_i$ with $\Theta_{i+N} = \Theta_i + 2\pi$, and choose the lifted multiplier argument $\vartheta \in (0, 2\pi)$. By lemma 2.5, $0 \in \operatorname{int} P$. Hence lemma 2.10 applies: after choosing the increasing lift, polar angle is strictly increasing along every positively oriented side. The side incidence first gives, for some integer $m_i$,

$$\Theta_{i-1} < \Theta_{i-\kappa} + \vartheta + 2\pi m_i \le \Theta_i. \tag{4.14}$$

The integer $m_i$, which accounts for the possible addition of $2\pi m_i$ to the chosen lift, is necessarily zero. If $1 \le \kappa < N$, then

$$0 \le \Theta_{i-1} - \Theta_{i-\kappa} < \Theta_i - \Theta_{i-\kappa} < 2\pi,$$

because this interval spans $\kappa$ consecutive angular gaps and omits at least one positive gap. If $\kappa = N$, then

$$\Theta_{i-1} - \Theta_{i-N} = 2\pi - (\Theta_i - \Theta_{i-1}), \qquad \Theta_i - \Theta_{i-N} = 2\pi.$$

Thus the possible difference interval in (4.14) is contained in $[0, 2\pi)$ in the first case and in $(0, 2\pi]$ in the second. Since $\vartheta \in (0, 2\pi)$, the only possible integer is $m_i = 0$. Hence (4.12) gives the genuine lifted inequality

$$\Theta_{i-1} < \Theta_{i-\kappa} + \vartheta \le \Theta_i.$$

Summing for one complete period, say over $i = 1, \dots, N$, and using $\sum_{i=1}^N \Theta_{i-\kappa} = \sum_{i=1}^N \Theta_i - 2\pi\kappa$ and $\sum_{i=1}^N \Theta_{i-1} = \sum_{i=1}^N \Theta_i - 2\pi$ yields

$$2\pi(\kappa - 1) < N\vartheta \le 2\pi\kappa.$$

The final equality cannot occur. Equality in the sum would force equality in every right-hand inequality, so every contact would be an endpoint and $\mu x_{i-\kappa} = x_i$ for all $i$. Iteration around a shift orbit gives $x_i = \mu^{N/\gcd(N,\kappa)} x_i$, impossible because $|\mu| < 1$ and no vertex is 0. This proves (4.13).

If $\kappa = N$, then the source of contact $i$ is $x_i$ itself. An endpoint contact would give $\mu x_i = x_i$, again impossible. Hence every contact lies in the relative interior of its assigned side. $\square$

From this point through the end of the first-return analysis, fix the contact data supplied by lemma 4.13 and denote its complex multiplier by $\lambda$. Put

$$\theta = \arg_+(\lambda) \in (0, 2\pi), \qquad x = \frac{\theta}{2\pi}.$$

The change from the symbol $\mu$ in lemma 4.13 to $\lambda$ is only a local renaming of the multiplier in these already fixed data; it is never undone. The chosen orientation, half-open assignment, cyclic side permutation, vertex-replacement class, first-return data, and projective deformation are not changed anywhere in the vertex-replacement, finite-rotation, or adjacent-return arguments. If the final finite chain of recurrence relations is more naturally read in the opposite orientation, that reversal is performed only after those arguments have ended, by the explicit reflection lemma lemma 8.3; the resulting selected eigenvalue then receives the new symbol $\mu$.

More precisely, the *standing contact data* are a tuple

$$(P, \lambda, \kappa; (\alpha_i, \beta_i)_{i \in \mathbb{Z}/N\mathbb{Z}})$$

satisfying the following five assumptions:

(A0) multiplication by $\lambda$ is an $N$-critical elliptic contraction;

(A1) $P$ is a positively indexed invariant $N$-gon, with its complete cyclic vertex list, for this multiplication;

(A2) every side of $P$ intersects $\lambda P$, and every vertex of $\lambda P$ lies on $\partial P$;

(A3) the fixed representative $\kappa \in \{1, \ldots, N\}$ and the coefficients $(\alpha_i, \beta_i)$ satisfy (4.12), so every right-half-open side $E_i^+$ contains its assigned image vertex; and

(A4) $S = \{i : \beta_i > 0\}$ is exactly the set of indices whose assigned image vertices lie in the relative interiors of their sides.

Later references to the standing assumptions mean precisely (A0)–(A4).

**Lemma 4.14** (Iteration of endpoint equalities)**.** *For contact data satisfying* (A0)–(A4)*, extend the vertex angles by $\Theta_{i+N} = \Theta_i + 2\pi$. For every integer side index $j$,*

$$\Theta_{j-1} < \Theta_{j-\kappa} + \theta \le \Theta_j. \tag{4.15}$$

*If the contact at index $j$ is an endpoint contact, equality holds on the right. Hence, if the side indices $a + \kappa, \ldots, a + t\kappa$ are all endpoint contacts, then*

$$\Theta_a + t\theta = \Theta_{a+t\kappa}. \tag{4.16}$$

*If the first $t - 1$ destination indices are endpoint contacts and the contact at index $a + t\kappa$ lies in the relative interior of its assigned side, then instead*

$$\Theta_{a+t\kappa-1} < \Theta_a + t\theta < \Theta_{a+t\kappa}. \tag{4.17}$$

*Proof.* The side incidence gives an integer $m_j$ such that

$$\Theta_{j-1} < \Theta_{j-\kappa} + \theta + 2\pi m_j \le \Theta_j.$$

Exactly as in the lift check in lemma 4.13, the interval of possible differences is contained in $[0, 2\pi)$ when $1 \le \kappa < N$ and in $(0, 2\pi]$ when $\kappa = N$. Since the chosen multiplier argument satisfies $0 < \theta < 2\pi$, this forces $m_j = 0$ and proves (4.15). At an endpoint contact the image is $x_j$, forcing equality with $\Theta_j$. Iteration gives (4.16); replacing the final endpoint by a relative-interior contact gives (4.17). □

For the vertex-replacement and return theory we now assume

$$1 \le \kappa < N. \tag{4.18}$$

The identity cyclic side permutation $\kappa = N$ is treated directly in lemma 8.4; it requires no vertex replacement or projective deformation.

# 5 Vertex replacement and interval reduction

Call contact $i$ a *relative-interior contact* if $\beta_i > 0$ and an *endpoint contact* if $\beta_i = 0$.

Write

$$S = \{i \in \mathbb{Z}/N\mathbb{Z} : \beta_i > 0\}$$

for the set of indices whose assigned image points lie in the relative interiors of their sides. When $i \in S$ and $i+1 \notin S$, clipping the corner $x_i$ replaces the index $i$ in this set by $i+\kappa$:

$$S' = (S \setminus \{i\}) \cup \{i+\kappa\}.$$

If $i+\kappa$ already belongs to $S$, this union has one fewer element. This is only a statement about the index set: no geometric image vertices coalesce. The old target image is replaced when $x_i$ is removed, and multiplication by $\lambda \neq 0$ keeps the $N$ images of the $N$ primed vertices distinct. The next proposition proves that this set update is realized by an actual vertex replacement and preserves the complete cyclic vertex-list structure, invariance, the half-open assignment, and every unchanged contact.

**Coinciding-index cases.** The vertex-replacement theorem is used only in the proper-shift range (4.18). Put $j_0 = i+\kappa$ in $\mathbb{Z}/N\mathbb{Z}$. The complete list of possible index coincidences is

$$j_0 = i+1 \iff \kappa = 1, \qquad j_0 - 1 = i+1 \iff \kappa = 2, \qquad j_0 = i-1 \iff \kappa = N-1.$$

The first equality is the only overlap between the changed source index and the changed target-side index. The second is the only case in which the preceding image $A$ below is the unchanged endpoint $x_{i+1}$ of a changed side. In the third case the changed image is assigned to the preceding side index $i-1$, but neither endpoint of that target side is the replaced vertex $x_i$; hence it belongs to the generic incidence row. For $N \geq 4$, the cases $\kappa = 1$, $\kappa = 2$, $\kappa = N-1$, and $3 \leq \kappa \leq N-2$ are disjoint and exhaustive, with the last two sharing the same algebraic formula. When $N = 3$, the cases $\kappa = 2$ and $\kappa = N-1$ coincide, and the $\kappa = 2$ row applies. All equalities and side labels in the proposition below are interpreted modulo $N$, so wrap-around creates no additional case.

**Proposition 5.1** (Exact vertex replacement and its set update)**.** *Let*

$$(P, \lambda, \kappa; (\alpha_j, \beta_j)_{j\in\mathbb{Z}/N\mathbb{Z}})$$

*be contact data satisfying* (A0)–(A4) *with* $1 \leq \kappa < N$*, and put*

$$S = \{j : \beta_j > 0\}.$$

*Suppose* $i \in S$ *and* $i+1 \notin S$*. Thus*

$$\xi_i = \beta_i x_{i-1} + \alpha_i x_i \in \operatorname{relint} E_i, \qquad \xi_{i+1} = x_{i+1}.$$

*Set*

$$x'_i = \xi_i, \qquad x'_j = x_j \quad (j \neq i), \qquad P' = \operatorname{conv}\{x'_0, \dots, x'_{N-1}\},$$

*and put* $\eta_j = \lambda x'_{j-\kappa}$*. Then* $P'$ *is an invariant* $N$*-gon with the displayed complete cyclic vertex list, and every* $\eta_j$ *belongs to* $E'^{+}_j$*. Consequently there are unique coefficients* $(\alpha'_j, \beta'_j)$ *satisfying*

$$\eta_j = \beta'_j x'_{j-1} + \alpha'_j x'_j, \qquad \alpha'_j > 0, \quad \beta'_j \geq 0, \quad \alpha'_j + \beta'_j = 1,$$

*and the primed tuple*

$$(P', \lambda, \kappa; (\alpha'_j, \beta'_j)_{j\in\mathbb{Z}/N\mathbb{Z}})$$

*again satisfies* (A0)–(A4)*. The chosen representative* $\kappa \in \{1, \dots, N-1\}$ *of the cyclic shift is unchanged. More precisely, for some* $t \in (0, 1]$ *the new coefficients are*

$$\alpha'_i = 1, \quad \beta'_i = 0, \qquad \alpha'_{i+\kappa} = \alpha_i t, \quad \beta'_{i+\kappa} = 1 - \alpha_i t, \tag{5.1}$$

*and* $(\alpha'_j, \beta'_j) = (\alpha_j, \beta_j)$ *for* $j \notin \{i, i+\kappa\}$*. The corresponding index set after replacement is*

$$S' = (S \setminus \{i\}) \cup \{i + \kappa\}. \tag{5.2}$$

*In particular, if* $i + \kappa \in S$*, the number of relative-interior contacts decreases by one.*

*Proof.* All indices are modulo $N$. Write

$$D(a, b, c) = \det(b - a, c - a).$$

For distinct vertices in the complete cyclic list of a positively oriented polygon, $D(x_a, x_b, x_c) > 0$ whenever $x_a, x_b, x_c$ occur in positive cyclic order.

Put $\alpha = \alpha_i$ and $\beta = \beta_i$. The relative-interior condition at contact $i$ gives

$$\xi_i = \beta x_{i-1} + \alpha x_i, \qquad 0 < \alpha, \beta < 1, \qquad \alpha + \beta = 1.$$

*Step 1: intersect with the retained half-plane bounded by an image edge.* The global half-open assignment puts no image vertex in the boundary arc from $\xi_i$ to $\xi_{i+1} = x_{i+1}$, apart from its endpoints. Since the cyclic order of the vertices of $Q = \lambda P$ agrees with their order on $\partial P$, the two points $\xi_i$ and $\xi_{i+1}$ are consecutive vertices of $Q$. Hence, for

$$H = \{z : D(\xi_i, x_{i+1}, z) \geq 0\},$$

one has $Q \subseteq H$. The sign of every displayed vertex of $P$ relative to the cutting line is explicit:

$$D(\xi_i, x_{i+1}, x_i) = -\beta D(x_{i-1}, x_i, x_{i+1}) < 0, \tag{5.3}$$

$$D(\xi_i, x_{i+1}, x_k) = \beta D(x_{i-1}, x_{i+1}, x_k) + \alpha D(x_i, x_{i+1}, x_k) > 0 \tag{5.4}$$

for $k \notin \{i, i+1\}$, while $D(\xi_i, x_{i+1}, x_{i+1}) = 0$. In (5.4), convexity makes the first summand nonnegative and the second positive; this also covers $k = i - 1$. Thus the cutting line meets $\partial P$ exactly at $\xi_i$ and $x_{i+1}$, and $x_i$ is the only discarded displayed vertex. Consequently

$$P' = P \cap H = \operatorname{conv}(x_0, \dots, x_{i-1}, \xi_i, x_{i+1}, \dots, x_{N-1}). \tag{5.5}$$

Since $Q \subseteq P' \subseteq P$,

$$\lambda P' \subseteq \lambda P = Q \subseteq P'. \tag{5.6}$$

For completeness, the only changed consecutive-turn determinants in the displayed boundary list are

$$\begin{aligned}
D(x_{i-2}, x_{i-1}, \xi_i) &= \alpha D(x_{i-2}, x_{i-1}, x_i) > 0, \\
D(x_{i-1}, \xi_i, x_{i+1}) &= \alpha D(x_{i-1}, x_i, x_{i+1}) > 0, \\
D(\xi_i, x_{i+1}, x_{i+2}) &= \beta D(x_{i-1}, x_{i+1}, x_{i+2}) + \alpha D(x_i, x_{i+1}, x_{i+2}) > 0.
\end{aligned}$$

All other consecutive-turn determinants are unchanged. Together with the convex intersection description (5.5), these inequalities show that the displayed list consists of $N$ distinct extreme points in positive cyclic order. Hence $P'$ is an invariant $N$-gon with the displayed complete cyclic vertex list.

Because the multiplier is the same $N$-critical map, theorem 3.2 additionally gives full side and vertex touching for $P'$. Thus the only remaining part of the standing assumptions is the labeled right-half-open incidence.

*Step 2: all unchanged sources and target sides are accounted for.* Put $j_0 = i + \kappa$. Apart from the location of the changed image $\eta_{j_0}$, the complete incidence table is

| side index $j$ | new image | target side | contact type |
|---|---|---|---|
| $i$ | $\eta_i = \xi_i = x'_i$ | $E'_i = [x_{i-1}, \xi_i]$ | endpoint |
| $i+1$, $\kappa \neq 1$ | $\eta_{i+1} = \xi_{i+1} = x_{i+1}$ | $E'_{i+1} = [\xi_i, x_{i+1}]$ | endpoint |
| $j_0$ | $\eta_{j_0} = \lambda\xi_i$ | $E'_{j_0}$ | checked below |
| $j \notin \{i, i+1, j_0\}$ | $\eta_j = \xi_j$ | $E'_j = E_j$ | unchanged |

The rows are disjoint: if $\kappa = 1$, the second row is absent and $j_0 = i+1$; otherwise $i, i+1, j_0$ are distinct. Thus only index $j_0$ remains to be located.

*Step 3: the changed image has exactly one possible side.* Set

$$A = \lambda x_{i-1} = \xi_{j_0-1}, \qquad B = \lambda x_i = \xi_{j_0}.$$

Then

$$\eta_{j_0} = \lambda\xi_i = \beta A + \alpha B \in \operatorname{relint}[A, B]. \tag{5.7}$$

The endpoint incidences after replacement are exhausted by

| $\kappa$ | $A$ | $B$ |
|---|---|---|
| 1 | $A = x'_i \in E'^{+}_i = E'^{+}_{j_0-1}$ | $B = x_{i+1} \in E'^{+}_{i+1} = E'^{+}_{j_0}$ |
| 2 | $A = x_{i+1} \in E'^{+}_{i+1} = E'^{+}_{j_0-1}$ | $B = \xi_{i+2} \in E^{+}_{i+2} = E'^{+}_{j_0}$ |
| $\kappa \notin \{1,2\}$ | $A = \xi_{j_0-1} \in E^{+}_{j_0-1} = E'^{+}_{j_0-1}$ | $B = \xi_{j_0} \in E^{+}_{j_0} = E'^{+}_{j_0}$ |

Thus, uniformly in $\kappa$,

$$A \in E'^{+}_{j_0-1}, \qquad B \in E'^{+}_{j_0}. \tag{5.8}$$

The point $x'_i$ is an extreme point of $P'$, and multiplication by the nonzero scalar $\lambda$ maps extreme points bijectively to extreme points of $\lambda P'$. Thus $\eta_{j_0} = \lambda x'_i$ is a vertex of $\lambda P'$. Full vertex touching gives $\eta_{j_0} \in \partial P'$. Since $\eta_{j_0} = \beta A + \alpha B$ with $A \neq B$ and $0 < \alpha, \beta < 1$, the hypotheses of lemma 4.5 apply to the primed polygon, using (5.8). Hence

$$A = x'_{j_0-1}, \qquad [A, B] \subseteq E'_{j_0},$$

Notice that when $\kappa \geq 2$, contact $j_0 - 1 = i + \kappa - 1$ was already an endpoint contact before the replacement. For $\kappa = 2$, this is the permitted-replacement hypothesis $i+1 \notin S$. For $\kappa \geq 3$, the vertex in question is unchanged, and the identities

$$A = \xi_{j_0-1} = x'_{j_0-1} = x_{j_0-1}$$

imply $\beta_{j_0-1} = 0$. Thus the "unchanged" contact-type entry for this index in Step 2 is consistent; the endpoint property is a consequence of the existing hypotheses, not an additional assumption.

Because $B \in E'^{+}_{j_0}$ and $B \neq A$, there is a unique $t \in (0, 1]$ such that

$$B = A + t(x'_{j_0} - A).$$

Substitution in (5.7) now yields

$$\eta_{j_0} = A + \alpha t(x'_{j_0} - A), \qquad 0 < \alpha t < 1.$$

Therefore $\eta_{j_0} \in \operatorname{relint} E'_{j_0}$; equivalently, its new barycentric coefficients are

$$\alpha'_{j_0} = \alpha_i t \in (0, 1), \qquad \beta'_{j_0} = 1 - \alpha_i t \in (0, 1).$$

*Step 4: complete assignment and check every side inequality.* Thus contact $i$ changes from relative-interior to endpoint contact, contact $i+\kappa$ lies in the relative interior after replacement, and every other contact retains its type. This proves the coefficient update (5.1) and the set update (5.2). We now verify all labelled side incidences, not merely the local ones shown in the construction.

For the polygon $P'$ put

$$C'_{r,k} = \det(x'_r - x'_{r-1}, x'_k - x'_{r-1})$$

with cyclic indices, and extend this notation by $C'_{r,r-1} = C'_{r,r} = 0$. Since $P'$ has the displayed complete vertex list and is positively oriented,

$$P' = \{z : \det(x'_r - x'_{r-1}, z - x'_{r-1}) \geq 0 \text{ for every } r\},$$

and the half-open sides $E'^{+}_r$ form a disjoint partition of $\partial P'$. The following list exhausts the incidences.

(a) $\eta_i = x'_i \in E'^{+}_i$.
(b) If $\kappa \neq 1$, then $\eta_{i+1} = x'_{i+1} \in E'^{+}_{i+1}$.
(c) If $j \notin \{i, i+1, i+\kappa\}$, then the source and the target side are unchanged, so $\eta_j = \xi_j \in E^{+}_j = E'^{+}_j$.
(d) For $j_0 = i + \kappa$, the preceding step proved $\eta_{j_0} \in \operatorname{relint} E'_{j_0}$.

These alternatives are disjoint and cover all $j \in \mathbb{Z}/N\mathbb{Z}$; when $\kappa = 1$ the second item is intentionally absent and index $i+1$ is covered by the final item. Hence each image vertex lies on its specified half-open side, and the disjoint half-open partition precludes any conflict between the assignments.

Equivalently, lemma 4.6, applied to the primed vertices and the just-listed points $\eta_j$, gives the complete side matrix. In expanded form: if $\eta_j = x'_j$, then

$$\det(x'_r - x'_{r-1}, \eta_j - x'_{r-1}) = C'_{r,j} \geq 0;$$

if $\eta_j = (1 - t_j)x'_{j-1} + t_j x'_j$ with $0 < t_j < 1$, then for every $r$

$$\det(x'_r - x'_{r-1}, \eta_j - x'_{r-1}) = (1 - t_j)C'_{r,j-1} + t_j C'_{r,j} \geq 0.$$

The formula gives the exact zero on the assigned side and strict positivity away from incident sides. Thus the labeled incidences independently imply $\lambda P' \subseteq P'$; the earlier inclusion (5.6) was not being used to infer the side assignment. Finally, the algebraic identities remain

$$\eta_j = \lambda x'_{j-\kappa} \qquad (j \in \mathbb{Z}/N\mathbb{Z}),$$

with the same chosen representative $\kappa \in \{1, \ldots, N-1\}$ of the cyclic shift. Since the target side index of the source $j - \kappa$ is still exactly $j$, this chosen representative, not only its residue class, is preserved. Together with the conclusion of theorem 3.2 above, this proves that the primed tuple satisfies (A0)–(A4). □

**Corollary 5.2** (Equivariance under the label-preserving bijection between old and new side sets)**.** *Let $(P, T, \chi)$ be the half-open contact bijection obtained in the orientation selected by lemma 4.13, and write its positive indexed form as in lemma 4.13. Thus*

$$E_i = [x_{i-1}, x_i], \qquad \chi(x_{i-\kappa}) = E_i.$$

*Then*

$$\operatorname{succ}(E_i) = E_{i+1}, \qquad \sigma(E_i) = E_{i+\kappa}. \tag{5.9}$$

*Let a permitted vertex replacement at $e = E_i$ produce the primed half-open contact assignment, write $E'_j = [x'_{j-1}, x'_j]$, and let*

$$b = b_e : \mathcal{E}(P) \longrightarrow \mathcal{E}(P'), \qquad b(E_j) = E'_j,$$

*be the label-preserving bijection between the old and new side sets. Denote the primed successor, the set of sides with relative-interior contact, and the induced cyclic permutation of side labels by* $\mathrm{succ}'$, $I'$, *and* $\sigma'$, *respectively. The permitted replacement condition is exactly* $i \in S$, $i+1 \notin S$, *and the vertex-replacement proposition gives the well-typed identities*

$$\begin{aligned} b \circ \mathrm{succ} &= \mathrm{succ}' \circ b, \\ I' &= b\big((I \setminus \{e\}) \cup \{\sigma(e)\}\big), \\ \sigma' &= b \circ \sigma \circ b^{-1}. \end{aligned} \tag{5.10}$$

*Under an invertible real-linear conjugacy, with the transported boundary orientation, these identities are carried by the induced old and new side bijections. In particular, the earlier choice between the two ambient orientations introduces no additional sign or index convention into* (5.10)*.*

*Proof.* The successor identity in (5.9) is the definition of the positive side indexing. Since the head of $E_i$ is $x_i$, its image under the induced cyclic permutation is

$$\sigma(E_i) = \chi(\mathrm{head}(E_i)) = \chi(x_i) = E_{i+\kappa},$$

because $x_i = x_{(i+\kappa)-\kappa}$. The sides in $I$ are precisely those whose assigned image points lie in their relative interiors, and the condition $\mathrm{succ}(E_i) \notin I$ is precisely $i+1 \notin S$. The indexed update (5.2), together with $b(E_j) = E'_j$, gives

$$I' = b\big((I \setminus \{e\}) \cup \{\sigma(e)\}\big).$$

Moreover,

$$\mathrm{succ}'(b(E_j)) = E'_{j+1} = b(\mathrm{succ}(E_j)),$$

and preservation of the chosen representative $\kappa$ of the cyclic shift in proposition 5.1 gives

$$\sigma'(b(E_j)) = E'_{j+\kappa} = b(\sigma(E_j)).$$

These are exactly the three identities in (5.10). Real-linear covariance follows from (2.3) in proposition 2.3. The chosen orientation is already the positively indexed orientation used above, including when it arose from the conjugate alternative in lemma 4.11; no subsequent orientation reversal occurs in the vertex-replacement module. □

**Corollary 5.3** (Geometric realization of every permitted update sequence)**.** *Starting from the half-open contact assignment, every finite sequence of updates* $S \mapsto (S \setminus \{i\}) \cup \{i+\kappa\}$ *for which* $i \in S$ *and* $i+1 \notin S$ *at that step is realized by successive invariant* $N$*-gons satisfying* (A0)–(A4)*, with the same chosen representative* $\kappa$ *of the cyclic shift. The index set* $S$ *after each step is exactly the set produced by* (5.2)*.*

*Proof.* Apply proposition 5.1 inductively. Its conclusion includes preservation of the right-half-open assignment and of $\kappa$, so the hypotheses and the description by the corresponding index set $S$ remain valid at every later step. □

**Corollary 5.4** (Right-to-left set updates are geometrically realizable)**.** *Every right-to-left sequence of set updates used in the proof of lemma 5.5 is a sequence of actual vertex replacements. More precisely, start from the current index set* $S^{(0)}$ *and define*

$$S^{(r)} = (S^{(r-1)} \setminus \{i_r\}) \cup \{i_r + \kappa\} \qquad (1 \le r \le m).$$

*If* $i_r \in S^{(r-1)}$ *and* $i_r + 1 \notin S^{(r-1)}$ *at every step, then there are invariant* $N$*-gons*

$$P = P^{(0)}, P^{(1)}, \ldots, P^{(m)}$$

*with the same multiplier and the same chosen representative $\kappa$ of the cyclic shift whose sets of indices with relative-interior contact are exactly the sets produced after the corresponding updates.*

*Proof.* This is a finite induction on the length of the sequence. At each stage the permitted-replacement condition means precisely that the contact at the current index lies in the relative interior and the contact at the next index is an endpoint, which is the hypothesis of proposition 5.1. That proposition returns an invariant polygon satisfying (A0)–(A4), with the corresponding index set $S$ updated by (5.2). Since the standing assumptions and the chosen representative $\kappa$ of the cyclic shift are part of the conclusion, the next permitted set update is again realized by a vertex replacement. No separate geometric-reachability assumption enters lemma 5.5. $\square$

## 5.1 Reduction to one cyclic interval

Let $C_N$ be the cycle graph on $\mathbb{Z}/N\mathbb{Z}$, with consecutive residues joined. For $S \subseteq \mathbb{Z}/N\mathbb{Z}$, write $\operatorname{comp}(S)$ for the number of connected components of the induced subgraph $C_N[S]$. When $S$ is proper, each such component is a maximal cyclic interval contained in $S$.

The reduction minimizes the pair $(|S|, \operatorname{comp}(S))$ lexicographically. An update whose target already belongs to the current set decreases the first coordinate, while a completed translation that joins two components decreases the second. Neither can occur from a minimizing set. Because the $\kappa$-orbits are exactly the residue classes modulo $\gcd(N, \kappa)$, every such class must meet $S$; otherwise a full orbit of endpoint equalities would imply $x_j = \lambda^{N/\gcd(N,\kappa)} x_j$. These two facts force a single cyclic interval of length at least $\gcd(N, \kappa)$.

**Lemma 5.5** (Reduction to one cyclic interval and a first-entrance identity)**.** *Put*

$$\delta = \gcd(N, \kappa). \tag{5.11}$$

*Starting from the half-open contact assignment fixed in lemma 4.13, consider only tuples satisfying* (A0)–(A4) *that are obtainable by a finite sequence of permitted vertex replacements. Choose one for which the following pair is lexicographically least:*

$$(|S|,\ \operatorname{comp}(S)). \tag{5.12}$$

*This minimization is only among such tuples reached by finitely many vertex replacements; it is neither a new vertex-count minimization nor an area minimization. Write $\varphi = |S|$. After a cyclic relabelling, its indices with relative-interior contact form one cyclic interval*

$$\{1, 2, \dots, \varphi\}, \qquad \varphi \geq \delta. \tag{5.13}$$

*If $\varphi < N$, then for some $h > 0$*

$$[h\kappa]_N = N - \varphi, \qquad [m\kappa]_N < N - \varphi \quad (0 \leq m < h), \tag{5.14}$$

*where $[\cdot]_N$ denotes the representative in $\{0, \dots, N-1\}$. For $\varphi = N$, the corresponding case is the time-zero identity $[0\kappa]_N = 0$.*

*Proof.* Let $\mathcal{R}$ be the family of corresponding index sets of reachable representatives. It is a nonempty subset of the finite set $2^{\mathbb{Z}/N\mathbb{Z}}$, so a lexicographic minimizer $S \in \mathcal{R}$ exists. If $S = \mathbb{Z}/N\mathbb{Z}$, the conclusion holds with $\varphi = N$ and the stated time-zero identity. Assume henceforth that $S \neq \mathbb{Z}/N\mathbb{Z}$; every connected component of $C_N[S]$ is then a proper cyclic interval.

We first record the complete transition table for updating one connected component from right to left. Let $G = \{a, a+1, \dots, b\}$ be a connected component of $C_N[S]$, let $\ell = |G| < N$, and put $R = S \setminus G$. Starting from $S^{(0)} = S$, consider, in order, the updates at

$$j_r = b - r, \qquad 0 \le r < \ell,$$

and denote the resulting sets by

$$S^{(r+1)} = (S^{(r)} \setminus \{j_r\}) \cup \{j_r + \kappa\}.$$

Until the first step for which $j_r + \kappa \in S^{(r)}$, every source $j_r$ belongs to $S^{(r)}$ when used: it belonged to $G$ initially, has not yet appeared as a source, and earlier updates remove only their distinct sources. For $r = 0$, its right neighbor $b+1$ does not belong to $S$ by maximality of $G$. For $r > 0$, the right neighbor $j_r + 1 = j_{r-1}$ was removed in the preceding update. Before the first target-overlap step it cannot have been reinserted by an earlier target: if $j_s + \kappa = j_{r-1}$ for some $s < r-1$, then $j_{r-1}$ still belonged to the current set when the update at $j_s$ was made, so that earlier target would already have belonged to the current set; the case $s = r-1$ would force $\kappa \equiv 0 \pmod N$, contrary to $1 \le \kappa < N$. Hence every update up to and including the first target-overlap step satisfies the permitted-replacement condition $j_r \in S^{(r)}$, $j_r + 1 \notin S^{(r)}$. Before that step, direct induction gives the disjoint-union invariant

$$S^{(r)} = R \,\dot\cup\, \{a, \dots, b-r\} \,\dot\cup\, (\{b-r+1, \dots, b\} + \kappa). \tag{5.15}$$

The pair $(|S^{(r)}|, \mathrm{comp}(S^{(r)}))$ at the first target-overlap step, or at the end when no target overlap occurs, is therefore exactly as follows.

| event | cardinality | component count |
|---|---|---|
| $j_r + \kappa \in S^{(r)}$ for the first time | $\lvert S^{(r+1)}\rvert = \lvert S\rvert - 1$ | irrelevant |
| no target overlap and $G + \kappa$ is not adjacent to $R$ | $\lvert S^{(\ell)}\rvert = \lvert S\rvert$ | same as $\mathrm{comp}(S)$ |
| no target overlap and $G + \kappa$ is adjacent to $R$ | $\lvert S^{(\ell)}\rvert = \lvert S\rvert$ | at most $\mathrm{comp}(S) - 1$ |

Indeed, in the last two rows

$$S^{(\ell)} = R \,\dot\cup\, (G + \kappa). \tag{5.16}$$

This table also makes the minimality consequence and the repeatability precise. During a partial right-to-left update, an adjacency of the translated suffix to a component of $R$ does not obstruct the next permitted replacement: the next source's right neighbor does not belong to the current set, by the right-to-left order. If such a partial adjacency ever lowers the component count, the reachable intermediate state already contradicts lexicographic minimality; otherwise it is ignored, and only the completed translate is used in the repeatability invariant. For later use, after $m$ completed right-to-left update sequences write

$$G_m = G + m\kappa, \qquad T_m = R \,\dot\cup\, G_m.$$

The repeatability invariant is that $T_m$ is reachable with

$$(|T_m|, \mathrm{comp}(T_m)) = (|S|, \mathrm{comp}(S)),$$

while $G_m$ is a proper connected component of $C_N[T_m]$, disjoint from and nonadjacent to every connected component contained in $R$. It holds at $m = 0$: $G$ is a proper connected component of $C_N[S]$ and $R = S \setminus G$.

Suppose it holds at $m$. Apply the same transition table to the connected component $G_m$ in $T_m$. A target already belonging to the current set would produce a reachable set of smaller cardinality, and an overlap-free sequence whose terminal translate is adjacent to $R$ would produce one with the same cardinality and fewer connected components. Both contradict the lexicographic minimality of $S$. Thus no target overlap occurs, the completed translate is nonadjacent to $R$, and

$$T_{m+1} = R \,\dot{\cup}\, G_{m+1}.$$

Translation preserves $|G_{m+1}| = |G| < N$, so $G_{m+1}$ is proper; its nonadjacency to the unchanged set $R$ makes it a connected component of $C_N[T_{m+1}]$. The minimizing pair is unchanged, and hence the invariant holds at $m+1$. Therefore the right-to-left update sequence may be repeated as many times as required. No assertion about the number of connected components during a partial sequence is being used.

Every residue class modulo $\delta$ meets $S$. If one did not, all $L = N/\delta$ indices in the corresponding $\kappa$-orbit would be endpoint contacts. Iterating their identities $\lambda x_{j-\kappa} = x_j$ gives

$$x_j = \lambda^L x_j,$$

contrary to $x_j \neq 0$ and $|\lambda| < 1$.

Distinct connected components of $C_N[S]$ have disjoint residue sets modulo $\delta$. For if $p \in G$ and $q \in H$, where $G \neq H$ are connected components and $p \equiv q \pmod{\delta}$, there is $t \in \{1, \dots, L-1\}$ such that

$$p + t\kappa = q \pmod{N}.$$

Translate $G$ through $t$ complete right-to-left update sequences, leaving all other connected components fixed. The preceding transition argument permits each repetition unless it has already produced a set with smaller cardinality or a completed translate with fewer connected components; either outcome contradicts minimality. Otherwise, during the $t$-th sequence, the index that started at $p$ reaches the index $q$, which already belongs to $H$. The corresponding update would reduce the cardinality, again contradicting minimality. This proves residue disjointness.

If some connected component has at least $\delta$ consecutive indices, its residue set is all of $\mathbb{Z}/\delta\mathbb{Z}$, and residue disjointness excludes every other connected component. Suppose instead that every connected component has length less than $\delta$. The residue set of each component is then a proper cyclic interval in $\mathbb{Z}/\delta\mathbb{Z}$. These intervals are pairwise disjoint and, by residue coverage, cover the cyclic group $\mathbb{Z}/\delta\mathbb{Z}$. Two of them are therefore consecutive. Write the corresponding connected components as $G = \{a, \dots, b\}$ and $H = \{c, \dots, d\}$, choosing their terminal and initial indices so that

$$\delta \mid (c - b - 1).$$

Because the subgroup generated by $\kappa$ is $\delta\mathbb{Z}/N\mathbb{Z}$, there is $t \in \{0, \dots, L-1\}$ with

$$t\kappa \equiv c - b - 1 \pmod{N}. \tag{5.17}$$

Here $t \neq 0$, since $t = 0$ would make $b$ and $c$ adjacent indices and would therefore place them in the same connected component. Translate $G$ through $t$ complete right-to-left update sequences. If a target already belongs to the current set, the resulting set has smaller cardinality. Otherwise the terminal index of the final translate is $c-1$, so the final translate is adjacent to $H$. The third row of the transition table then gives the same cardinality and fewer connected components. Both alternatives contradict minimality. Hence $C_N[S]$ has one connected component. Since it meets every residue class, the length $\varphi$ of this cyclic interval is at least $\delta$. A cyclic relabelling gives (5.13).

Assume now that $\varphi < N$. Relabel temporarily so that

$$S = F \,\dot{\cup}\, \{0\}, \qquad F = \{N - \varphi + 1, \dots, N-1\}.$$

Put

$$a_m = [m\kappa]_N, \qquad I = \{N-\varphi, \dots, N-1\}.$$

The orbit of 0 consists of the multiples of $\delta$ modulo $N$. Because $|I| = \varphi \geq \delta$, the interval $I$ meets this orbit; because $0 \notin I$, there is a least $h > 0$ such that $a_h \in I$.

The exact single-index induction is

$$S_m = F \,\dot\cup\, \{a_m\} \qquad (0 \leq m \leq h), \tag{5.18}$$

where $S_m$ is reachable from $S$ by successively updating only the displayed index. For $m = 0$ the formula reads $S_0 = F \,\dot\cup\, \{0\} = S$, so it holds. If $m < h$, minimality of $h$ gives $a_m < N - \varphi$, and therefore

$$a_m + 1 \notin F \cup \{a_m\};$$

the update at $a_m$ is permitted. If its target $a_{m+1}$ belonged to $F$, the updated set would have cardinality $\varphi - 1$, contrary to minimality. Thus $a_{m+1}$ does not belong to the current set, and (5.18) follows at $m+1$. At the first entrance time one consequently has

$$a_h \in I \setminus F = \{N - \varphi\},$$

whereas $a_m \notin I$, hence $a_m < N - \varphi$, for every $m < h$. This is exactly (5.14). The computation used displacements from the terminal index, so the final common cyclic relabelling to (5.13) changes neither $\kappa$ nor the residue statement. For $\varphi = N$, the time-zero identity $[0\kappa]_N = 0$ was already handled at the start of the proof. □

# 6 Finite rotation and first-return decomposition from the lattice sail

The cyclic arithmetic is isolated in this section. It depends only on the rotation of a finite cyclic set; no polygon is used until the final tower identities are read through endpoint contacts.

Fix integers

$$N \geq 2, \qquad 1 \leq \kappa < N, \qquad \delta = \gcd(N, \kappa),$$

and write $[a]_N$ for the representative of $a$ in $\{0, \dots, N-1\}$. Whenever an interval $\{1, \dots, \nu\}$ is used as a set of cyclic labels, its entries are read modulo $N$. In the full-length case $\nu = N$, the terminal label $N$ therefore denotes the residue 0; the displayed list records one complete circuit and does not introduce an additional label. Put

$$L(a, b) = a\kappa - bN. \tag{6.1}$$

A time $h \geq 0$ is an *upper-record time* when

$$[h\kappa]_N > [t\kappa]_N \qquad (0 \leq t < h).$$

Its deficit is $\nu = N - [h\kappa]_N$, with the convention that time zero has deficit $N$. Equivalently, with

$$b = \begin{cases} 1, & h = 0, \\ \lceil h\kappa/N \rceil, & h > 0, \end{cases} \qquad V = (h, b),$$

one has

$$L(V) = -\nu, \qquad [t\kappa]_N < N - \nu = [h\kappa]_N \quad (0 \leq t < h). \tag{6.2}$$

The terminal record has deficit $\delta$, because the attainable residues are exactly the multiples of $\delta$ and $N - \delta$ is the largest one. The upper-record vectors are ordered by their time coordinates. Two of them are called consecutive when no upper-record time lies strictly between their time coordinates; the next upper-record vector is the latter member of such a pair.

**Theorem 6.1** (Finite rotation and first-return decomposition). *Let $V = (h, b)$ be an upper-record vector with deficit $\nu > \delta$, and let $V' = (h', b')$ be the next upper-record vector. Define*

$$U = V' - V = (q, p), \qquad \Delta = \nu - \nu'.$$

*Since the records are listed chronologically, $h' > h$ and therefore $q = h' - h > 0$. Then*

$$q\kappa - pN = \Delta > 0, \qquad h\kappa - bN = -\nu, \tag{6.3}$$

$$0 < \Delta < \nu, \qquad \det(U, V) = 1, \tag{6.4}$$

$$q\nu + h\Delta = N, \qquad \gcd(\Delta, \nu) = \delta. \tag{6.5}$$

*The initial record $\nu = N$ has $V = (0, 1)$. Moreover $h = 0$ if and only if $\nu = N$.*

*Let*

$$q\kappa - pN = \Delta, \qquad h\kappa - bN = -\nu, \qquad q\nu + h\Delta = N, \tag{6.6}$$

*and define*

$$H_i = \begin{cases} q, & 1 \le i \le \nu - \Delta, \\ q + h, & \nu - \Delta < i \le \nu. \end{cases}$$

*Then*

$$\mathcal{D} = \{(t, i) : 1 \le i \le \nu,\ 0 \le t < H_i\} \longrightarrow \mathbb{Z}/N\mathbb{Z}, \qquad F(t, i) = [i + t\kappa]_N \tag{6.7}$$

*is a bijection. The first-return map on the base interval $\{1, \ldots, \nu\}$ is addition of $\Delta$ modulo $\nu$ and has the two return times displayed above.*

*If $\Delta = 1$, put*

$$w = \lfloor h/q \rfloor, \qquad E = V - wU = (e, c), \qquad d = -L(E) = \nu + w.$$

*Then $E$ is an upper-record vector, and*

$$q\kappa - pN = 1, \qquad e\kappa - cN = -d, \qquad qd + e = N, \qquad 0 \le e < q. \tag{6.8}$$

*In particular $d = \lfloor N/q \rfloor \ge \nu$, and*

$$E, E + U, \ldots, E + (w + 1)U$$

*are consecutive upper-record vectors.*

*Finally, suppose a polygonal contact assignment satisfying the standing conditions* (A0)–(A4) *has no relative-interior contact index outside the base interval $\{1, \ldots, \nu\}$ and has the record data above. Then every internal tower state has an assigned contact at a side endpoint, and*

$$\lambda^t x_i = x_{F(t,i)} \qquad (0 \le t < H_i). \tag{6.9}$$

*The top and closure relations are*

$$\lambda^q x_i = \xi_{i+\Delta} \qquad (1 \le i \le \nu - \Delta), \tag{6.10}$$

$$\lambda^{q+h} x_i = \xi_{i+\Delta-\nu} \qquad (\nu - \Delta < i \le \nu), \tag{6.11}$$

$$\lambda^h x_\nu = x_0, \tag{6.12}$$

$$\lambda^q x_0 = \xi_\Delta. \tag{6.13}$$

*Proof.* We proceed in four stages.

*1. Consecutive record vectors are unimodular.* List the upper-record times chronologically. The list is finite and ends at deficit $\delta$. Indeed,

$$\gcd(\kappa/\delta, N/\delta) = 1,$$

so multiplication by $\kappa/\delta$ permutes the residue classes modulo $N/\delta$. Hence there is an $h$ such that

$$h\kappa/\delta \equiv -1 \pmod{N/\delta},$$

equivalently $h\kappa \equiv N-\delta \pmod N$. Thus the attainable residue $N-\delta$ occurs, and the upper-record list ends at deficit $\delta$. The time-zero record vector $V_0=(0,1)$ is primitive. Every other record vector $V=(h,b)$ has $h>0$ and is primitive as well. Indeed, if $V=gW$ with an integer $g>1$, then $L(W)=-\nu/g<0$, while the time coordinate of $W$ is a nonnegative integer strictly smaller than $h$. Its residue is $N-\nu/g>N-\nu$, contradicting the record property.

Let $V=(h,b)$ and $V'=(h',b')$ be consecutive records, with deficits $\nu>\nu'>0$. Since

$$b=\frac{h\kappa+\nu}{N}, \qquad b'=\frac{h'\kappa+\nu'}{N},$$

we have

$$bh'-hb'=\frac{h'\nu-h\nu'}{N}>0.$$

Thus $\det(V,V')<0$. Suppose $D=-\det(V,V')>1$. By lemma A.6, the half-open fundamental parallelogram generated by $V,V'$ contains a nonzero lattice point $W=\alpha V+\beta V'$ with $0\le\alpha,\beta<1$. Primitivity excludes a nonzero point on either radial edge. Replacing $W$ by $V+V'-W$ when $\alpha+\beta>1$, we obtain a lattice point in the triangle $\operatorname{conv}\{0,V,V'\}$ different from its three vertices. Hence

$$W=\alpha V+\beta V', \qquad \alpha,\beta>0, \qquad \alpha+\beta\le 1.$$

Write $W=(t,c)$. Its time $t=\alpha h+\beta h'$ is an integer with $0<t<h'$, while

$$-L(W)=\alpha\nu+\beta\nu'<\nu.$$

Since $0<-L(W)<\nu\le N$, one has $c=\lceil t\kappa/N\rceil$ and therefore

$$[t\kappa]_N=N+L(W)>N-\nu.$$

This contradicts that $V$ and $V'$ are consecutive upper-record vectors. Therefore $D=1$ and

$$\det(V,V')=-1.$$

For $U=V'-V$, this is $\det(U,V)=1$. Linearity of $L$ gives $L(U)=\nu-\nu'=\Delta$, so $0<\Delta<\nu$. Expanding the determinant,

$$q\nu+h\Delta=q(bN-h\kappa)+h(q\kappa-pN)=N(qb-ph)=N.$$

Because $U,V$ are a unimodular basis, the subgroup generated by $L(U)=\Delta$ and $L(V)=-\nu$ equals $L(\mathbb{Z}^2)$. The latter is $\delta\mathbb{Z}$. Hence $\gcd(\Delta,\nu)=\delta$. The time-zero record has $h=0$, $V=(0,1)$, and deficit $N$. Conversely, suppose $\nu=N$. Then $[h\kappa]_N=N-\nu=0$. If $h>0$, the upper-record inequality compared with time zero would require

$$[h\kappa]_N>[0\kappa]_N=0,$$

which is impossible. Hence $h=0$.

*2. The tower map is bijective.* Give $\mathcal{D}$ the successor map

$$\sigma(t,i)=\begin{cases}(t+1,i), & t+1<H_i,\\ (0,i+\Delta), & t=q-1,\ i\le\nu-\Delta,\\ (0,i+\Delta-\nu), & t=q+h-1,\ i>\nu-\Delta.\end{cases}$$

The congruences in (6.6) give

$$F(\sigma(t,i)) = F(t,i) + \kappa \pmod N. \tag{6.14}$$

On the bases, $\sigma$ induces addition of $\Delta$ modulo $\nu$. It has $\delta = \gcd(\Delta, \nu)$ cycles, namely the residue classes modulo $\delta$. Each base cycle contains $\nu/\delta$ bases. The long-base labels form the block of $\Delta$ consecutive integers $\nu - \Delta + 1, \dots, \nu$. Since $\delta$ divides both $\nu$ and $\Delta$, this block contains exactly $\Delta/\delta$ representatives of each residue class modulo $\delta$. Hence each base cycle contains $\Delta/\delta$ long bases, so its total number of states is

$$q\frac{\nu}{\delta} + h\frac{\Delta}{\delta} = \frac{N}{\delta}.$$

Addition of $\kappa$ on $\mathbb{Z}/N\mathbb{Z}$ also has $\delta$ cycles of length $N/\delta$. Since $F(t,i) \equiv i \pmod \delta$, distinct base cycles map into distinct $\kappa$-orbits. Equation (6.14) makes the restriction to each cycle an equivariant map between two cycles of the same finite length, hence a bijection. This proves (6.7) and the return-time statement.

*3. When $\Delta = 1$, the preceding record vectors form the stated arithmetic progression.* Assume $\Delta = 1$. By construction $E = V - wU = (e,c)$ has $0 \le e < q$ and $L(E) = -(\nu + w) = -d$. The determinant remains $\det(U,E) = 1$, so

$$qd + e = N.$$

It remains to prove that $E$ and the displayed arithmetic progression are exactly consecutive record vectors.

For a putative earlier improving time $t$ for $E$, put

$$z = \lceil t\kappa/N \rceil, \qquad Z = (t,z).$$

If $[t\kappa]_N > N - d$, then $-d < L(Z) < 0$; a tie $[t\kappa]_N = N - d$ is represented by choosing the integer second coordinate so that $L(Z) = -d$ (with the declared time-zero convention when $d = N$). Thus excluding integer vectors in these ranges excludes every earlier improvement and tie. The same observation applies below with $d$ replaced by $d - j$.

Because $U, E$ are a basis, write any lattice vector $Z = (t,z)$ uniquely as $Z = AE + BU$. If $e = 0$, write $E = (0,c)$. The identity $\det(U,E) = 1$ becomes $qc = 1$. Since $q > 0$, it follows that $q = c = 1$, and hence

$$E = (0,1), \qquad d = -L(E) = N.$$

Thus $E$ is precisely the declared time-zero record vector.

If $e > 0$, suppose that $0 \le t < e$ and $-d < L(Z) < 0$. The inequalities are

$$-d < -Ad + B < 0.$$

If $A \le 0$, then $B < Ad \le 0$ and $t = Ae + Bq < 0$. If $A = 1$, then $B \ge 1$ and $t \ge e + q$. If $A \ge 2$, then $B > (A-1)d \ge 1$ and again $t = Ae + Bq > e$. All alternatives contradict $0 \le t < e$. Hence $E$ is an upper record unless an earlier time has the same residue. Equality with the residue of $E$ would mean $L(Z) = -d$, so $B = (A-1)d$ and

$$t = Ae + (A-1)dq = e + (A-1)N,$$

where $qd + e = N$. No number of this form lies in $0 \le t < e$. Thus an earlier tie is impossible, and $E$ is an upper record when $e > 0$. Together with the time-zero case, this proves that $E$ is an upper record in all cases.

More generally, fix $j$ with $0 \le j \le w$. Since $d - w = \nu > \delta \ge 1$, one has $d - j \ge \nu > 1$. Suppose $0 \le t < e + (j+1)q$ while

$$-(d-j) < L(Z) < 0.$$

Equivalently,

$$-(d-j) < -Ad + B < 0.$$

If $A \le 0$, then $t < 0$. If $A = 1$, the left inequality gives $B > j$, so $B \ge j+1$ and $t \ge e+(j+1)q$. If $A \ge 2$, it gives $B > (A-1)d + j$, hence $B \ge j+1$ and $t = Ae + Bq \ge e + (j+1)q$. Thus no improved residue occurs before the time of $E + (j+1)U$, whose deficit is one less than that of $E + jU$. Induction shows that the displayed vectors are consecutive records. Since $V = E + wU$, this includes the given pair $V, V+U$. Finally $N = qd + e$ with $0 \le e < q$ gives $d = \lfloor N/q \rfloor$.

*4. Geometric tower identities.* Assume now that every relative-interior contact index belongs to $\{1, \dots, \nu\}$. If an internal state $F(t,i)$ with $1 \le t < H_i$ belonged to the base block, then it and the corresponding level-zero state would have the same image under the bijection $F$, impossible. Hence every internal destination index has its assigned contact at a side endpoint. If $j = F(t,i)$, the endpoint-contact identity at index $j$ is

$$\lambda x_{F(t-1,i)} = x_{F(t,i)}.$$

Induction gives (6.9). One further multiplication at a short or long top gives (6.10) and (6.11). Since $F(h,\nu) \equiv \nu + h\kappa \equiv 0$ and $h < H_\nu$, the internal identity gives (6.12); combining it with the long top at base $\nu$ gives the additional identity (6.13) at base 0. □

**Corollary 6.2** (Upper-record intervals extended by endpoint contacts)**.** *Let $S$ be the set of relative-interior contact indices of a contact assignment satisfying the standing conditions* (A0)–(A4)*. Suppose $U = (q,p)$ and $E = (e,c)$ are integer vectors for which $E$ and $E+U$ are consecutive upper-record vectors and*

$$q\kappa - pN = 1, \qquad e\kappa - cN = -d, \qquad qd + e = N, \qquad 0 \le e < q, \qquad d > 1. \tag{6.15}$$

*Assume*

$$S \subseteq J := \{1, \dots, d\}. \tag{6.16}$$

*Then the first-return decomposition over the base interval $J$ has height $q$ at bases $1, \dots, d-1$, height $q+e$ at base $d$, and first return equal to the successor. Its tower states form a bijection with the $N$ polygon labels. Every internal destination index lies outside $J$ and therefore has its assigned contact at a side endpoint, and*

$$\lambda^q x_{j-1} = \xi_j \qquad (1 \le j \le d), \tag{6.17}$$

$$\lambda^e x_d = x_0. \tag{6.18}$$

*For every $j \in J \setminus S$, one has $\xi_j = x_j$. Thus extending a consecutive block of relative-interior contact indices to such an upper-record interval adds only indices whose assigned contacts are side endpoints. At each added index one has $\beta_j = 0$ and $\alpha_j = 1$; no further return datum is needed in this topic.*

*Proof.* Since $q\kappa - pN = 1$, every common divisor of $N$ and $\kappa$ divides 1. Hence $\delta = \gcd(N,\kappa) = 1$, and the assumption $d > 1$ gives $d > \delta$. We may therefore apply theorem 6.1 to the consecutive record pair $V = E$, $V' = E + U$. In the notation of that theorem,

$$\nu = d, \qquad \Delta = 1, \qquad h = e.$$

The arithmetic assumptions in (6.15) are precisely (6.6); hence the tower map is bijective, its return map is addition of one modulo $d$, and its heights are the stated two values.

The geometric conclusion of theorem 6.1 requires only that there be no relative-interior contact index outside the chosen base interval. This is exactly (6.16). More explicitly, if an internal tower state had destination label $j \in J$, then it would have the same image under the tower bijection as the level-zero state $(0, j)$, which is impossible. Thus every internal destination lies outside $J$ and, since all relative-interior contact indices belong to $J$, its assigned contact is a side endpoint. The short-return identities for bases $1, \dots, d-1$ give $\lambda^q x_i = \xi_{i+1}$; the additional relation at base 0 gives $\lambda^q x_0 = \xi_1$; and the tower closure gives $\lambda^e x_d = x_0$. These are (6.17)–(6.18). Finally, in a contact assignment satisfying (A0)–(A4), an index outside $S$ has $\beta_j = 0$ in (4.12), so $\xi_j = x_j$ and the asserted endpoint-extension statement follows. □

*Remark* 6.3. Let $C$ be the closed cone bounded by the positive $b$-axis and the ray $L(h, b) = 0$. The Klein sail of $C$ is the boundary visible from the origin of

$$\operatorname{conv}\big((C \cap \mathbb{Z}^2) \setminus \{0\}\big).$$

By theorem 6.1, every record vector is primitive and every consecutive pair is unimodular. These are the only lattice-geometric properties used below. No identification of the record chain with the sail, or of every record vector with a sail vertex, is asserted or needed. The sail and continued fractions remain useful geometric context for the arithmetic.

**Why the projective proof of $\Delta = 1$ starts at four vertices.** The finite-rotation theorem above remains valid for $N \geq 2$, but the projective proof that the first return reaches the cyclic successor assumes $N \geq 4$. This restriction is essential. For $1/2 < a < 1$, the doubly stochastic matrix

$$B_a = \begin{pmatrix} 0 & 1-a & a \\ a & 0 & 1-a \\ 1-a & a & 0 \end{pmatrix}$$

has eigenvalues, counted with algebraic multiplicity, 1, $\lambda_a = -1/2 + i\sqrt{3}(2a-1)/2$, and $\overline{\lambda_a}$. It preserves the plane $u_0 + u_1 + u_2 = 0$ and restricts there to the elliptic contraction $T_a$ with eigenvalues $\lambda_a$ and $\overline{\lambda_a}$. The centred standard simplex $P$ is independent of $a$ and is an invariant triangle with $(\varphi, \kappa, \Delta) = (3, 2, 2)$. No invariant segment is possible for a nonreal planar map, while any invariant triangle for a radial enlargement $tT_a$, $t > 1$, would yield a nonnegative stochastic $3 \times 3$ realization with trace $1 + 2t \operatorname{Re} \lambda_a = 1 - t < 0$, which is impossible. Consequently $\nu_{\text{poly}}(T_a) = 3$ and $\nu_{\text{poly}}(tT_a) > 3$ for every $t > 1$: in the terminology of definition 1.1, each $T_a$ is 3-critical. Thus the conclusion $\Delta = 1$ is false for $N = 3$. The small orders, including order three, are treated directly in proposition II.9.2.

# 7 Why the first return cannot skip the cyclic successor

Throughout this section assume $N \geq 4$. The positive integer $\Delta = \nu - \nu'$ from theorem 6.1 is the advance of its first-return map on the base interval. Assume $\varphi > \delta$, so the interval of relative-interior contact indices contains more labels than the number $\delta$ of $\kappa$-orbits. The first-return map is the translation $i \mapsto i + \Delta$ on that interval. The proof that it lands at the next base label, equivalently $\Delta = 1$, is begun in this topic and completed in Topic VI. It is separated into a preparatory part and a deformation part. The results through proposition 7.5 establish only the preparation: the return towers produce consecutive boundary vertices while omitting at least one side, account for every nonclosing source–target pair, and provide an affine chart adapted to the resulting composition of perspectivities. They do not yet rule out $\Delta > 1$. The subsequent projective deformation opens the single unconstrained closing contact and completes the contradiction. The contradiction is with the literal image-vertex boundary conclusion of theorem 3.2: every vertex of $TR$ must lie on $\partial R$ for every invariant polygon $R$ with at most $N$ vertices.

Under this standing assumption $\varphi > \delta$, lemma 5.5 gives

$$[h\kappa]_N = N - \varphi, \qquad [t\kappa]_N < N - \varphi \quad (0 \le t < h)$$

when $\varphi < N$. Hence $h > 0$ is an upper-record time and its deficit is $\varphi$. When $\varphi = N$, the same role is played by the time-zero record $h = 0$, whose deficit is $N = \varphi$ by convention. Thus in both cases theorem 6.1 applies with $\nu = \varphi$ and gives integers $q, h, p, b, \Delta$ such that

$$q\kappa - pN = \Delta, \qquad h\kappa - bN = -\varphi, \qquad q\varphi + h\Delta = N, \qquad 0 < \Delta < \varphi. \tag{7.1}$$

We shall prove that the alternative $\Delta > 1$ is impossible. For $1 \le i \le \varphi$, define the return height $H_i$ and return target index $r(i)$ by

$$H_i = \begin{cases} q, & 1 \le i \le \varphi - \Delta, \\ q + h, & \varphi - \Delta < i \le \varphi, \end{cases} \qquad r(i) = \begin{cases} i + \Delta, & 1 \le i \le \varphi - \Delta, \\ i + \Delta - \varphi, & \varphi - \Delta < i \le \varphi. \end{cases} \tag{7.2}$$

Thus the two top relations (6.10)–(6.11) are the single identity

$$\lambda^{H_i} x_i = \xi_{r(i)} \in \operatorname{relint} E_{r(i)} \qquad (1 \le i \le \varphi). \tag{7.3}$$

The additional relation at base index 0 is

$$\lambda^q x_0 = \xi_\Delta. \tag{7.4}$$

For the remainder of the finite-return construction, through proposition 7.3, assume toward a contradiction that $\Delta > 1$. The projective statements beginning with definition 7.4 do not retain this arithmetic assumption. Because $\Delta > 1$, the base $\varphi - 1$ belongs to the long-return range, so

$$H_{\varphi-1} = q + h, \qquad h < H_{\varphi-1},$$

where the strict inequality uses $q > 0$. We may therefore apply the internal-tower identity at level $h$ to this base. Together with (6.12), this records the transported terminal side:

$$\lambda^h x_\varphi = x_0, \qquad \lambda^h x_{\varphi-1} = x_{-1}, \qquad \lambda^h \operatorname{relint} E_\varphi = \operatorname{relint} E_0. \tag{7.5}$$

If $h = 0$, then theorem 6.1 gives $\varphi = N$, and the same display is just the cyclic identity $E_\varphi = E_0$.

For a base $i$, put

$$\ell_i := \lambda^{-H_i} \operatorname{aff} E_{r(i)}. \tag{7.6}$$

Since $\lambda^{H_i} P \subseteq P$ and (7.3) places $\lambda^{H_i} x_i$ in the relative interior of the side $E_{r(i)}$, the line $\ell_i$ supports $P$ at $x_i$.

**Lemma 7.1** (Pulled-back supporting lines expose one vertex)**.** *Assume $\Delta > 1$, and set*

$$m_+ = \Delta, \qquad m_- = \varphi - \Delta + 1, \qquad m = \min\{m_+, m_-\}. \tag{7.7}$$

*If $m = m_+$, then $\ell_i$ supports $P$ and exposes only $x_i$ for $2 \le i \le \Delta$. If $m = m_-$, then $\ell_i$ supports $P$ and exposes only $x_i$ for $\Delta - 1 \le i \le \varphi - 2$. When $2\Delta = \varphi + 1$, one has $m_+ = m_-$, so both conclusions are valid; the orientation table below assigns this equality case to the forward orientation.*

*Proof.* By the support-face test lemma 2.9, it is enough to show that neither neighboring vertex of $x_i$ lies on $\ell_i$. Since multiplication by $\lambda^{H_i}$ is invertible, this is equivalent to showing that $\lambda^{H_i} x_{i-1}$ and $\lambda^{H_i} x_{i+1}$ avoid aff $E_{r(i)}$.

If a neighboring base index has the same height as $i$, its image is the relative-interior contact point on the adjacent target side:

$$\lambda^{H_i} x_{i-1} = \xi_{r(i)-1}, \qquad \lambda^{H_i} x_{i+1} = \xi_{r(i)+1},$$

with cyclic side labels. A relative-interior point of an adjacent side of a polygon is not on the line of $E_{r(i)}$. Thus only the interface after the last base index of height $q$,

$$i_0 = \varphi - \Delta$$

and before the first base index of height $q + h$, namely $i_0 + 1$, needs separate checking.

For the base index $i_0$, the target side is $E_\varphi$. Its outgoing neighbor has height $q + h$. If $h > 0$, then level $q$ in that tower is internal, and

$$\lambda^q x_{i_0+1} = x_{\varphi+1} \notin \operatorname{aff} E_\varphi.$$

If $h = 0$, then $\varphi = N$ and the long top relation gives $\lambda^q x_{i_0+1} = \xi_1 \in \operatorname{relint} E_1$, again not on the line of $E_\varphi$.

For the base index $i_0 + 1$, the target side is $E_1$. Its incoming neighbor is $i_0$, and (7.5) gives

$$\lambda^{q+h} x_{i_0} = \lambda^h \xi_\varphi \in \operatorname{relint} E_0.$$

The outgoing neighbor returns to $\xi_2 \in \operatorname{relint} E_2$. Neither point is on the line of $E_1$.

The two index ranges in the statement meet the short–long interface at most once and contain no other height transition. Hence both adjacent vertices avoid every displayed pulled-back line, so each such supporting line exposes only the indicated vertex. ☐

The shorter of the two cyclic arcs is now read as a single boundary chain. The following table gives both orientation cases; after this point the projective argument is orientation-free.

| | forward orientation | reverse orientation |
|---|---|---|
| condition | $2\Delta \le \varphi + 1$ | $2\Delta > \varphi + 1$ |
| length | $m = \Delta$ | $m = \varphi - \Delta + 1$ |
| chain vertices | $X_i = x_i$ | $X_i = x_{\varphi-i}$ |
| chain contacts | $C_i = \xi_i$ | $C_i = \xi_{\varphi-i+1}$ |
| transport exponent | $a = q$ | $a = q + h$ |
| base subset $M$ | $\{1, \ldots, \Delta\}$ | $\{\Delta - 1, \ldots, \varphi - 1\}$ |
| target $c$ | $\Delta + 1$ | $\Delta - 1$ |

Here $0 \le i \le m + 1$ for the chain vertices, while the contact formula is used only for $2 \le i \le m + 1$. In either column,

$$C_i \in \operatorname{relint}[X_{i-1}, X_i] \qquad (2 \le i \le m + 1), \tag{7.8}$$

and the initial side satisfies

$$\lambda^a X_1 = C_{m+1}, \qquad \lambda^a X_0 = C_m. \tag{7.9}$$

In the forward orientation the first identity is the short-top relation for base 1, and the second is (7.4). In the reverse orientation they are the long-top relations for bases $\varphi - 1$ and $\varphi$.

For $2 \le i \le m$, define

$$\mathcal{L}_i = \ell_{b_i}, \qquad b_i = \begin{cases} i, & \text{in the forward orientation,} \\ \varphi - i, & \text{in the reverse orientation.} \end{cases} \tag{7.10}$$

By lemma 7.1, $\mathcal{L}_i$ supports $P$ and exposes only $X_i$.

**Lemma 7.2** (The selected boundary arc omits at least one side)**.** *Assume $N \geq 4$, $1 < \Delta < \varphi \leq N$, and choose the branch in the preceding table. Then $m \geq 2$ and*

$$m + 1 < N. \tag{7.11}$$

*Consequently the displayed vertices $X_0, \ldots, X_{m+1}$ are pairwise distinct consecutive vertices in the chosen cyclic orientation and the chain omits at least one side of $P$.*

*This conclusion includes all boundary values used later: if $\Delta = 2$, the forward branch has $m = 2$; if $2\Delta = \varphi + 1$, the equality belongs to the forward branch and does not identify two chain labels; and if $h = 0$, then $\varphi = N$ but the same strict inequality (7.11) still holds.*

*Proof.* In the forward branch, $m = \Delta$. If $\varphi < N$, then

$$m + 1 = \Delta + 1 \leq \varphi < N$$

because $\Delta < \varphi$. If $\varphi = N$ and equality $m + 1 = N$ were to hold, then $\Delta = N - 1$; the branch inequality $2\Delta \leq \varphi + 1 = N + 1$ would give $2N - 2 \leq N + 1$, hence $N \leq 3$, a contradiction. Thus (7.11) holds. The labels $0, 1, \ldots, \Delta + 1$ therefore represent distinct vertices modulo $N$. The special equality $2\Delta = \varphi + 1$ changes the return-edge wrap but not this label interval. When $\Delta = 2$, the same calculation gives $m = 2$ and $m + 1 = 3 < N$.

In the reverse branch, $m = \varphi - \Delta + 1 \geq 2$. If $\varphi < N$, then $\Delta \geq 2$ gives

$$m + 1 = \varphi - \Delta + 2 \leq \varphi < N.$$

If $\varphi = N$, the strict branch inequality $2\Delta > N + 1$ forces $\Delta \geq 3$ because $N \geq 4$. Hence

$$m + 1 = N - \Delta + 2 < N.$$

The reverse labels run through the integer interval $\varphi, \varphi - 1, \ldots, \Delta - 2$, whose length is $m + 2$ and whose total index variation is $m + 1 < N$; they are therefore distinct modulo $N$. When $\varphi = N$, the first label $x_\varphi = x_0$ is not repeated because $\Delta - 2 \geq 1$. Finally, theorem 6.1 gives $h = 0 \iff \varphi = N$, so the preceding two $\varphi = N$ arguments also cover the zero-height case. In either branch, $m + 1 < N$ means that the consecutive chain uses fewer than all $N$ sides and is proper. ☐

**Proposition 7.3** (Partition of the return source–target pairs)**.** *Let*

$$\mathcal{B} = \{1, \ldots, \varphi\}, \qquad s = r^{-1},$$

*where $r$ is addition of $\Delta$ modulo $\varphi$ in the representative set $\mathcal{B}$. Assume $\Delta > 1$ and choose the forward or reverse orientation from the preceding table. Define the subset $M$ of base indices, the distinguished base index $b_*$, the three target-index sets $D, R, A$, and the distinguished target index $c$ by the following exact formulas. We classify the pairs $(i, r(i))$ consisting of a base index and its return target.*

*An integer interval $\{u, \ldots, v\}$ is understood to be empty when $u > v$. In the forward branch $2\Delta \leq \varphi + 1$, put*

$$\begin{aligned}
M &= \{1, \ldots, \Delta\}, & b_* &= 1, \\
M \setminus \{b_*\} &= \{2, \ldots, \Delta\}, & & \qquad (7.12) \\
D &= \{2, \ldots, \Delta\}, & c &= \Delta + 1, \\
R &= r(M \setminus \{b_*\}), & & \\
A &= \mathcal{B} \setminus (D \cup R \cup \{c\}). & & \qquad (7.13)
\end{aligned}$$

*More explicitly,*

$$R = \begin{cases} \{\Delta+2,\dots,2\Delta\}, & 2\Delta \le \varphi, \\ \{\Delta+2,\dots,\varphi\} \cup \{1\}, & 2\Delta = \varphi+1. \end{cases} \tag{7.14}$$

*In the reverse branch $2\Delta > \varphi+1$, equivalently $2\Delta \ge \varphi+2$, put*

$$\begin{aligned} M &= \{\Delta-1,\dots,\varphi-1\}, & b_* &= \varphi-1, \\ M \setminus \{b_*\} &= \{\Delta-1,\dots,\varphi-2\}, & & \quad (7.15) \\ D &= \{\Delta,\dots,\varphi-1\}, & c &= \Delta-1, \\ R &= r(M \setminus \{b_*\}), \\ A &= \mathcal{B} \setminus (D \cup R \cup \{c\}). & & \quad (7.16) \end{aligned}$$

*In this branch*

$$R = \{2\Delta-\varphi-1,\dots,\Delta-2\}. \tag{7.17}$$

*For either branch the four sets*

$$\mathcal{B} = D \,\dot\cup\, R \,\dot\cup\, \{c\} \,\dot\cup\, A \tag{7.18}$$

*are pairwise disjoint, and*

$$r(M) = R \,\dot\cup\, \{c\}, \qquad r(b_*) = c, \qquad r(M \setminus \{b_*\}) = R. \tag{7.19}$$

*Consequently*

$$s(D) \cap M = \varnothing, \qquad s(R) = M \setminus \{b_*\}, \qquad s(c) = b_*, \qquad s(A) \cap M = \varnothing. \tag{7.20}$$

*Thus every source–target pair $(i, r(i))$ is classified exactly once by its target in $D$, $R$, $\{c\}$, or $A$, while (7.20) records whether its source lies outside $M$, in $M \setminus \{b_*\}$, or at the distinguished base index $b_*$. No deformation is being assumed in this proposition.*

*Proof.* Because $r$ is a translation of the finite cyclic set $\mathcal{B}$, it is a bijection. It is therefore enough to compute the images of the displayed integer intervals and to verify the stated disjointness.

*Forward branch.* For $2 \le i \le \Delta-1$ one has

$$i+\Delta \le 2\Delta-1 \le \varphi,$$

so $r(i) = i+\Delta$. For $i = \Delta$, either $2\Delta \le \varphi$ and $r(\Delta) = 2\Delta$, or $2\Delta = \varphi+1$ and $r(\Delta) = 1$. This proves (7.14). In particular,

$$R \subseteq \{\Delta+2,\dots,\varphi\} \cup \{1\},$$

which is disjoint from $D = \{2,\dots,\Delta\}$ and from $c = \Delta+1$. Since $r(1) = \Delta+1 = c$ and $r$ is injective, (7.18) and (7.19) follow.

For $j \in D$, the inverse formula is

$$s(j) = \varphi - \Delta + j.$$

The branch inequality gives

$$s(j) \ge \varphi - \Delta + 2 \ge \Delta + 1,$$

so $s(D) \cap M = \varnothing$. The assertions for $R$, $c$, and $A$ now follow from bijectivity and (7.19).

*Reverse branch.* For every $i \in M \setminus \{b_*\}$,

$$i+\Delta \ge 2\Delta-1 \ge \varphi+1, \qquad i+\Delta \le \varphi+\Delta-2 < 2\varphi,$$

so there is exactly one wrap and

$$r(i) = i + \Delta - \varphi.$$

As $i$ runs from $\Delta - 1$ to $\varphi - 2$, this gives precisely (7.17). Its lower endpoint is at least 1 because $2\Delta \geq \varphi + 2$. Hence

$$R \subseteq \{1, \ldots, \Delta - 2\},$$

which is disjoint from $c = \Delta - 1$ and from $D = \{\Delta, \ldots, \varphi - 1\}$. Also $r(\varphi - 1) = \Delta - 1 = c$. Injectivity of $r$ again proves (7.18) and (7.19).

For $j = \Delta$ one has $s(j) = \varphi$. For $\Delta < j \leq \varphi - 1$ one has $s(j) = j - \Delta$, and

$$1 \leq j - \Delta \leq \varphi - \Delta - 1 \leq \Delta - 3.$$

Thus $s(D) \cap M = \varnothing$, and the remaining assertions in (7.20) again follow from bijectivity. This completes the reverse branch and the proof. $\square$

### 7.1 From the boundary chain to a projectivity

The following construction and proposition are purely projective. They contain no stochastic matrices, no Euclidean algorithm, and no modular labels.

**Definition 7.4** (Projectivity associated with a boundary-contact chain)**.** Let $P$ be a polygon with its complete cyclic vertex list. Consider consecutive boundary vertices

$$X_0, X_1, \ldots, X_{m+1}, \qquad m \geq 2,$$

read in either cyclic orientation, which are pairwise distinct and whose displayed sides do not exhaust the sides of $P$; points

$$C_i \in \operatorname{relint}[X_{i-1}, X_i] \qquad (2 \leq i \leq m + 1),$$

and supporting lines $\mathcal{L}_i$ of $P$ that expose only $X_i$ for $2 \leq i \leq m$. Regard every affine line below as its projective completion. For distinct projective points $A, B$, write $\langle A, B\rangle$ for their join. Put

$$\Lambda_1 = \operatorname{aff}(X_0, X_1), \qquad \Lambda_i = \mathcal{L}_i \quad (2 \leq i \leq m), \qquad K = \operatorname{aff}(C_m, C_{m+1}).$$

Thus $\Lambda_i$ and $K$ denote the projective completions of the displayed affine lines. Starting with $Y_1 \in \Lambda_1$, define successively

$$Y_i = \Lambda_i \cap \langle C_i, Y_{i-1}\rangle \quad (2 \leq i \leq m), \qquad Y_{m+1} = K \cap \langle X_{m+1}, Y_m\rangle.$$

Here every projection is well defined. For the first step, $C_2 \notin \Lambda_1 = \operatorname{aff}(X_0, X_1)$ because the three consecutive vertices $X_0, X_1, X_2$ are not collinear, while $C_2 \notin \Lambda_2$ because $\Lambda_2$ meets $P$ only at $X_2$. The lines $\Lambda_1$ and $\Lambda_2$ are distinct for the same noncollinearity reason. For $3 \leq i \leq m$, the center $C_i \in \operatorname{relint}[X_{i-1}, X_i]$ is different from both endpoints. Since the source line $\Lambda_{i-1}$ and target line $\Lambda_i$ expose only $X_{i-1}$ and $X_i$, respectively, $C_i$ lies on neither line; those two lines are also distinct, since a common line would meet $P$ at both named vertices.

For the closing step, $X_{m+1} \notin \Lambda_m$ because $\Lambda_m$ exposes only $X_m$. Also $X_{m+1} \notin K$. Indeed, if $X_{m+1}$ lay on $K$, then $K$, which contains $C_{m+1} \in \operatorname{relint}[X_m, X_{m+1}]$, would be the last side line $\operatorname{aff}(X_m, X_{m+1})$. It would then also contain $C_m \in \operatorname{relint}[X_{m-1}, X_m]$, forcing the two adjacent side lines to coincide, which would make three consecutive vertices collinear. Finally, $\Lambda_m \neq K$ because $K$ contains $C_m \in P \setminus \{X_m\}$ whereas $\Lambda_m \cap P = \{X_m\}$. Thus each projection center lies on neither its source nor its target line, and each source line is distinct from its target line. Hence

$$\Pi : \Lambda_1 \longrightarrow K, \qquad \Pi(Y_1) = Y_{m+1},$$

is a projectivity, being a composition of perspectivities. It is a map between two generally different projective lines. Only after an explicit projective identification $\Theta : K \to \Lambda_1$ has been specified does $\Theta \circ \Pi : \Lambda_1 \to \Lambda_1$ become a self-map of one projective line.

**Proposition 7.5** (Affine chart with parallel endpoint supporting lines)**.** *For every boundary-contact chain in definition 7.4, there exist supporting lines $M_0, M_1$ of $P$ such that*

$$P \cap M_0 = \{X_0\}, \qquad P \cap M_1 = \{X_{m+1}\},$$

*and there is an affine chart of the projective plane in which the image of $P$ is a bounded convex polygon, the images of the same labelled vertices and sides have the same incidences and cyclic boundary order up to a simultaneous reversal, $M_0$ and $M_1$ are parallel, no side of the displayed chain and no line $\mathcal{L}_i$ is parallel to them, and the displayed chain is the graph of a convex piecewise-linear function whose consecutive edge slopes are strictly increasing over a strictly increasing coordinate.*

*Proof.* At a vertex of a polygon, the supporting lines that expose that vertex form a nonempty open interval in the pencil of all lines through the vertex. Let $\mathcal{F}$ be the finite family consisting of all side lines of $P$ and all lines $\mathcal{L}_i$. Choose a supporting line $M_0$ that meets $P$ only at $X_0$. It is distinct from every line in $\mathcal{F}$: a side line in $\mathcal{F}$ meets $P$ along an entire side, while a line in $\mathcal{F}$ exposing another displayed vertex cannot contain $X_0$.

Now vary a supporting line $M_1$ that meets $P$ only at $X_{m+1}$. Exclude first the single direction parallel to $M_0$, so that $O = M_0 \cap M_1$ is a finite point. For a fixed $F \in \mathcal{F}$, the condition $O \in F$ excludes at most one further member of the pencil. Indeed, if $M_0$ is not parallel to $F$, put $K_F = M_0 \cap F$; then $O \in F$ is equivalent to $M_1 = \mathrm{aff}(X_{m+1}, K_F)$. If $M_0$ is parallel to $F$, no finite point of $M_0$ lies on $F$. Removing these finitely many exceptional lines from the open interval of such supporting lines leaves a valid choice of $M_1$. For this choice,

$$O = M_0 \cap M_1 \notin F \qquad (F \in \mathcal{F}). \tag{7.21}$$

The point $O$ lies outside $P$. Otherwise the defining contact property of $M_0$ would give $O = X_0$, while that of $M_1$ would give $O = X_{m+1}$, contradicting that $X_0$ and $X_{m+1}$ are distinct displayed vertices. Since $P$ is compact and convex, strict separation of $O$ from $P$ gives an affine functional $\ell$ such that

$$\ell(z) < \ell(O) \qquad (z \in P).$$

Hence the line $J = \{z : \ell(z) = \ell(O)\}$ passes through $O$ and is disjoint from $P$. Apply a projective automorphism $\Phi$ sending $J$ to the line at infinity. In affine coordinates it has the form

$$\Phi(z) = \frac{Az + b}{\omega(z)},$$

where $\omega$ is affine and vanishes precisely on $J$. It has constant sign on the connected set $P$; after multiplying the homogeneous matrix by $-1$ if necessary, $\omega > 0$ there. Compactness therefore gives a constant $\varepsilon_0 > 0$ such that $\omega(z) \geq \varepsilon_0$ for every $z \in P$. For $x, y \in P$ and $0 \leq t \leq 1$, direct substitution gives

$$\Phi((1-t)x + ty) = \frac{(1-t)\omega(x)}{(1-t)\omega(x) + t\omega(y)}\Phi(x) + \frac{t\omega(y)}{(1-t)\omega(x) + t\omega(y)}\Phi(y). \tag{7.22}$$

The two coefficients are nonnegative and sum to one, and they are both strictly positive when $0 < t < 1$. Thus $\Phi$ maps every segment in $P$ onto the segment joining the endpoint images and maps relative interiors to relative interiors. Applying the same formula to $\Phi^{-1}$ in the image chart shows that no additional points enter those segments. Consequently $\Phi(P)$

is a compact, hence bounded, convex polygon, and the images of the labelled vertices and sides have the same incidences and cyclic boundary order, up to a simultaneous reversal of orientation. Supporting lines that expose one vertex are also preserved.

The images of $M_0$ and $M_1$ are distinct parallel supporting lines, because their projective closures meet the new line at infinity at the common point which is the image of $O$. By (7.21), no side line of $P$ and no $\mathcal{L}_i$ has that point at infinity; equivalently, none is parallel to the two endpoint supporting lines.

Choose an affine coordinate $t$ constant on lines parallel to the two endpoint supporting lines, with $t(X_0) < t(X_{m+1})$. These are the two opposite supporting lines of $P$ in this direction and, by construction, meet $P$ only at $X_0$ and $X_{m+1}$. For every intermediate value of $t$, the corresponding transverse line cuts the convex polygon in a nondegenerate segment; its two endpoints lie on the two boundary arcs joining $X_0$ to $X_{m+1}$. Thus each arc is a graph over $t$. Since no side of the displayed arc is parallel to the endpoint supporting lines, no graph segment is vertical and $t$ is strictly increasing along the displayed chain. After reflecting the second affine coordinate if necessary, this chain is the lower graph of the polygon. Convexity makes its consecutive slopes nondecreasing. Equality of two consecutive slopes would make the corresponding three consecutive displayed vertices collinear, contrary to the vertex-list convention in definition 1.2. Hence the slopes are strictly increasing. □

**Lemma 7.6** (Location of the final intersection when $Z_1 = X_0$)**.** *Regard every affine line below as its projective completion, and write $\overline{AB}$ for the projective line through two distinct points $A, B$. Set $Z_1 = X_0$ and recursively define the projective points*

$$Z_i = \mathcal{L}_i \cap \overline{Z_{i-1}C_i} \qquad (2 \le i \le m). \tag{7.23}$$

*There exists an admissible affine chart as in proposition 7.5 in which all the points $Z_i$ are finite. The two projective lines in the final intersection are distinct, and the projectively defined point*

$$W_* := \overline{Z_m X_{m+1}} \cap \overline{C_m C_{m+1}} \in \operatorname{relint}[C_{m+1}, C_m] \tag{7.24}$$

*in the original affine chart. The same assertion holds when the original boundary chain is read in the opposite cyclic orientation.*

*Proof.* Apply a projective transformation supplied by proposition 7.5, and in its target affine chart use the same symbols for the transformed points and lines. The recursion (7.23) uses only joins and intersections, so projective transformations preserve all its incidence relations. Write

$$X_i = (t_i, f_i), \qquad t_0 < t_1 < \cdots < t_{m+1},$$

and let $m_i$ be the slope of the edge $[X_{i-1}, X_i]$. Since the chain is the lower graph of a polygon with its complete vertex list,

$$m_1 < m_2 < \cdots < m_{m+1}. \tag{7.25}$$

Because $\mathcal{L}_i$ is a supporting line of $P$ satisfying $P \cap \mathcal{L}_i = \{X_i\}$, its slope lies strictly between the slopes of the two incident sides. Hence

$$m_i < \operatorname{slope}(\mathcal{L}_i) < m_{i+1} \qquad (2 \le i \le m). \tag{7.26}$$

Write $C_i = (c_i, \zeta_i)$. Because $C_i \in \operatorname{relint}[X_{i-1}, X_i]$,

$$t_{i-1} < c_i < t_i, \qquad \zeta_i = f_i + m_i(c_i - t_i). \tag{7.27}$$

We prove inductively that every projectively defined point $Z_i$ is affine, say $Z_i = (z_i, g_i)$, and that, for $2 \le i \le m$,

$$c_i < z_i < t_i, \qquad \rho_i < m_i, \tag{7.28}$$

where $\rho_i$ is the slope of $\Gamma_i = \overline{Z_{i-1}C_i}$. When $i < m$, we also prove that $Z_i$ lies strictly above the backward extension of the next edge $[X_i, X_{i+1}]$.

For $i = 2$, the point $Z_1 = X_0$, and the slope of $\Gamma_2$ is

$$\rho_2 = \frac{(t_1 - t_0)m_1 + (c_2 - t_1)m_2}{c_2 - t_0}. \tag{7.29}$$

Both coefficients $t_1 - t_0$ and $c_2 - t_1$ are positive, and their sum is the denominator $c_2 - t_0$. Thus $\rho_2$ is a strict convex combination of $m_1$ and $m_2$, so $m_1 < \rho_2 < m_2$, in particular $\rho_2 < m_2$.

Assume now that $i \geq 3$ and the assertion has been proved at $i - 1$. The previous "above the next edge" statement says

$$g_{i-1} > f_{i-1} + m_i(z_{i-1} - t_{i-1}).$$

Since $C_i$ lies on that same edge line, $\zeta_i = f_{i-1} + m_i(c_i - t_{i-1})$. Also $z_{i-1} < t_{i-1} < c_i$. Subtracting the two displayed relations gives

$$\zeta_i - g_{i-1} < m_i(c_i - z_{i-1}),$$

and division by the positive denominator $c_i - z_{i-1}$ yields

$$\rho_i = \frac{\zeta_i - g_{i-1}}{c_i - z_{i-1}} < m_i. \tag{7.30}$$

Regard $\Gamma_i$ and $\mathcal{L}_i$ as affine functions of the coordinate $t$. At $t = c_i$, using (7.27),

$$\begin{aligned} \Gamma_i(c_i) - \mathcal{L}_i(c_i) &= \zeta_i - \big(f_i + \operatorname{slope}(\mathcal{L}_i)(c_i - t_i)\big) \\ &= (\operatorname{slope}(\mathcal{L}_i) - m_i)(t_i - c_i) > 0. \end{aligned} \tag{7.31}$$

At $t = t_i$,

$$\Gamma_i(t_i) - \mathcal{L}_i(t_i) = (\rho_i - m_i)(t_i - c_i) < 0. \tag{7.32}$$

Since $\rho_i < \operatorname{slope}(\mathcal{L}_i)$, the two lines have a unique intersection, and the opposite signs at $c_i$ and $t_i$ locate it at $c_i < z_i < t_i$. Because $Z_i \in \mathcal{L}_i$, its vertical difference from the backward extension of $[X_i, X_{i+1}]$ equals

$$g_i - (f_i + m_{i+1}(z_i - t_i)) = (\operatorname{slope}(\mathcal{L}_i) - m_{i+1})(z_i - t_i) > 0. \tag{7.33}$$

Here both factors on the right are negative. This completes the induction.

Let $L = \operatorname{aff}(Z_m, X_{m+1})$, and denote its slope by $\eta$. Since $Z_m \in \mathcal{L}_m$, direct subtraction of the endpoint ordinates gives

$$\eta = \frac{(t_m - z_m)\operatorname{slope}(\mathcal{L}_m) + (t_{m+1} - t_m)m_{m+1}}{t_{m+1} - z_m}. \tag{7.34}$$

The two coefficients in the numerator are positive and sum to the denominator. Therefore

$$\operatorname{slope}(\mathcal{L}_m) < \eta < m_{m+1}. \tag{7.35}$$

The line $\Gamma_m = \operatorname{aff}(C_m, Z_m)$ has slope $\rho_m < m_m < \operatorname{slope}(\mathcal{L}_m) < \eta$. Since $c_m < z_m$,

$$L(c_m) - \zeta_m = (\eta - \rho_m)(c_m - z_m) < 0. \tag{7.36}$$

On the other hand, $L$ passes through $X_{m+1}$, while $C_{m+1}$ lies on the last chain edge of slope $m_{m+1}$; hence

$$L(c_{m+1}) - \zeta_{m+1} = (\eta - m_{m+1})(c_{m+1} - t_{m+1}) > 0. \tag{7.37}$$

Let $K = \operatorname{aff}(C_m, C_{m+1})$. The affine function $L - K$ is negative at $c_m$ and positive at $c_{m+1}$. Because $c_m < t_m < c_{m+1}$, it has a zero at an abscissa strictly between $c_m$ and $c_{m+1}$. This zero is exactly the transformed point corresponding to $W_*$, and it lies in $\operatorname{relint}[C_m, C_{m+1}]$ in the admissible chart. The projective transformation used above and its inverse map every segment contained in the polygon onto the corresponding segment and preserve relative interiors, as proved in proposition 7.5. Applying the inverse transformation therefore gives (7.24) in the original affine chart. Distinctness of the two final projective lines is likewise preserved.

If the boundary chain is read in the opposite cyclic orientation, apply proposition 7.5 to that oriented chain and repeat the same calculation; no sign or index change is required inside the argument. $\square$

**Lemma 7.7** (Sign of $t-u(t)$ near a fixed point of a real projectivity)**.** *Let $u$ be a real projective automorphism, written in an affine coordinate near $0$, such that*

$$u(0) = 0, \qquad 0 < u(1) < 1.$$

*Then there are arbitrarily small nonzero real $\tau$ for which*

$$\tau - u(\tau) > 0. \tag{7.38}$$

*Proof.* A real projective automorphism fixing 0 has, on an open neighborhood of 0, the normal form

$$u(\tau) = \frac{a\tau}{1 + c\tau}, \qquad a \neq 0. \tag{7.39}$$

Consequently

$$\tau - u(\tau) = \frac{\tau(1 - a + c\tau)}{1 + c\tau}. \tag{7.40}$$

If $a \neq 1$, choose the sign of $\tau$ so that $\tau(1 - a) > 0$, and then choose $|\tau|$ small enough that $1 + c\tau > 0$ and $1 - a + c\tau$ has the sign of $1 - a$. The right side of (7.40) is then positive. If $a = 1$, the condition $0 < u(1) < 1$ gives $0 < 1/(1 + c) < 1$, hence $c > 0$; for every sufficiently small nonzero $\tau$,

$$\tau - u(\tau) = \frac{c\tau^2}{1 + c\tau} > 0.$$

In either case the admissible $\tau$'s can be chosen arbitrarily close to 0. $\square$

**Theorem 7.8** (A small deformation across the final supporting line)**.** *Let the vertices, side-interior points, and exposing supporting lines of definition 7.4 be given, and define the moving chain by*

$$X_1(\tau) = (1 - \tau)X_1 + \tau X_0, \qquad X_i(\tau) = \mathcal{L}_i \cap \operatorname{aff}(X_{i-1}(\tau), C_i) \quad (2 \leq i \leq m). \tag{7.41}$$

*Put*

$$Y(\tau) = (1 - \tau)C_{m+1} + \tau C_m.$$

*The recursive intersections are finite and continuous on some open interval about $0$. There are arbitrarily small nonzero real parameters $\tau$ for which $Y(\tau)$ lies in the open half-plane bounded by $\operatorname{aff}(X_m(\tau), X_{m+1})$ that contains $X_{m-1}(\tau)$. For every parameter in a sufficiently small interval about $0$, the displayed vertices remain distinct, in the same cyclic order, and in convex position.*

*Proof.* Put

$$\Lambda_1 = \overline{\operatorname{aff}(X_0, X_1)}, \qquad \Lambda_i = \overline{\mathcal{L}_i} \quad (2 \leq i \leq m), \qquad K = \overline{\operatorname{aff}(C_m, C_{m+1})},$$

where the bars denote projective completion. We first verify every perspectivity in the construction.

For $i = 2$, the center $C_2 \in \operatorname{relint}[X_1, X_2]$ is not on $\Lambda_1$, and $\mathcal{L}_2 \cap P = \{X_2\}$ gives $C_2 \notin \Lambda_2$. For $3 \leq i \leq m$, the point $C_i \in \operatorname{relint}[X_{i-1}, X_i]$ lies on neither $\Lambda_{i-1}$ nor $\Lambda_i$, because the corresponding affine supporting lines meet $P$ only at $X_{i-1}$ and $X_i$, respectively. Hence projection from center $C_i$ defines a projective isomorphism $\Lambda_{i-1} \to \Lambda_i$. For the final projection, $\mathcal{L}_m \cap P = \{X_m\}$ gives $X_{m+1} \notin \Lambda_m$. Also $X_{m+1} \notin K$: if it were, then $K$, which already contains $C_{m+1} \in \operatorname{relint}[X_m, X_{m+1}]$, would equal the side line $\operatorname{aff}(X_m, X_{m+1})$; this would put $C_m \in \operatorname{relint}[X_{m-1}, X_m]$ on that adjacent side line, forcing three consecutive vertices of $P$ to be collinear. Thus projection from $X_{m+1}$ is a projective isomorphism $\Lambda_m \to K$.

The affine parametrization of the distinguished starting point

$$\tau \longmapsto X_1(\tau) = (1-\tau)X_1 + \tau X_0$$

extends to a projective isomorphism from the compactified parameter line to $\Lambda_1$. Composing it with the verified perspectivities gives the following projective isomorphism from the compactified parameter line to the final contact line:

$$\mathcal{H} : \mathbb{P}^1(\mathbb{R}) \longrightarrow K. \tag{7.42}$$

Let

$$\upsilon : K \longrightarrow \mathbb{P}^1(\mathbb{R})$$

be the projective coordinate induced by the original affine line $\operatorname{aff}(C_m, C_{m+1})$, normalized by $\upsilon(C_{m+1}) = 0$ and $\upsilon(C_m) = 1$, and define the real projective automorphism

$$u = \upsilon \circ \mathcal{H}.$$

Both $\mathcal{H}$ and $\upsilon$ are projective isomorphisms; hence $u$ is a nonconstant real projectivity, as required by lemma 7.7. At $\tau = 0$, the projective recursion gives $\mathcal{H}(0) = C_{m+1}$, so $u(0) = 0$. At $\tau = 1$, the distinguished starting point has moved to $X_0$; preservation of joins and intersections under projective transformations gives $\mathcal{H}(1) = W_*$, where $W_*$ is the point of lemma 7.6. Since $W_* \in \operatorname{relint}[C_{m+1}, C_m]$,

$$0 < u(1) < 1. \tag{7.43}$$

At $\tau = 0$, every recursive intersection is the original vertex $X_i$, and the closing intersection is $C_{m+1}$. Therefore none of the intermediate affine coordinate maps has a pole at 0. Since each has at most one pole, there is an open interval $J$ about 0 on which all points $X_i(\tau)$ and the closing intersection

$$W(\tau) = \operatorname{aff}(X_m(\tau), X_{m+1}) \cap \operatorname{aff}(C_m, C_{m+1}) \tag{7.44}$$

are finite and continuous. In particular, the two lines in (7.44) are distinct throughout $J$. On this interval the local affine recursion agrees with the global line-to-line projectivity: $W(\tau) = \mathcal{H}(\tau)$, and $u(\tau)$ is precisely the normalized affine coordinate of $W(\tau)$.

By lemma 7.7, the set of nonzero parameters satisfying $\tau - u(\tau) > 0$ accumulates at 0. We postpone the choice of such a parameter until all remaining open conditions have been fixed.

Set

$$z(t) = (1-t)C_{m+1} + tC_m.$$

Choose $\varepsilon \in \{1, -1\}$ so that

$$\varepsilon \det(X_{m+1} - X_m, X_{m-1} - X_m) > 0,$$

and define the affine determinant function $\mathcal{S}$ together with its restriction $d$ to the parametrized line through $C_m, C_{m+1}$:

$$\begin{aligned} \mathcal{S}(x,\tau) &= \varepsilon \det\bigl(X_{m+1} - X_m(\tau), x - X_m(\tau)\bigr) \quad (x \in V), \\ d(t,\tau) &= \mathcal{S}(z(t),\tau). \end{aligned} \tag{7.45}$$

Thus $\mathcal{S}(x,\tau) > 0$ selects one of the two open half-planes bounded by the moving line $\operatorname{aff}(X_m(\tau), X_{m+1})$, and the choice of sign places the preceding chain vertex in that half-plane at $\tau = 0$. For $\tau \in J$, the point $z(u(\tau)) = W(\tau)$ is the unique intersection of that closing line with $\operatorname{aff}(C_m, C_{m+1})$. Since $d$ is affine in $t$,

$$d(t,\tau) = \gamma(\tau)(t - u(\tau)), \qquad \gamma(\tau) = \varepsilon \det\bigl(X_{m+1} - X_m(\tau), C_m - C_{m+1}\bigr). \tag{7.46}$$

The coefficient $\gamma$ is continuous. Write $C_m = (1-\alpha)X_{m-1} + \alpha X_m$ with $0 < \alpha < 1$. The defining choice of $\varepsilon$ gives

$$\gamma(0) = d(1,0) = (1-\alpha)\varepsilon \det(X_{m+1} - X_m, X_{m-1} - X_m) > 0. \tag{7.47}$$

After shrinking $J$, if necessary, $\gamma(\tau) > 0$ throughout $J$.

Fix the orientation of the original displayed chain. By lemma 2.6, every cyclically ordered triple of its displayed vertices has a nonzero determinant with one common sign at $\tau = 0$. There are finitely many such determinants, and all vertices depend continuously on $\tau$ on $J$. Shrink $J$ once more so that every one of these determinants keeps its sign throughout $J$. The triple-sign criterion, lemma 2.7, then shows that the displayed vertices remain distinct, in the same cyclic order, and in convex position for every $\tau \in J$. In particular,

$$\varepsilon \det\bigl(X_{m+1} - X_m(\tau), X_{m-1}(\tau) - X_m(\tau)\bigr) > 0$$

throughout $J$. Hence the half-plane described by $\mathcal{S}(x,\tau) > 0$ is exactly the open half-plane bounded by $\operatorname{aff}(X_m(\tau), X_{m+1})$ that contains $X_{m-1}(\tau)$.

Now use the accumulation statement from lemma 7.7 to choose a nonzero $\tau \in J$ with $\tau - u(\tau) > 0$. Evaluating (7.46) at $t = \tau$ gives

$$d(\tau,\tau) = \gamma(\tau)(\tau - u(\tau)) > 0.$$

Since $Y(\tau) = z(\tau)$, we have $\mathcal{S}(Y(\tau),\tau) = d(\tau,\tau) > 0$. Thus $Y(\tau)$ lies in the open half-plane bounded by $\operatorname{aff}(X_m(\tau), X_{m+1})$ that contains $X_{m-1}(\tau)$. Because such parameters accumulate at 0, the chosen parameter can be made arbitrarily small. □

## 7.2 Global realization of the boundary-chain motion

The local projective motion is useful only after it has been extended to all $N$ polygon vertices and all return incidences. The next lemma makes every required assignment and incidence explicit.

**Lemma 7.9** (Extension of the local deformation to all polygon vertices)**.** *Assume $\varphi > \delta$ and $\Delta > 1$, choose the forward or reverse boundary-contact chain from the orientation table, and suppose that its displayed boundary chain omits at least one side of $P$. Let $M, b_*, D, R, c, A$ be the data of proposition 7.3; the moved bases other than the distinguished base $b_*$ are $M \setminus \{b_*\}$. For $0 \le i \le m+1$, define the chain label*

$$\iota_i = \begin{cases} i, & \textit{in the forward branch,} \\ \varphi - i, & \textit{in the reverse branch,} \end{cases}$$

*so that $X_i = x_{\iota_i}$. For $2 \le i \le m+1$, define the side index*

$$k_i = \begin{cases} i, & \text{in the forward branch,} \\ \varphi - i + 1, & \text{in the reverse branch,} \end{cases}$$

*so that $C_i = \xi_{k_i}$. Then the map $i \mapsto \iota_i$ is a bijection $\{1, \dots, m\} \to M$, the map $i \mapsto k_i$ is a bijection $\{2, \dots, m\} \to D$, and*

$$b_* = \iota_1, \qquad c = k_{m+1} = r(b_*), \qquad a = H_{b_*}.$$

*In particular, the chain and contact labels have no repetitions. Moreover, $a = q$ in the forward branch while $a = q + h$ in the reverse branch.*

*Define the moving chain by*

$$X_1(\tau) = (1-\tau)X_1 + \tau X_0, \qquad X_i(\tau) = \mathcal{L}_i \cap \operatorname{aff}\big(X_{i-1}(\tau), C_i\big) \quad (2 \le i \le m). \tag{7.48}$$

*There is an open interval $U_0$ about $0$ on which all these points are finite and continuous. On $U_0$, define one base function for each $j \in \mathcal{B} = \{1, \dots, \varphi\}$ by*

$$B_j(\tau) = \begin{cases} X_i(\tau), & j = \iota_i \text{ for some } 1 \le i \le m, \\ x_j, & j \notin M, \end{cases} \tag{7.49}$$

*and define all $N$ polygon-vertex functions by the tower formula*

$$\widehat{x}_{F(t,j)}(\tau) = \lambda^t B_j(\tau) \qquad (1 \le j \le \varphi,\ 0 \le t < H_j). \tag{7.50}$$

*For $k \in \mathbb{Z}/N\mathbb{Z}$, put*

$$\widehat{E}_k(\tau) = [\widehat{x}_{k-1}(\tau), \widehat{x}_k(\tau)], \qquad P_\tau = \operatorname{conv}\{\widehat{x}_k(\tau) : k \in \mathbb{Z}/N\mathbb{Z}\}.$$

*Then, after shrinking $U_0$ to an open interval $U$ about $0$, the following statements hold for every $\tau \in U$.*

(i) Unique global assignment and cyclic order. *Equation (7.50) assigns exactly one function to each polygon label, every function is continuous, $\widehat{x}_k(0) = x_k$, and the $N$ points $\widehat{x}_k(\tau)$ are distinct extreme points in their original positive cyclic order. Hence $P_\tau$ is an $N$-gon with that complete vertex list and sides $\widehat{E}_k(\tau)$.*

(ii) Exact internal propagation. *For every tower state with $t + 1 < H_j$,*

$$\lambda \widehat{x}_{F(t,j)}(\tau) = \widehat{x}_{F(t+1,j)}(\tau). \tag{7.51}$$

*Thus every internal tower image is exactly a polygon vertex, with no side or endpoint convention involved.*

(iii) Final images assigned to every side. *For each $k \in \mathcal{B}$, let*

$$\widehat{V}_k(\tau) = \lambda^{H_{s(k)}} B_{s(k)}(\tau). \tag{7.52}$$

*For every $k \in \mathcal{B} \setminus \{c\}$,*

$$\widehat{V}_k(\tau) \in \operatorname{relint}\big(\widehat{E}_k(\tau)\big). \tag{7.53}$$

*The only final-image incidence not constrained to its assigned side is the return from the distinguished base $b_*$ to the side labelled $c$:*

$$Y(\tau) := \widehat{V}_c(\tau) = \lambda^a X_1(\tau) = (1-\tau)C_{m+1} + \tau C_m. \tag{7.54}$$

*Its assigned side is exactly*

$$\widehat{E}_c(\tau) = [X_m(\tau), X_{m+1}], \tag{7.55}$$

*where the endpoints may appear in the reverse order relative to the positive boundary orientation in the reverse branch.*

(iv) All nonclosing side inequalities for the closing image. *For every polygon side label $k$, define*

$$G_k(\tau) = \det\big(\widehat{x}_k(\tau) - \widehat{x}_{k-1}(\tau), Y(\tau) - \widehat{x}_{k-1}(\tau)\big). \tag{7.56}$$

*Then*

$$G_k(\tau) > 0 \qquad (k \neq c,\ \tau \in U). \tag{7.57}$$

*Thus the closing side is the only side inequality not fixed by openness.*

(v) Invariant-polygon criterion and the only unchecked side inequality. *For every $\tau \in U$, all image vertices other than $Y(\tau)$ lie on $\partial P_\tau$, while*

$$Y(\tau) \in \operatorname{Ext}(\lambda P_\tau).$$

*If, in addition, $G_c(\tau) > 0$, then*

$$Y(\tau) \in \operatorname{int} P_\tau, \qquad \lambda P_\tau \subseteq P_\tau, \qquad \operatorname{Ext}(\lambda P_\tau) \cap \operatorname{int} P_\tau = \{Y(\tau)\}. \tag{7.58}$$

*Proof.* We prove the five conclusions in six explicit stages.

*1. The local chain and the fixed chain endpoints.* By lemma 7.1, every line $\mathcal{L}_i$ in (7.48) supports $P$ and satisfies $P \cap \mathcal{L}_i = \{X_i\}$. Together with the vertices and side-interior points in (7.8), these are precisely the data required in definition 7.4. The first assertion of theorem 7.8 therefore supplies an open interval $U_0$ on which every relevant denominator is nonzero and the recursion is finite and continuous. At $\tau = 0$, induction gives $X_i(0) = X_i$: for $i = 1$ this is immediate, and for $i \geq 2$ both lines in the defining intersection pass through $X_i$ and are distinct because $P \cap \mathcal{L}_i = \{X_i\}$ while $C_i \neq X_i$ lies on the incident side.

The endpoints $X_0$ and $X_{m+1}$ are fixed global tower vertices. In the forward branch, $X_{m+1} = x_{\Delta+1}$ is an unmoved base. Also $F(h, \varphi) = 0$ and $h < H_\varphi = q + h$; when $h = 0$, this statement is the same identity with $F(0, \varphi) = 0$ and $\varphi = N$. Hence

$$\widehat{x}_0(\tau) = \lambda^h B_\varphi(\tau) = \lambda^h x_\varphi = x_0 = X_0.$$

In the reverse branch, $X_0 = x_\varphi$ and $X_{m+1} = x_{\Delta-2}$ are unmoved bases. The reverse inequality forces $\Delta \geq 3$, so the latter label belongs to $\mathcal{B}$. Thus in both branches the constants appearing in the local boundary-contact chain are exactly the corresponding vertices in the global tower assignment.

*2. The tower formula is a unique continuous labeling.* The branch formulas show directly that $i \mapsto \iota_i$ is a bijection from $\{1, \dots, m\}$ onto $M$. Consequently (7.49) assigns exactly one formula to every base label: a moved formula for $j \in M$ and the fixed formula $x_j$ for $j \notin M$.

The map $F$ in (6.7) is a bijection from the finite state set

$$\mathcal{D} = \{(t, j) : 1 \leq j \leq \varphi,\ 0 \leq t < H_j\}$$

onto $\mathbb{Z}/N\mathbb{Z}$. Therefore every polygon label has exactly one preimage $(t, j)$, so (7.50) gives one and only one definition for every $\widehat{x}_k(\tau)$. In particular, no moved formula and no fixed formula can assign different points to the same polygon label. Each base function is continuous and multiplication by the fixed linear map $\lambda^t$ is continuous, hence all global vertex functions are continuous.

At $\tau = 0$, the geometric tower identity (6.9) gives

$$\widehat{x}_{F(t,j)}(0) = \lambda^t B_j(0) = \lambda^t x_j = x_{F(t,j)}.$$

Since $F$ is onto, $\widehat{x}_k(0) = x_k$ for every polygon label $k$.

Apply the simultaneous convex-admissibility lemma lemma 2.8 first with no auxiliary test points. It gives an open interval $U_1 \subseteq U_0$ on which the global vertex list consists of distinct points in convex position and retains its positive cyclic order. This proves part (i) on $U_1$.

Equation (7.51) is now an identity:

$$\lambda \widehat{x}_{F(t,j)}(\tau) = \lambda^{t+1} B_j(\tau) = \widehat{x}_{F(t+1,j)}(\tau).$$

This proves part (ii).

*3. Supporting lines for the indices in $\mathcal{B}$.* Write

$$L_k(\tau) = \operatorname{aff}\bigl(\widehat{x}_{k-1}(\tau), \widehat{x}_k(\tau)\bigr) \qquad (k \in \mathcal{B}).$$

We identify every one of these lines.

The branch formulas also show that $i \mapsto k_i$ maps $\{2, \dots, m\}$ bijectively onto $D$ and sends $m+1$ to $c$.

If $k \in D$, there is therefore a unique chain index

$$\psi(k) = \begin{cases} k, & \text{forward branch,} \\ \varphi - k + 1, & \text{reverse branch,} \end{cases} \qquad 2 \le \psi(k) \le m,$$

for which $C_{\psi(k)} = \xi_k$. The two endpoints of $\widehat{E}_k(\tau)$ are $X_{\psi(k)-1}(\tau)$ and $X_{\psi(k)}(\tau)$, possibly in reverse order. By the recursion,

$$L_k(\tau) = \operatorname{aff}\bigl(X_{\psi(k)-1}(\tau), C_{\psi(k)}\bigr). \tag{7.59}$$

We now verify directly, using the vertex functions already defined in (7.49)–(7.50), that every line with label in $R \cup A$ is fixed. In the forward branch,

$$R \cup A = \{1\} \cup \{\Delta + 2, \dots, \varphi\}.$$

The tower based at $\varphi \notin M$ is fixed, and the tower identity $F(h, \varphi) = 0$ gives $\widehat{x}_0(\tau) = \lambda^h B_\varphi(\tau) = x_0$. The distinguished starting point $X_1(\tau)$ moves on the original line $\operatorname{aff} E_1$, so $L_1(\tau) = L_1(0)$. If $\Delta + 2 \le k \le \varphi$, both endpoints of $E_k$ have base labels outside $M$, except that when $k = \varphi = N$ the second endpoint is the already fixed vertex $x_0$. Hence $L_k(\tau) = L_k(0)$ throughout this range.

In the reverse branch,

$$R \cup A = \{1, \dots, \Delta - 2\} \cup \{\varphi\}.$$

Here $B_\varphi(\tau) = x_\varphi$, while the distinguished starting point $X_1(\tau) = \widehat{x}_{\varphi-1}(\tau)$ moves on the original line $\operatorname{aff} E_\varphi$; thus $L_\varphi(\tau) = L_\varphi(0)$. For $2 \le k \le \Delta - 2$, both endpoints have base labels outside $M$, so $L_k(\tau) = L_k(0)$. Finally, $\widehat{x}_1(\tau) = x_1$ and $\widehat{x}_0(\tau) = \lambda^h B_\varphi(\tau) = x_0$ by (7.5), so $L_1(\tau) = L_1(0)$ as well.

The one remaining side with label in $\mathcal{B}$ has label $c$, and direct reading of the branch table gives (7.55). Thus the partition

$$\mathcal{B} = D \,\dot\cup\, R \,\dot\cup\, \{c\} \,\dot\cup\, A$$

accounts for every index in $\mathcal{B}$ exactly once.

*4. Four-case partition and line membership.* First partition the index pairs in the return-time tower into those whose next iterate remains in the same orbit segment and those that index its final point:

$$\begin{aligned} \mathcal{D}_{\text{int}} &= \{(t, j) \in \mathcal{D} : t + 1 < H_j\}, \\ \mathcal{D}_{\text{top}} &= \{(H_j - 1, j) : j \in \mathcal{B}\}. \end{aligned} \tag{7.60}$$

These sets are disjoint, their union is $\mathcal{D}$, and

$$|\mathcal{D}_{\text{int}}| = N - \varphi, \qquad |\mathcal{D}_{\text{top}}| = \varphi.$$

Because $F$ is bijective, they index disjoint sets of polygon vertices whose union is the complete set of $N$ source vertices. Part (ii) has already accounted for every source in $\mathcal{D}_{\mathrm{int}}$. The image of the final point $(H_j - 1, j)$ in the orbit segment based at $j$ is $\lambda^{H_j} B_j(\tau) = \widehat{V}_{r(j)}(\tau)$. Since $r$ is a bijection of $\mathcal{B}$, every such image is therefore indexed exactly once by its assigned side label $k$, with unique source base $s(k)$.

We now apply the inverse-source partition (7.20) to those $\varphi$ final images according to their assigned sides. The four cases are

| side class | return index $s(k)$ | line containing $\widehat{E}_k(\tau)$ | reason for line membership |
|---|---|---|---|
| $D$ | fixed, since $s(D) \cap M = \varnothing$ | varies by the projection recursion | the fixed final image $\xi_k = C_{\psi(k)}$ is built into that line |
| $R$ | $s(k) \in M \setminus \{b_*\}$ and moves | fixed | the base remains on the line $\lambda^{-H_{s(k)}} L_k(0)$ |
| $A$ | fixed | fixed | the original incidence is unchanged |
| $\{c\}$ | the distinguished index $b_*$ | $\mathrm{aff}(X_m(\tau), X_{m+1})$ | the only final incidence not imposed in advance, namely $Y(\tau)$ |

The row sets are disjoint and exhaustive by (7.18); the source assertions are exactly (7.20).

If $k \in D$, then $s(k) \notin M$, so its source base is fixed. The original top relation gives

$$\widehat{V}_k(\tau) = \lambda^{H_{s(k)}} x_{s(k)} = \xi_k = C_{\psi(k)}.$$

Equation (7.59) puts this fixed point on the moving line $L_k(\tau)$.

If $k \in R$, then $s(k) \in M \setminus \{b_*\}$. Let $i$ be the unique chain index with $\iota_i = s(k)$; necessarily $2 \le i \le m$. The recursion puts $B_{s(k)}(\tau) = X_i(\tau)$ on

$$\mathcal{L}_i = \lambda^{-H_{s(k)}} L_k(0).$$

The target line is fixed, so

$$\widehat{V}_k(\tau) = \lambda^{H_{s(k)}} B_{s(k)}(\tau) \in L_k(0) = L_k(\tau).$$

If $k \in A$, both its source base and its target side line are fixed. Hence

$$\widehat{V}_k(\tau) = \xi_k \in L_k(0) = L_k(\tau).$$

The only remaining side label is $c$. Here $s(c) = b_*$, so

$$\widehat{V}_c(\tau) = \lambda^{H_{b_*}} B_{b_*}(\tau) = \lambda^a X_1(\tau).$$

Using the interpolation of the distinguished starting point and (7.9),

$$\lambda^a X_1(\tau) = (1-\tau)\lambda^a X_1 + \tau \lambda^a X_0 = (1-\tau) C_{m+1} + \tau C_m.$$

This proves (7.54). The partition from proposition 7.3 proves simultaneously that every final image has been treated and that no final image belongs to two cases.

For every nonclosing side label $k$, exact collinearity has now been proved, and at $\tau = 0$ the point $\widehat{V}_k(0) = \xi_k$ belongs to $\operatorname{relint} E_k$. Apply lemma 2.8 to the finite family

$$w_k(\tau) = \widehat{V}_k(\tau), \qquad k \in \mathcal{B} \setminus \{c\}.$$

After shrinking $U_1$ to an interval $U_2$, every one of these points lies in the relative interior of its assigned moving side. This proves (7.53) and part (iii).

*5. All nonclosing inequalities for the image of the distinguished base.* At $\tau = 0$,

$$Y(0) = C_{m+1} = \xi_c \in \operatorname{relint} E_c.$$

By (2.11) in lemma 2.8, this point satisfies the strict side inequality for every $k \neq c$. Apply the same lemma to the finite family of test pairs

$$(Y(\tau), k), \qquad k \in \mathbb{Z}/N\mathbb{Z} \setminus \{c\}.$$

After shrinking $U_2$ to an interval $U \ni 0$, the functions (7.56) satisfy (7.57). This is a global statement: it includes sides whose endpoints are internal tower vertices, not merely the displayed boundary-chain sides. This proves part (iv).

*6. Invariance and the only remaining unchecked side inequality.* Let $k \in \mathbb{Z}/N\mathbb{Z}$ and let $(t, j)$ be its unique tower preimage under $F$. If $t + 1 < H_j$, then (7.51) gives $\lambda\widehat{x}_k(\tau)$ as another polygon vertex. If $t = H_j - 1$ and $r(j) \neq c$, then

$$\lambda\widehat{x}_k(\tau) = \lambda^{H_j} B_j(\tau) = \widehat{V}_{r(j)}(\tau) \in \operatorname{relint}(\widehat{E}_{r(j)}(\tau)).$$

Thus every image vertex other than the closing image lies on $\partial P_\tau$ for every $\tau \in U$.

Consecutive record times satisfy $q > 0$ and $h \geq 0$, so $a = H_{b_*} \geq 1$. Put

$$k_* = F(a - 1, b_*).$$

Then

$$\widehat{x}_{k_*}(\tau) = \lambda^{a-1} B_{b_*}(\tau), \qquad \lambda\widehat{x}_{k_*}(\tau) = Y(\tau).$$

The point $\widehat{x}_{k_*}(\tau)$ is an extreme point of the polygon $P_\tau$. Since multiplication by $\lambda \neq 0$ is invertible, it maps extreme points bijectively to extreme points; hence $Y(\tau) \in \operatorname{Ext}(\lambda P_\tau)$ for every $\tau \in U$.

Now fix any $\tau \in U$ for which $G_c(\tau) > 0$. Together with (7.57), this gives all $N$ strict side inequalities for $Y(\tau)$. The final assertion of lemma 2.8 therefore yields $Y(\tau) \in \operatorname{int} P_\tau$. Every vertex image belongs to $P_\tau$: internal images are polygon vertices, nonclosing top images lie in relative side interiors, and the closing top image is $Y(\tau)$. Convexity and linearity give

$$\lambda P_\tau = \operatorname{conv}\{\lambda\widehat{x}_k(\tau) : k \in \mathbb{Z}/N\mathbb{Z}\} \subseteq P_\tau.$$

All image vertices except $Y(\tau)$ lie on the boundary, while $Y(\tau)$ is interior. This proves (7.58) and part (v). ☐

**Theorem 7.10** (Existence of an invariant deformation with one interior image vertex)**.** *Under the hypotheses and notation of lemma 7.9, every neighborhood of* 0 *contains a nonzero parameter $\tau \in U$ for which*

$$Y(\tau) \in \operatorname{int} P_\tau, \qquad \lambda P_\tau \subseteq P_\tau, \qquad \operatorname{Ext}(\lambda P_\tau) \cap \operatorname{int} P_\tau = \{Y(\tau)\}. \tag{7.61}$$

*In particular, $Y(\tau) \in \operatorname{Ext}(\lambda P_\tau) \cap \operatorname{int} P_\tau$, while every other vertex of $\lambda P_\tau$ lies on $\partial P_\tau$.*

*Proof.* Theorem 7.8 supplies, arbitrarily close to 0, a nonzero parameter for which $Y(\tau)$ lies in the open half-plane bounded by $\operatorname{aff}(X_m(\tau), X_{m+1})$ that contains $X_{m-1}(\tau)$. Choose that parameter inside the common admissibility interval $U$ from lemma 7.9. By part (i) of that lemma, the vertices of $P_\tau$ are distinct, in convex position, and retain their original cyclic order. Hence the half-plane bounded by $\operatorname{aff}(\widehat{E}_c(\tau)) = \operatorname{aff}(X_m(\tau), X_{m+1})$ that contains the nonincident vertex $X_{m-1}(\tau)$ is the polygon-interior half-plane. The theorem therefore gives $G_c(\tau) > 0$. Part (v) of lemma 7.9 now gives (7.61). ☐

## 7.3 The first-return step

**Theorem 7.11** (The first-return step satisfies $\Delta = 1$)**.** *Assume $N \geq 4$. If $\varphi > \delta$, then $\Delta = 1$.*

*Proof.* Assume $\Delta > 1$. By lemma 7.2, the selected branch has $m \geq 2$, its displayed vertices are pairwise distinct consecutive boundary vertices, and its displayed sides do not exhaust the sides of $P$. The same lemma verifies this directly at $\Delta = 2$, $2\Delta = \varphi + 1$, $h = 0$, and $\varphi = N$. Together with lemma 7.1 and proposition 7.3, all hypotheses of lemma 7.9 and theorem 7.10 are satisfied.

Choose the nonzero parameter supplied by that theorem. Then $P_\tau$ is an invariant $N$-gon for the same $N$-critical map, while the vertex $Y(\tau)$ of $\lambda P_\tau$ lies in $\operatorname{int} P_\tau$. This contradicts the literal conclusion of theorem 3.2 that every extreme point of $\lambda P_\tau$ lies on $\partial P_\tau$. Hence $\Delta > 1$ is impossible. Since $\Delta$ is a positive integer, $\Delta = 1$. ☐

*Remark* 7.12 (Boundary cases in the proof that $\Delta = 1$)*.* The preceding proof includes the following boundary cases.

(a) If $\Delta = 2$, the reverse inequality is impossible because $\Delta < \varphi$; the forward boundary-contact chain has $m = 2$, one supporting line $\mathcal{L}_2$ satisfying $P \cap \mathcal{L}_2 = \{X_2\}$, one side index in $D$, and the same exhaustive partition of the side indices.

(b) The equality $2\Delta = \varphi + 1$ belongs to the forward branch. It is exactly the reduction modulo $\varphi$ expressed by $r(\Delta) = 1$ in (7.14); side label 1 remains on the fixed supporting line.

(c) The equality $h = 0$ is equivalent to $\varphi = N$ by theorem 6.1. Then $\lambda^h E_\varphi = E_0$ is the cyclic identity, and every formula in the boundary-contact chain and global tower construction remains literal.

(d) The orbit number $\delta$, including $\delta = 1$, enters this argument only through the hypothesis $\varphi > \delta$ and the existence of the first-return decomposition. It creates no additional deformation case.

*Remark* 7.13 (Local and global data used in the deformation argument)*.* The local choice of the deformation parameter depends only on the behavior near 0 of the normalized projective map $u$. The preceding extension lemma supplies the separate global assertions needed to use that local choice: every polygon index is assigned exactly once, every nonclosing image lies on the line containing its assigned side and remains in the relative interior of that side, and every side inequality except $G_c(\tau) > 0$ persists for sufficiently small $\tau$.

*Proof of theorem 1.3.* Choose an adapted complex coordinate by proposition 2.1. The boundary conclusion for every invariant polygon with at most $N$ vertices is theorem 3.2. The least-area selection and cyclic interlacing in lemma 4.13 give a half-open contact assignment after choosing the appropriate orientation. Write the resulting positive indexing as in lemma 4.13, so that $E_i = [x_{i-1}, x_i]$ and $\chi(x_{i-\kappa}) = E_i$. If $\kappa = N$, then

$$\sigma(E_i) = \chi(\operatorname{head}(E_i)) = \chi(x_i) = E_{i+\kappa} = E_i,$$

so $\sigma$ is the identity. The final assertion of lemma 4.13 says that every contact lies in the relative interior of its assigned side, and hence $I = \mathcal{E}(P)$. There are therefore no permitted vertex replacements, so assertion (ii) is vacuous; the only relative-interior contact set obtainable by such replacements is $I$ itself, which is a cyclic interval with $\varphi = N = \delta$. Assertions (iii) and (iv) follow immediately. This completes the identity case.

Henceforth assume $1 \leq \kappa < N$. The permitted vertex replacement is proposition 5.1, and corollary 5.2 translates its set update $i \mapsto i + \kappa$ exactly into $e \mapsto \sigma(e)$. Under the induced side bijection $b$, the new contact permutation is $\sigma' = b \circ \sigma \circ b^{-1}$, and the construction is covariant under real-linear conjugacy. Apply lemma 5.5 to a lexicographically minimal relative-interior contact set obtainable by permitted vertex replacements. It is one interval of length $\varphi \geq \delta$, where $\delta$ is the number of contact-permutation orbits. If $\varphi = \delta$, the interval meets every orbit and has exactly one point in each; hence it is a complete orbit transversal, and the common return time is the orbit length $N/\delta$.

Suppose $\varphi > \delta$. The last assertion of lemma 5.5 makes $N - \varphi$ an upper-record residue. Apply theorem 6.1 with $\nu = \varphi$. It gives a two-height tower partition and a first-return translation by $\Delta$. Theorem 7.11 gives $\Delta = 1$. Thus the first return is the cyclic successor on the relative-interior contact interval. All parts of the theorem have now been proved. □

## 8 Consecutive Farey fractions and finite product equations

Let $F_n$ denote the Farey sequence of order $n$ on $[0,1]$. Multiplying the recurrence relations along a finite closed chain and using its closing relation to cancel the initial and terminal nonzero vertices gives the finite product equations below. Every equality of arguments in this section is an equality between specified real lifts, not merely a congruence modulo $2\pi$.

**Lemma 8.1** (Farey adjacency criterion). *Two reduced fractions $a/b < c/d$ in $F_n$ are consecutive if and only if*

$$bc - ad = 1, \qquad b + d > n. \tag{8.1}$$

*Proof.* Assume first that $bc - ad = 1$. If a reduced fraction $h/k$ lies strictly between $a/b$ and $c/d$, define

$$m = ck - dh, \qquad \ell = bh - ak.$$

Cross multiplication shows that $m$ and $\ell$ are positive integers. The determinant-one relation gives the exact decomposition

$$(k,h) = m(b,a) + \ell(d,c).$$

Looking at the first coordinate yields $k = mb + \ell d \geq b + d$. Hence, if $b + d > n$, no fraction of denominator at most $n$ can lie between the two endpoints, so they are consecutive in $F_n$.

Conversely, suppose the two fractions are consecutive in $F_n$ and put $D = bc - ad$. The determinant is a positive integer. If $D > 1$, the half-open parallelogram spanned by the primitive integer vectors $u = (b,a)$ and $v = (d,c)$ contains a nonzero integer point

$$w = \alpha u + \beta v, \qquad 0 \leq \alpha, \beta < 1.$$

Indeed, lemma A.6 says that the half-open fundamental parallelogram contains exactly $D = |\det(u,v)|$ representatives of the quotient group $\mathbb{Z}^2/(\mathbb{Z}u + \mathbb{Z}v)$. When $D > 1$, one of these representatives is different from the origin. If $\alpha = 0$, then $w = \beta v$ is a nonzero integer point in the relative interior of $[0,v]$, which is impossible because $v$ is primitive. Likewise, $\beta = 0$ would give a nonzero integer point in the relative interior of $[0,u]$. Thus $0 < \alpha, \beta < 1$. Both $w$ and $u+v-w$ are positive combinations of $u, v$, so their slopes lie strictly between $a/b$ and $c/d$. Their first coordinates are positive integers $k$ and $b+d-k$; one is at most $(b+d)/2 \leq n$ because $b, d \leq n$. After reducing that intermediate slope, its denominator can only decrease. This produces an element of $F_n$ strictly between the alleged consecutive fractions, a contradiction. Therefore $D = 1$. Finally, if $b+d \leq n$, the reduced mediant $(a+c)/(b+d)$ has denominator at most $n$ and lies strictly between the endpoints. Thus consecutiveness also forces $b+d > n$. □

**Lemma 8.2** (Reflection preserves Farey adjacency). *Let $a/b$ and $c/d$ be consecutive fractions of $F_N$, with $a/b < c/d$. Then $(d-c)/d$ and $(b-a)/b$ are consecutive fractions of $F_N$, and whenever $a/b < x < c/d$ one has*

$$\frac{d-c}{d} < 1 - x < \frac{b-a}{b}. \tag{8.2}$$

*The reflected open interval is therefore again bounded by consecutive fractions of $F_N$. The reflection exchanges the endpoint denominators: the left and right denominators in (8.2) are $d$ and $b$, respectively. In particular, if $d \le b$, the reflected ordered interval has the smaller denominator first.*

*Proof.* Reducedness is preserved because

$$\gcd(d-c, d) = \gcd(c, d) = 1, \qquad \gcd(b-a, b) = \gcd(a, b) = 1.$$

The inequalities follow by subtracting the original inequalities from 1. Moreover

$$(b-a)d - (d-c)b = bc - ad = 1,$$

and the sum of the reflected denominators is the unchanged number $b + d > N$. The Farey adjacency criterion, lemma 8.1, therefore proves consecutiveness. The denominator statement is immediate from the displayed fractions. $\square$

**Lemma 8.3** (Conjugation and reversal of a finite recurrence)**.** *Let $\lambda \in \mathbb{C}\backslash\mathbb{R}$ satisfy $0 < |\lambda| < 1$, let $\theta = \arg_+(\lambda)$, and let $q, d \ge 1$ and $0 \le h < dq$. Suppose nonzero points $z_0, \dots, z_d$ and numbers*

$$0 \le b_i < 1, \qquad a_i > 0 \qquad (1 \le i \le d),$$

*are given. Choose lifted arguments $\Theta_i \in \mathbb{R}$ such that*

$$z_i = |z_i| e^{i\Theta_i} \qquad (0 \le i \le d).$$

*Assume*

$$\begin{aligned}
(\lambda^q - b_i) z_i &= a_i z_{i-1} && (1 \le i \le d), && (8.3)\\
\lambda^h z_d &= z_0, && && (8.4)\\
0 < \Theta_i - \Theta_{i-1} &< \pi && (1 \le i \le d), && (8.5)\\
\Theta_d + h\theta &= \Theta_0 + 2\pi m && && (8.6)
\end{aligned}$$

*for an integer $m$.*

*Put*

$$\mu = \overline{\lambda}, \qquad \vartheta = \arg_+(\mu) = 2\pi - \theta, \qquad e = -h, \qquad s = dq - h,$$

*and, for $1 \le j \le d$, define*

$$w_j = \overline{z}_{d-j}, \qquad w_0 = \overline{z}_d, \qquad \beta'_j = b_{d-j+1}, \qquad \alpha'_j = a_{d-j+1}.$$

*Then*

$$\begin{aligned}
(\mu^q - \beta'_j) w_{j-1} &= \alpha'_j w_j && (1 \le j \le d), && (8.7)\\
\mu^e w_d &= w_0. && && (8.8)
\end{aligned}$$

*Consequently*

$$\mu^e \prod_{j=1}^{d} (\mu^q - \beta'_j) = \prod_{j=1}^{d} \alpha'_j, \tag{8.9}$$

$$\mu^s \prod_{j=1}^{d} (\mu^q - \beta'_j) = \mu^{dq} \prod_{j=1}^{d} \alpha'_j. \tag{8.10}$$

*Put*

$$u_j = \Theta_{d-j+1} - \Theta_{d-j} \in (0, \pi),$$

*for* $1 \le j \le d$*. Then* (8.7) *implies that* $\mu^q - \beta'_j$ *has argument* $u_j$ *modulo* $2\pi$*. Because* $u_j \in (0, \pi)$*, it is the unique argument of that factor in* $(0, \pi)$*, and*

$$e\vartheta + \sum_{j=1}^{d} u_j = 2\pi(m - h). \tag{8.11}$$

*Proof.* Conjugate (8.3) with $i = d - j + 1$. Since

$$\overline{z}_{d-j+1} = w_{j-1}, \qquad \overline{z}_{d-j} = w_j,$$

the conjugated equation is exactly (8.7). Conjugating (8.4) gives

$$\mu^h \overline{z}_d = \overline{z}_0.$$

Because $w_0 = \overline{z}_d$ and $w_d = \overline{z}_0$, this is equivalent, using $\mu \neq 0$, to (8.8) with $e = -h$. Multiplying the forward recurrence equations and cancelling the nonzero points $w_0, w_d$ by the signed closure proves (8.9). Multiplication by $\mu^{dq}$ gives (8.10); the exponent on its left is $dq - h = s > 0$ by hypothesis.

Choose lifted arguments for the reflected points by

$$\Phi_j = -\Theta_{d-j}.$$

Because $w_j = \overline{z}_{d-j}$, these choices satisfy $w_j = |w_j| e^{i\Phi_j}$. Moreover,

$$\Phi_j - \Phi_{j-1} = \Theta_{d-j+1} - \Theta_{d-j} = u_j \in (0, \pi).$$

Since $\alpha'_j > 0$, equation (8.7) shows that $u_j$ is congruent modulo $2\pi$ to the argument of $\mu^q - \beta'_j$. Its membership in $(0, \pi)$ selects the oriented branch uniquely. Finally,

$$\sum_{j=1}^{d} u_j = \Theta_d - \Theta_0 = 2\pi m - h\theta$$

by (8.6); therefore

$$e\vartheta + \sum_{j=1}^{d} u_j = -h(2\pi - \theta) + (2\pi m - h\theta) = 2\pi(m - h),$$

which is (8.11). $\square$

**Lemma 8.4** (The case $\kappa = N$)**.** *Assume the integer lift in lemma 4.13 is* $\kappa = N$*. For the data in theorem 1.4, take the eigenvalue in the opposite orientation:*

$$\mu = \overline{\lambda}, \qquad y = 1 - x.$$

*For this eigenvalue the ordered Farey interval is*

$$\frac{0}{1} < y < \frac{1}{N},$$

*so* $p = 0$*,* $q = 1$*,* $r = 1$*,* $s = N$*,* $d = N$*, and* $e = 0$*. There are parameters satisfying* (1.5) *for which* (1.6)*–*(1.9) *hold.*

*Proof.* Every contact lies in the relative interior of its assigned side by lemma 4.13; hence $0 < \alpha_i, \beta_i < 1$. Extend the indices periodically and set $z_i = x_i$ for $0 \le i \le N$, with $z_N = z_0$ as a point and with lifted arguments $\Theta_N = \Theta_0 + 2\pi$. The contact equation for $\kappa = N$ is

$$\lambda x_i = \beta_i x_{i-1} + \alpha_i x_i.$$

After rearrangement it is the backward recurrence relation

$$(\lambda - \alpha_i) z_i = \beta_i z_{i-1} \qquad (1 \le i \le N),$$

where the coefficients at $i = N$ are the periodic coefficients at index 0. The inequalities required in (8.5) follow from lemma 2.10. Apply lemma 8.3 with

$$q = 1, \quad d = N, \quad h = 0, \quad m = 1, \quad b_i = \alpha_i, \quad a_i = \beta_i.$$

It produces the reflected recurrence relations with

$$\beta'_j = \alpha_{N-j+1}, \qquad \alpha'_j = \beta_{N-j+1}.$$

Because $\alpha_i + \beta_i = 1$, these coefficients satisfy $\alpha'_j = 1 - \beta'_j$ and hence (1.5). Relabel $(\alpha'_j, \beta'_j)$ as $(\alpha_j, \beta_j)$; these are the coefficients appearing in the final product equations and equation for the chosen real arguments. The reflection lemma also gives the polynomial and Laurent equations with $s = N$ and $e = 0$, and the equality

$$\sum_{j=1}^{N} u_j = 2\pi.$$

This is (1.9) because $r - dp = 1$.

It remains to identify the ordered Farey interval. The angle estimate (4.13) gives

$$\frac{N-1}{N} < x < 1.$$

The endpoints are consecutive in $F_N$, and lemma 8.2 gives

$$0 < 1 - x < \frac{1}{N}.$$

Thus the reflected orientation has exactly the Farey labels stated in the lemma, with the smaller denominator on the left. □

## 8.1 Recurrence relations and finite product equations

**Proposition 8.5** (More than one relative-interior contact in some orbit: $\varphi > \delta$)**.** *Assume $N \ge 4$ and $\varphi > \delta$. Then $\delta = 1$, and the data in theorem 1.4 use the eigenvalue in the fixed contact orientation:*

$$\mu = \lambda.$$

*There are integers $p, q, d, e$ with*

$$q\kappa - pN = 1, \qquad N = qd + e, \qquad 0 \le e < q, \qquad d = \lfloor N/q \rfloor, \tag{8.12}$$

*and parameters satisfying (1.5) such that*

$$(\lambda^q - \beta_j) x_{j-1} = \alpha_j x_j \qquad (1 \le j \le d), \tag{8.13}$$

$$\lambda^e x_d = x_0. \tag{8.14}$$

*The ordered Farey interval is*

$$\frac{p}{q} < x < \frac{\kappa}{N}. \tag{8.15}$$

*Thus $r = \kappa$, $s = N$, and $q < s$. For these data (1.6)–(1.9) hold.*

*Proof.* By theorem 7.11 and (6.5), $\Delta = 1$ and $\delta = \gcd(\Delta, \varphi) = 1$.

Suppose first that $\varphi = N$. The initial record pair is formed by the two lattice vectors $(1, 0)$ and $(0, 1)$, so $\Delta = \kappa$. Since $\Delta = 1$, this gives $\kappa = 1$. Put

$$q = 1, \quad p = 0, \quad d = N, \quad e = 0.$$

Every contact index is a relative-interior contact index and $\lambda x_{j-1} = \xi_j$. Since $q = 1$, this is (8.16) and hence (8.13); the identity $x_N = x_0$ is (8.14).

Now assume $\varphi < N$. By theorem 6.1, after $\Delta = 1$ there are integers $p, q, d, e$ satisfying (8.12), with $d \geq \varphi$. Because $\{1, \ldots, \varphi\} \subseteq \{1, \ldots, d\}$, corollary 6.2 gives

$$\lambda^q x_{j-1} = \xi_j \qquad (1 \leq j \leq d), \tag{8.16}$$

and its closing relation is (8.14). For $j \leq \varphi$, substitute the half-open contact equation for $\xi_j$. For $j > \varphi$, the same corollary gives $\xi_j = x_j$, so put $(\alpha_j, \beta_j) = (1, 0)$ and again obtain (8.13).

Thus, in both subcases $\varphi = N$ and $\varphi < N$, (8.12), (8.16), and (8.13)–(8.14) have been established. The rest of the proof applies to both subcases.

Multiplying (8.13) gives

$$\left( \prod_{j=1}^{d} (\lambda^q - \beta_j) \right) x_0 = \left( \prod_{j=1}^{d} \alpha_j \right) x_d.$$

No vertex is zero because $0 \in \operatorname{int} P$. Using $x_0 = \lambda^e x_d$ proves (1.7). Multiplying by $\lambda^{dq}$ and using $N = qd + e = s$ proves (1.6).

The return equality $q\varphi + h = N$ from (6.5) gives $q \leq N$. Equality is impossible, because $q = N$ would make $q\kappa - pN$ divisible by $N$, whereas (8.12) makes it equal to 1. Hence $q < N = s$. Since $\varphi > \delta$ and $\varphi \leq N$, one has $\delta < N$; thus $\kappa < N$, because $1 \leq \kappa \leq N$ and $\delta = \gcd(\kappa, N)$. The determinant equation $q\kappa - pN = 1$ implies

$$\gcd(p, q) = \gcd(\kappa, N) = 1.$$

Moreover $p = (q\kappa - 1)/N \geq 0$ and

$$0 \leq \frac{p}{q} < \frac{\kappa}{N} < 1.$$

Both fractions are therefore reduced elements of $F_N$. Since their denominator sum is $q + N > N$, the Farey adjacency criterion shows that they are consecutive in $F_N$. The upper inequality $x < \kappa/N$ follows from (4.13). To locate the lower endpoint, follow the $q$ successive $\kappa$-steps beginning at the lifted vertex $x_0$. Their total index advance is $q\kappa = pN + 1$, and (8.16) ends strictly on the lifted side $E_{pN+1}$. By lemma 4.14,

$$\Theta_0 + 2\pi p < \Theta_0 + q\theta < \Theta_1 + 2\pi p,$$

or equivalently

$$0 < q\theta - 2\pi p < \Theta_1 - \Theta_0.$$

Thus $p/q < x$.

Finally, in each recurrence relation choose the factor argument $u_j$ so that

$$u_j = \Theta_j - \Theta_{j-1}.$$

This is the unique argument in $(0, \pi)$ selected by the recurrence relation. The arithmetic equality

$$d + e\kappa = (\kappa - dp)N \tag{8.17}$$

follows from $q\kappa - pN = 1$ and $N = qd + e$. The closure path consists entirely of endpoint contacts and ends at lifted index $d + e\kappa = (\kappa - dp)N$; therefore lemma 4.14 gives

$$\Theta_d + e\theta = \Theta_0 + 2\pi(\kappa - dp).$$

Telescoping the $u_j$ gives

$$e\theta + \sum_{j=1}^{d} u_j = 2\pi(\kappa - dp),$$

which is (1.9) with $(r, s) = (\kappa, N)$. ☐

**Proposition 8.6** (Exactly one relative-interior contact in each orbit: $\kappa < N$ and $\varphi = \delta$)**.** *Assume $1 \le \kappa < N$, put*

$$\delta = \gcd(\kappa, N),$$

*and suppose $\varphi = \delta$. Put*

$$L = \frac{N}{\delta}, \qquad K = \frac{\kappa}{\delta},$$

*so $1 \le K < L$ and $\gcd(K, L) = 1$. Let $h \in \{1, \dots, L-1\}$ be the unique integer with $Kh \equiv -1 \pmod{L}$, and define*

$$b = \frac{Kh + 1}{L}, \qquad S = N - h, \qquad R = \kappa - b.$$

*Then*

$$KS - RL = 1, \qquad L + S > N, \qquad \frac{R}{S} < x < \frac{K}{L}. \tag{8.18}$$

*The data in theorem 1.4 are as follows.*

(a) *If $\delta \ge 2$, use the conjugate eigenvalue $\mu = \overline{\lambda}$. Then $y = 1 - x$ lies in the reflected Farey interval*

$$\frac{L-K}{L} < y < \frac{S-R}{S}, \tag{8.19}$$

*whose denominators satisfy $q = L < s = S$. With*

$$d = \delta, \qquad e = s - dq = -h,$$

*there are parameters satisfying (1.5) for which (1.6)–(1.9) hold.*

(b) *If $\delta = 1$, retain $\mu = \lambda$, the eigenvalue in the fixed contact orientation. Then the ordered interval in (8.18) has*

$$\frac{p}{q} = \frac{R}{S}, \qquad \frac{r}{s} = \frac{K}{L}, \qquad q = S < s = L = N.$$

*With $d = \lfloor N/q \rfloor$ and $e = s - dq$, there are parameters satisfying (1.5) for which the same equations hold.*

*Proof.* Because $1 \le K < L$ and $\gcd(K, L) = 1$, multiplication by $K$ permutes the nonzero residue classes modulo $L$. Hence the stated integer $h$ exists and is unique.

The $\kappa$-orbits are the $\delta$ residue classes modulo $\delta$, each of length $L$. The interval $\{1, \dots, \delta\}$ of indices whose contact points lie in the relative interiors of their sides contains exactly one index in each orbit; in standard orbit terminology, it is an orbit transversal. Starting at $x_i$, the indices $i + \kappa, \dots, i + (L-1)\kappa$ modulo $N$ are therefore endpoint-contact indices, while $i + L\kappa \equiv i \pmod{N}$ is the next index whose contact point lies in the relative interior of its side. Consequently

$$\lambda^L x_i = \xi_i = \beta_i x_{i-1} + \alpha_i x_i \qquad (1 \le i \le \delta). \tag{8.20}$$

Set

$$\widehat{\beta}_i = \alpha_i, \qquad \widehat{\alpha}_i = \beta_i.$$

Every contact in the interval lies in the relative interior of its side, so both numbers belong to $(0, 1)$, and

$$(\lambda^L - \widehat{\beta}_i)x_i = \widehat{\alpha}_i x_{i-1}. \tag{8.21}$$

The congruence $h\kappa \equiv -\delta \pmod N$ moves $x_\delta$ to $x_0$. None of the intermediate indices is an index whose contact point lies in a side's relative interior: the orbit remains in residue class 0 modulo $\delta$ and meets $\{1, \ldots, \delta\}$ again only after $L$ steps. Hence

$$\lambda^h x_\delta = x_0. \tag{8.22}$$

Multiplying (8.21), cancelling the nonzero vertices, and using (8.22) gives

$$\lambda^{-h} \prod_{i=1}^{\delta} (\lambda^L - \widehat{\beta}_i) = \prod_{i=1}^{\delta} \widehat{\alpha}_i. \tag{8.23}$$

We next prove the Farey assertions and the lifted closure needed for both cases. Since $1 \le K < L$ and $1 \le h < L$, the definition of $b$ gives

$$1 \le b = \frac{Kh + 1}{L} \le K.$$

Consequently

$$R = \delta K - b \ge 0, \qquad S = \delta L - h > 0.$$

The determinant computation is

$$KS - RL = K(N - h) - (\kappa - b)L = bL - Kh = 1,$$

so

$$0 \le \frac{R}{S} < \frac{K}{L} < 1.$$

In particular $R < S$. The same determinant equation gives $\gcd(R, S) = 1$, while $\gcd(K, L) = 1$ was established above. Since $L, S \le N$, both $R/S$ and $K/L$ are reduced elements of $F_N$. Moreover $L + S > N$ because $h < L$. The Farey adjacency criterion therefore shows that these two fractions are consecutive in $F_N$.

Let $\gamma_i = \Theta_i - \Theta_{i-1}$ and $\Gamma = \sum_{i=1}^{\delta} \gamma_i$. The strict $L$-return in (8.20) lies in relint $E_i$. Since $L\kappa = KN$, lemma 4.14 places its lift strictly between $\Theta_{i-1} + 2\pi K$ and $\Theta_i + 2\pi K$. Thus

$$\varepsilon = 2\pi K - L\theta \quad \text{satisfies} \quad 0 < \varepsilon < \gamma_i.$$

Likewise $h\kappa = bN - \delta$. The endpoint closure (8.22) therefore has the exact lifted form

$$\Theta_\delta + h\theta = \Theta_0 + 2\pi b, \qquad \Gamma = 2\pi b - h\theta. \tag{8.24}$$

Using $N = \delta L$ and $\kappa = \delta K$ gives

$$S\theta - 2\pi R = (N - h)\theta - 2\pi(\kappa - b) = \Gamma - \delta\varepsilon > 0,$$

whereas $L\theta < 2\pi K$. Hence $R/S < x < K/L$. Together with the Farey assertions proved above, this establishes (8.18).

Assume first that $\delta \ge 2$. Since $1 \le h < L$,

$$S = \delta L - h > L.$$

Apply lemma 8.2 to (8.18); this gives (8.19), with the smaller denominator $L$ on the left. Set

$$\frac{p}{q} = \frac{L-K}{L}, \qquad \frac{r}{s} = \frac{S-R}{S}, \qquad d = \delta, \qquad e = -h.$$

Then $d = \lfloor N/q \rfloor$ because $N = \delta L$, and

$$s - dq = S - \delta L = -h = e.$$

The inequalities $0 < \gamma_i < \pi$ required in (8.5) follow from lemma 2.10. Apply lemma 8.3 to (8.21)–(8.24) with

$$q = L, \quad d = \delta, \quad b_i = \widehat{\beta}_i, \quad a_i = \widehat{\alpha}_i, \quad m = b.$$

The hypothesis $h < dq$ is $h < N$, which is automatic. Because $\widehat{\alpha}_i + \widehat{\beta}_i = 1$, the reflected coefficients satisfy (1.5). Relabel the reflected coefficients $(\alpha'_j, \beta'_j)$ as $(\alpha_j, \beta_j)$. The lemma gives the forward recurrence relations for $\mu = \overline{\lambda}$, the Laurent equation with $e = -h$, and its polynomial form (1.6). It also gives

$$e \arg_+(\mu) + \sum_{j=1}^{d} u_j = 2\pi(b-h).$$

Finally,

$$r - dp = (S-R) - \delta(L-K) = b - h,$$

so this is exactly (1.9). This proves part (a).

Now assume $\delta = 1$. Then $L = N$ and $S = N - h < L$, so the original ordered interval already has the smaller denominator on the left. Put

$$q = S = N - h, \qquad s = N, \qquad p = R, \qquad r = K = \kappa.$$

Equation (8.23) becomes

$$\lambda^{-h}(\lambda^N - \widehat{\beta}_1) = \widehat{\alpha}_1,$$

or, after moving the second term,

$$\lambda^h(\lambda^{N-h} - \widehat{\alpha}_1) = \widehat{\beta}_1. \tag{8.25}$$

Let $d = \lfloor N/q \rfloor$ and $e = N - dq$. Define

$$\beta'_1 = \widehat{\alpha}_1, \quad \alpha'_1 = \widehat{\beta}_1, \qquad \beta'_j = 0, \quad \alpha'_j = 1 \quad (2 \le j \le d).$$

Since $e + (d-1)q = h$, equation (8.25) is exactly

$$\lambda^e \prod_{j=1}^{d} (\lambda^q - \beta'_j) = \prod_{j=1}^{d} \alpha'_j.$$

Multiplying by $\lambda^{dq}$ gives (1.6).

The first factor has a genuine consecutive-vertex relation. Indeed, $\lambda^h x_1 = x_0$ and $\lambda^N x_1 = \widehat{\alpha}_1 x_0 + \widehat{\beta}_1 x_1$ imply

$$(\lambda^q - \beta'_1)x_0 = \alpha'_1 x_1. \tag{8.26}$$

Let $u_1 = \Theta_1 - \Theta_0 \in (0, \pi)$ be its chosen factor argument. The lifted closure (8.24) is

$$h\theta + u_1 = 2\pi b. \tag{8.27}$$

For $2 \le j \le d$, assign the factor with $\beta'_j = 0$ the argument $u_j = q\theta - 2\pi p$. The Farey inequalities and $q\kappa = pN + 1$ show that this is the unique argument in $(0, \pi)$ of $\lambda^q$. Because $e + (d-1)q = h$,

$$e\theta + \sum_{j=1}^{d} u_j = 2\pi\big(b - (d-1)p\big).$$

Here $p = R = \kappa - b$ and $r = K = \kappa$, so $b - (d-1)p = r - dp$. Relabel $(\alpha'_j, \beta'_j)$ as $(\alpha_j, \beta_j)$; these are the coefficients asserted in the statement. This proves (1.9) and part (b). □

The finite product equation alone determines factor arguments only modulo $2\pi$. The recurrence relations remove that ambiguity: consecutive lifted vertex angles select the unique argument in $(0, \pi)$ of each factor, and every factor with $\beta_j = 0$ is assigned the left endpoint argument $A$. The following lemma records the factor-argument bounds before convexity is applied.

**Lemma 8.7** (Bounds for the arguments of $\mu^q - \beta_j$)**.** *Assume $N \ge 4$. For every set of data constructed in lemma 8.4 and propositions 8.5 and 8.6, let $\vartheta = \arg_+(\mu)$ and put*

$$A = q\vartheta - 2\pi p.$$

*Put*

$$M = \arg_+(\mu^q - 1).$$

*Then*

$$0 < A < M < \pi, \qquad M > \frac{\pi}{2}.$$

*Every chosen argument $u_j$ in (1.9) is the unique argument in $(0, \pi)$ of $\mu^q - \beta_j$, and*

$$u_j \in [A, M) \qquad (1 \le j \le d). \tag{8.28}$$

*Proof.* Because this is an order-$N$ Farey interval, $q + s > N \ge 4$; with $q < s$ this gives $s \ge 3$. The determinant equation and $p/q < y < r/s$ therefore give

$$0 < A < \frac{2\pi}{s} < \pi.$$

Thus

$$\operatorname{Im}(\mu^q - \beta) = |\mu|^q \sin A > 0 \qquad (0 \le \beta \le 1),$$

and

$$\operatorname{Re}(\mu^q - 1) = |\mu|^q \cos A - 1 < 0.$$

Thus $M \in (\pi/2, \pi)$. As $\beta$ increases from 0 to 1, the point $\mu^q - \beta$ moves strictly to the left along a horizontal line in the open upper half-plane, so its argument increases continuously and strictly from $A$ to $M$. In particular, $A < M$.

It remains only to identify the chosen arguments from the recurrence relations. In the case $\varphi > \delta$, (8.13) has the form

$$(\mu^q - \beta_j)x_{j-1} = \alpha_j x_j, \qquad \alpha_j > 0,$$

where $x_{j-1}, x_j$ are consecutive vertices. Their lifted angular gap lies in $(0, \pi)$ and is congruent modulo $2\pi$ to the factor argument; it is therefore the unique upper-half-plane argument. In both reflected cases—$\kappa = N$ and the one-contact-per-orbit case $\delta \ge 2$—this conclusion is exactly the final assertion of lemma 8.3. In the one-contact-per-orbit case $\delta = 1$ it applies to (8.26), while every factor with $\beta_j = 0$ was explicitly assigned the argument $A$. These cases exhaust the constructions listed in the statement, and the range assertion follows from the preceding monotonicity in $\beta$. □

*Proof of theorem 1.4.* Apply lemma 4.13; throughout the contact-return analysis its selected eigenvalue has been denoted by $\lambda$. Determine the selected eigenvalue only after the return case is known. The identity case is $\kappa = N$. If $\kappa < N$, lemma 5.5 gives $\varphi \geq \delta$, so exactly one of $\varphi > \delta$ and $\varphi = \delta$ holds. In the equality case, exactly one of $\delta \geq 2$ and $\delta = 1$ holds. Hence the following four cases are exhaustive and mutually exclusive:

| return case | selected eigenvalue | denominator order | integer exponent $e$ | source |
|---|---|---|---|---|
| $\kappa = N$ | $\mu = \overline{\lambda}$ | $1 < N$ | $0$ | lemma 8.4 |
| $\kappa < N$, $\varphi > \delta$ | $\mu = \lambda$ | $q < N$ | $0 \leq e < q$ | proposition 8.5 |
| $\kappa < N$, $\varphi = \delta \geq 2$ | $\mu = \overline{\lambda}$ | $L < S$ | $e = -h < 0$ | proposition 8.6, case (a) |
| $\kappa < N$, $\varphi = \delta = 1$ | $\mu = \lambda$ | $S < N$ | $0 \leq e < S$ | proposition 8.6, case (b) |

If $\kappa = N$, choose $\mu = \overline{\lambda}$ and apply lemma 8.4. Its Farey interval has denominators $1 < N$, and it provides the polynomial equation, the equivalent Laurent equation, and the equation for the chosen real arguments.

Assume $\kappa < N$ and pass to the reduced interval of indices whose contact points lie in the relative interiors of their sides, as supplied by lemma 5.5. If $\varphi > \delta$, choose $\mu = \lambda$ and apply proposition 8.5, using the equality $\Delta = 1$ from theorem 7.11. Its right Farey endpoint is $\kappa/N$, so the ordered denominators satisfy $q < N = s$, and its Euclidean remainder $e$ is nonnegative.

If $\varphi = \delta$, apply proposition 8.6. That proposition makes the orientation choice explicitly: it chooses $\mu = \overline{\lambda}$ when $\delta \geq 2$ and $\mu = \lambda$ when $\delta = 1$. In the first subcase $e = -h < 0$ and the polynomial equation is the primary formulation; in the second subcase $e \geq 0$.

The denominator inequalities in the four rows of the table are all strict, so they give $q < s$ in every case. Thus in every case a specific eigenvalue $\mu \in \{\lambda, \overline{\lambda}\}$ has been selected, the ordered Farey interval has $q < s$, and (1.6)–(1.9) hold for that same $\mu$. Lemma 8.7 verifies that the factor arguments appearing in the case constructions have precisely the normalization and bounds asserted in the theorem. After this selection, the same $\mu$, Farey endpoints, exponents, coefficients, and chosen real arguments are used throughout the conclusion. $\square$

# 9 Stochastic-matrix corollary

For $n \geq 1$, let

$$\Theta_n = \bigcup \Big\{ \operatorname{spec} A : A \in \mathbb{R}_{\geq 0}^{n \times n},\ A\mathbf{1} = \mathbf{1} \Big\}; \tag{9.1}$$

thus $\Theta_n$ is the union of the spectra of all real row-stochastic $n \times n$ matrices.

**Proposition 9.1** (Compactness, conjugation, and disk bound)**.** *For every $n$, the set $\Theta_n$ is compact, invariant under complex conjugation, and contained in the closed unit disk.*

*Proof.* Let $\mathcal{S}_n$ be the set of real row-stochastic $n \times n$ matrices. Every entry lies in $[0, 1]$, and the row-sum equations define a closed subset of $[0, 1]^{n^2}$; hence $\mathcal{S}_n$ is compact. Write

$$\overline{\mathbb{D}} = \{ z \in \mathbb{C} : |z| \leq 1 \}.$$

Every eigenvalue lies in $\overline{\mathbb{D}}$: if $Av = \lambda v$, choose an index with $|v_i| = \|v\|_\infty > 0$; then

$$|\lambda| \|v\|_\infty = |(Av)_i| \leq \sum_{j=1}^{n} a_{ij} |v_j| \leq \|v\|_\infty \sum_{j=1}^{n} a_{ij} = \|v\|_\infty.$$

Consequently the eigenpair set

$$\mathcal{E}_n = \{ (A, \lambda) \in \mathcal{S}_n \times \overline{\mathbb{D}} : \det(\lambda I - A) = 0 \}$$

is closed in a compact space and is therefore compact. Its projection to the $\lambda$-coordinate is exactly $\Theta_n$, so $\Theta_n$ is compact. Real characteristic polynomials give conjugation symmetry. $\square$

**Theorem 9.2** (Invariant-polytope criterion)**.** *Let $n \geq 2$. For $\lambda \in \mathbb{C}$, the following are equivalent.*

(i) $\lambda \in \Theta_n$.

(ii) *There is a non-singleton convex polytope $P \subset \mathbb{C} \simeq \mathbb{R}^2$ with at most $n$ vertices such that*

$$\lambda P \subseteq P. \tag{9.2}$$

*Proof.* If $Av = \lambda v$, put $P = \operatorname{conv}\{v_1, \dots, v_n\}$. Every coordinate satisfies

$$\lambda v_i = \sum_j a_{ij} v_j \in P,$$

so (9.2) holds. If the hull is a singleton, all coordinates of the nonzero eigenvector are equal to the same nonzero number, so $\lambda = 1$; in that case any fixed segment may be used.

Conversely, let $x_1, \dots, x_m$ be the vertices of an invariant polytope, $m \leq n$. Choose barycentric coordinates

$$\lambda x_i = \sum_{j=1}^{m} a_{ij} x_j, \qquad a_{ij} \geq 0, \qquad \sum_j a_{ij} = 1.$$

Then $A = (a_{ij})$ is row-stochastic and $Ax = \lambda x$. If $m < n$, set

$$\widetilde{A} = A \oplus I_{n-m}, \qquad \widetilde{x} = (x_1, \dots, x_m, 0, \dots, 0)^T.$$

Then $\widetilde{A}$ is row-stochastic and $\widetilde{A}\widetilde{x} = \lambda \widetilde{x}$. □

The same block-diagonal construction gives the natural nesting

$$\Theta_m \subseteq \Theta_n \qquad (1 \leq m \leq n).$$

**Corollary 9.3** (Star-shapedness with respect to the origin)**.** *Let $n \geq 2$. If $\lambda \in \Theta_n$, then $t\lambda \in \Theta_n$ for every $0 \leq t \leq 1$.*

*Proof.* If $|\lambda| < 1$, take an invariant polytope $P$. Iteration gives $\lambda^k P \subseteq P$ and $\lambda^k x \to 0$, hence $0 \in P$; convexity then gives $t\lambda P \subseteq P$.

Suppose $|\lambda| = 1$. If $\lambda = 1$, the segment $[0, 1]$ satisfies $t[0, 1] \subseteq [0, 1]$. Otherwise take an invariant polytope $P$ with at most $n$ vertices. If $P$ has nonempty interior, area equality forces $\lambda P = P$, so multiplication by $\lambda$ permutes its vertices and $\lambda$ has finite order $k \leq n$. The regular orbit polygon

$$Q = \operatorname{conv}\{1, \lambda, \dots, \lambda^{k-1}\}$$

contains 0 because $1 + \lambda + \dots + \lambda^{k-1} = 0$, so 0 is the equal-weight average of its vertices. It satisfies $\lambda Q = Q$, and therefore $t\lambda Q = tQ \subseteq Q$. If $P$ is a segment, then $\lambda P \subseteq P$ and equality of lengths gives $\lambda P = P$. Multiplication by $\lambda$ therefore preserves the direction of its affine line, so $\lambda = \pm 1$. The case $\lambda = 1$ was already handled, and for $\lambda = -1$ the same conclusion follows from $Q = [-1, 1]$. The invariant-polytope criterion completes the proof. □

**Proposition 9.4** (Unit-circle points)**.** *Let $n \geq 2$. If $\lambda \in \Theta_n$ and $|\lambda| = 1$, then $\lambda$ is a root of unity of order at most $n$. Conversely every such root belongs to $\Theta_n$.*

*Proof.* Take an invariant polytope $P$ with at most $n$ vertices. If $P$ has nonempty interior, then $\lambda P \subseteq P$ and equality of areas forces $\lambda P = P$ by lemma A.5. Multiplication by $\lambda$ permutes the vertices, so a nonzero vertex has an orbit of length at most $n$ and $\lambda^k = 1$ for some $k \leq n$. If $P$ is a segment, then $\lambda P$ and $P$ have the same positive length; inclusion therefore gives $\lambda P = P$. Thus multiplication by $\lambda$ preserves the direction of the affine line containing $P$, so $\lambda = \pm 1$.

Conversely, if $\lambda$ has order $k \leq n$, the permutation matrix of a $k$-cycle is row-stochastic and has $\lambda$ as an eigenvalue. If $k < n$, take its direct sum with $I_{n-k}$. □

For $n \geq 2$ and $0 < \theta < \pi$, define

$$R_n(\theta) = \max \left\{ \rho \in [0,1] : \rho e^{i\theta} \in \Theta_n \right\}. \tag{9.3}$$

Indeed, $1 \in \Theta_n$, so corollary 9.3 gives $0 \in \Theta_n$; hence the set in (9.3) is nonempty. Compactness of $\Theta_n$ gives attainment of the maximum, while star-shapedness identifies the entire ray intersection as

$$\Theta_n \cap \{\rho e^{i\theta} : \rho \geq 0\} = \{\rho e^{i\theta} : 0 \leq \rho \leq R_n(\theta)\}.$$

**Corollary 9.5** (Farey equations for a non-inherited stochastic radial maximum)**.** *Let $N \geq 4$ and let*

$$\lambda = R_N(\theta) e^{i\theta} \in \Theta_N \setminus \Theta_{N-1}, \qquad 0 < |\lambda| < 1.$$

*Then one of $\lambda, \overline{\lambda}$ satisfies the complete conclusion of theorem 1.4; equivalently, it has two consecutive Farey fractions, the polynomial equation* (1.6)*, its equivalent Laurent equation* (1.7)*, the chosen factor arguments* (1.8) *with bounds $0 < A < M < \pi$ and $u_j \in [A, M)$, and the equation for the chosen real arguments* (1.9)*.*

*Proof.* Let $T_\lambda z = \lambda z$. Since $0 < \theta < \pi$, we have $\Im\lambda > 0$, and hence

$$(\operatorname{tr} T_\lambda)^2 - 4 \det T_\lambda = -4(\Im\lambda)^2 < 0.$$

Together with $0 < |\lambda| < 1$, this shows that $T_\lambda$ is an elliptic contraction. By the invariant-polytope criterion, membership in $\Theta_n$ is equivalent to the existence of a non-singleton invariant polytope with at most $n$ vertices. By lemma 2.5, every such polytope for $T_\lambda$ has nonempty interior, so its least possible number of vertices is exactly $\nu_{\text{poly}}(T_\lambda)$. The assumption $\lambda \in \Theta_N \setminus \Theta_{N-1}$ therefore gives $\nu_{\text{poly}}(T_\lambda) = N$. Radial maximality says that $t\lambda \notin \Theta_N$ for every $t > 1$; applying the same criterion to $tT_\lambda$ gives $\nu_{\text{poly}}(tT_\lambda) > N$. Thus $T_\lambda$ is $N$-critical. Apply theorem 1.4. Reversing the orientation labels the conjugate eigenvalue, which explains the choice between $\lambda$ and $\overline{\lambda}$. □

# 10 Conclusion

For $N \geq 4$, the parent theorem is a theorem about critical planar dynamics, not about stochastic matrices. Radial criticality forces every side to meet the image polygon and every image vertex to lie on the outer boundary; permitted vertex replacements preserve the contact hypotheses, and the finite set reduction makes the relative-interior contact set one cyclic interval. The lattice sail supplies the resulting unimodular first-return data, and the projectivity associated with the boundary-contact chain rules out a nonadjacent return. Farey adjacency and the varying-parameter product equation are scalar shadows of that adjacent return structure. Orders at most three are treated directly. The stochastic theorem uses none of the internal deformation or index bookkeeping: it follows from the invariant-polytope criterion in one paragraph.

# A Foundational finite-dimensional tools

This appendix proves the finite-dimensional ingredients invoked in the body of the paper. They are stated in the exact strength used there.

**Lemma A.1** (Weighted spectral-radius bound and nonnegative Perron vectors)**.** *Let $B = (b_{ij}) \in \mathbb{R}^{m \times m}$ be entrywise nonnegative.*

(i) *If $x > 0$ coordinatewise and $Bx \leq cx$, then* $\operatorname{spr}(B) \leq c$*.*

(ii) *There are nonzero vectors* $v, w \geq 0$ *such that*

$$Bv = \operatorname{spr}(B)v, \qquad B^T w = \operatorname{spr}(B)w.$$

*Proof.* For $x > 0$, define the weighted supremum norm

$$\|z\|_x = \max_i \frac{|z_i|}{x_i} \qquad (z \in \mathbb{C}^m).$$

If $Bx \leq cx$, then entrywise nonnegativity gives

$$|Bz| \leq B|z| \leq \|z\|_x Bx \leq c\|z\|_x x.$$

Hence the operator norm of $B$ in $\|\cdot\|_x$ is at most $c$. Every eigenvalue has modulus bounded by every operator norm, proving (i).

We first prove (ii) for a strictly positive matrix $A$. On the open probability simplex

$$\Sigma^\circ = \{x \in \mathbb{R}^m : x_i > 0,\ \sum_i x_i = 1\}$$

consider the function

$$\Phi(x) = \max_i \frac{(Ax)_i}{x_i}.$$

It satisfies $\Phi(tx) = \Phi(x)$ for every $t > 0$. Let $a_0 = \min_{i,j} a_{ij} > 0$. If a sequence in $\Sigma^\circ$ approaches the boundary, some coordinate $x_i$ tends to zero while $(Ax)_i \geq a_0 \sum_j x_j = a_0$; therefore $\Phi(x) \to \infty$. Fix one $x^{(0)} \in \Sigma^\circ$. The set $\{x \in \Sigma^\circ : \Phi(x) \leq \Phi(x^{(0)})\}$ stays a positive distance from the boundary and is closed in the compact probability simplex. It is therefore compact and nonempty, and continuity of $\Phi$ gives an interior minimizer $x$.

Put $M = \Phi(x)$. Then $Ax \leq Mx$. If the inequality were strict in at least one coordinate, the nonzero vector $Mx - Ax \geq 0$ would satisfy $A(Mx - Ax) > 0$ coordinatewise because every entry of $A$ is positive. Consequently

$$A^2 x < MAx$$

coordinatewise. With $y = Ax > 0$, this says $\Phi(y) < M$, contradicting the minimality of $x$ after normalizing $y$ back to the simplex. Thus $Ax = Mx$. Part (i), applied with equality, shows $\operatorname{spr}(A) \leq M$, while $M$ is itself an eigenvalue; hence $M = \operatorname{spr}(A)$. Applying the same argument to $A^T$ gives a strictly positive left Perron vector.

Now let $B \geq 0$ and put

$$A_\varepsilon = B + \varepsilon \mathbf{1}\mathbf{1}^T \qquad (\varepsilon > 0).$$

The preceding paragraph gives positive right and left Perron vectors $v_\varepsilon, w_\varepsilon$, normalized by $\sum_i (v_\varepsilon)_i = \sum_i (w_\varepsilon)_i = 1$, and a positive spectral radius $r_\varepsilon = \operatorname{spr}(A_\varepsilon)$. We show that $r_\varepsilon \to r := \operatorname{spr}(B)$.

First, $r \leq r_\varepsilon$. Choose an eigenvalue $\zeta$ of $B$ with $|\zeta| = r$ and a corresponding nonzero complex eigenvector $z$. Since

$$r|z| = |Bz| \leq B|z| \leq A_\varepsilon |z|,$$

let $C = \max_i |z_i|/(v_\varepsilon)_i$ and choose an index attaining the maximum. At that index,

$$rC(v_\varepsilon)_i \leq (A_\varepsilon |z|)_i \leq C(A_\varepsilon v_\varepsilon)_i = Cr_\varepsilon (v_\varepsilon)_i,$$

so $r \leq r_\varepsilon$.

For the converse bound, fix $c > r$. Choose $r_0$ with $r < r_0 < c$. In Jordan normal form, the $k$th power of a block $J_\zeta = \zeta I + N$ of size $m_\zeta$ is

$$J_\zeta^k = \sum_{j=0}^{m_\zeta - 1} \binom{k}{j} \zeta^{k-j} N^j.$$

Because $|\zeta| \le r < r_0$, this formula gives constants $C, M$ such that $\|B^k\| \le C k^M r_0^k$ for all $k$. Therefore the matrix series $\sum_{k \ge 0} c^{-k-1} B^k$ converges absolutely. Multiplying its partial sums by $cI - B$ and passing to the limit gives

$$(cI - B)^{-1} = \sum_{k=0}^{\infty} c^{-k-1} B^k.$$

Consequently

$$x_c = (cI - B)^{-1}\mathbf{1} = \sum_{k=0}^{\infty} c^{-k-1} B^k \mathbf{1} > 0, \qquad B x_c = c x_c - \mathbf{1}.$$

Therefore

$$A_\varepsilon x_c = c x_c - \mathbf{1} + \varepsilon(\mathbf{1}^T x_c)\mathbf{1} \le (c + \eta_\varepsilon) x_c,$$

where

$$\eta_\varepsilon = \varepsilon(\mathbf{1}^T x_c) \max_i (x_c)_i^{-1} \longrightarrow 0.$$

Part (i) gives $r_\varepsilon \le c + \eta_\varepsilon$. Given $\delta > 0$, choose $c$ with $r < c < r + \delta/2$. Since $\eta_\varepsilon \to 0$, for all sufficiently small $\varepsilon$ we have $r \le r_\varepsilon < c + \delta/2 < r + \delta$. Hence $r_\varepsilon \to r$.

The closed probability simplex is compact. Along a sequence $\varepsilon_k \downarrow 0$, the normalized vectors converge to nonzero vectors $v, w \ge 0$. Passing to the limit in

$$A_{\varepsilon_k} v_{\varepsilon_k} = r_{\varepsilon_k} v_{\varepsilon_k}, \qquad A_{\varepsilon_k}^T w_{\varepsilon_k} = r_{\varepsilon_k} w_{\varepsilon_k}$$

gives $Bv = rv$ and $B^T w = rw$. □

**Lemma A.2** (Strict separation from a compact convex set)**.** *Let $K$ be a nonempty compact convex subset of a Euclidean space and let $o \notin K$. There is a linear functional $\ell$ such that*

$$\ell(x) < \ell(o) \qquad (x \in K).$$

*Proof.* Choose $y \in K$ minimizing $|o - y|$ and put $v = o - y \ne 0$. For every $x \in K$, the segment $y + t(x - y)$, $0 \le t \le 1$, belongs to $K$. The function

$$f(t) = |o - y - t(x - y)|^2$$

has a minimum at $t = 0$, so $f'(0) \ge 0$. Thus $\langle v, x - y \rangle \le 0$. With $\ell(z) = \langle v, z \rangle$ we obtain

$$\ell(x) \le \ell(y) < \ell(o),$$

the last inequality being $\ell(o) - \ell(y) = |v|^2 > 0$. □

**Lemma A.3** (Polarity for planar polygons)**.** *Let $K \subset \mathbb{R}^2$ be a compact convex set with $0 \in \operatorname{int} K$, and define*

$$K^\circ = \{y : \langle y, x \rangle \le 1 \text{ for every } x \in K\}.$$

*Then $K^\circ$ is compact, convex, and contains $0$ in its interior. If $A$ is linear and $AK \subseteq K$, then*

$$A^* K^\circ \subseteq K^\circ.$$

*If $K$ is a polygon with $N$ vertices, then $K^\circ$ is a polygon with $N$ vertices. For every vertex $x$ of $K$, the set*

$$F_x = \{y \in K^\circ : \langle y, x\rangle = 1\}$$

*is a side of $K^\circ$, and every side arises uniquely in this way. Finally, if $x \in K$, $y \in K^\circ$, and $\langle y, x\rangle = 1$, then $x \in \partial K$ and $y \in \partial K^\circ$.*

*Proof.* Write

$$\overline{\mathbb{D}} = \{z \in \mathbb{R}^2 : |z| \le 1\}.$$

Choose radii $0 < r < R$ with $r\overline{\mathbb{D}} \subseteq K \subseteq R\overline{\mathbb{D}}$. The defining inequalities show

$$R^{-1}\overline{\mathbb{D}} \subseteq K^\circ \subseteq r^{-1}\overline{\mathbb{D}},$$

so $K^\circ$ has the asserted compactness and interior properties. If $y \in K^\circ$ and $x \in K$, then $Ax \in K$ and

$$\langle A^*y, x\rangle = \langle y, Ax\rangle \le 1;$$

therefore $A^*y \in K^\circ$.

Let the complete cyclic vertex list of the polygon be $x_1, \dots, x_N$. Since a linear functional reaches its maximum over a convex hull at a vertex,

$$K^\circ = \bigcap_{i=1}^{N} \{y : \langle y, x_i\rangle \le 1\}.$$

Every displayed inequality is irredundant. Indeed, at the vertex $x_i$ choose a supporting functional $u_i$ that exposes $x_i$. Its support value $h_K(u_i)$ is positive because 0 is interior. After replacing $u_i$ by $u_i/h_K(u_i)$, one has

$$\langle u_i, x_i\rangle = 1, \qquad \langle u_i, x_j\rangle < 1 \quad (j \ne i).$$

On the line $\langle y, x_i\rangle = 1$, all other inequalities remain strict in a small interval about $u_i$. Hence $F_{x_i}$ is a nondegenerate side of $K^\circ$. The intersection description has only the $N$ displayed half-planes, so it has at most $N$ sides, while the preceding argument has produced $N$ distinct sides. It therefore has exactly $N$ sides. Traversing the boundary of a bounded full-dimensional polygon alternates sides and vertices, so it also has exactly $N$ vertices. Since no displayed side is redundant or degenerate, the polar polygon has the complete cyclic vertex list prescribed in definition 1.2. The same argument also proves the asserted bijection between vertices of $K$ and sides of $K^\circ$.

If $x$ were interior to $K$ and $y \ne 0$, then $x + \varepsilon y \in K$ for all sufficiently small $\varepsilon > 0$, giving

$$1 \ge \langle y, x + \varepsilon y\rangle = 1 + \varepsilon |y|^2,$$

a contradiction. Thus equality forces $x \in \partial K$. The defining inequality $\langle \cdot, x\rangle \le 1$ for $K^\circ$ is attained at $y$, so $y \in \partial K^\circ$. □

**Lemma A.4** (Hausdorff limits of finitely generated polygons)**.** *Fix $N$ and let*

$$P_k = \operatorname{conv}\{z_{k,1}, \dots, z_{k,N}\}$$

*with all points in one compact subset of $\mathbb{R}^2$. If $z_{k,j} \to z_j$ for every $j$, then*

$$P_k \longrightarrow P := \operatorname{conv}\{z_1, \dots, z_N\}$$

*in the Hausdorff metric. If $P$ has nonempty interior, then* $\operatorname{area}(P_k) \to \operatorname{area}(P)$*. Moreover, for every fixed linear map $A$, the conditions $AP_k \subseteq P_k$ pass to the limit, and*

$$\max_{z \in P_k} |z| \longrightarrow \max_{z \in P} |z|.$$

*Proof.* For nonempty compact sets $K, L \subseteq \mathbb{R}^2$, their Hausdorff distance is

$$d_H(K,L) = \max\left\{\sup_{x\in K}\operatorname{dist}(x,L), \sup_{y\in L}\operatorname{dist}(y,K)\right\},$$

and we write $\overline{\mathbb{D}} = \{z \in \mathbb{R}^2 : |z| \le 1\}$ for the closed unit disk.

Put $\varepsilon_k = \max_j |z_{k,j} - z_j|$. If $z = \sum_j t_j z_{k,j} \in P_k$, then $z' = \sum_j t_j z_j \in P$ and $|z - z'| \le \varepsilon_k$; the converse estimate is identical. Hence the Hausdorff distance is at most $\varepsilon_k$.

All polygons lie in one large disk. We prove convergence of their indicator functions away from $\partial P$. If $z \notin P$, its positive distance from $P$ and Hausdorff convergence imply $z \notin P_k$ for large $k$. If $z \in \operatorname{int} P$, choose $r > 0$ with $z + r\overline{\mathbb{D}} \subseteq P$. Hausdorff convergence implies uniform convergence of support functions:

$$|h_{P_k}(u) - h_P(u)| \le d_H(P_k, P) \qquad (|u| = 1).$$

The contained disk gives the uniform support margin

$$h_P(u) \ge \langle u, z\rangle + r \qquad (|u| = 1).$$

For large $k$, when $d_H(P_k, P) < r$, these two estimates give

$$h_{P_k}(u) > \langle u, z\rangle \qquad (|u| = 1).$$

If $z \notin P_k$, lemma A.2 would produce a unit vector $u$ with $\langle u, z\rangle > h_{P_k}(u)$, a contradiction. Thus $z \in P_k$ eventually. All generating points lie in one compact set, so one fixed disk $R\overline{\mathbb{D}}$ contains every $P_k$ and $P$. The indicators of the polygons are therefore dominated by the integrable function $\mathbf{1}_{R\overline{\mathbb{D}}}$. Since the boundary of a polygon is a finite union of line segments and has planar measure zero, dominated convergence of the indicator functions gives area convergence.

If $AP_k \subseteq P_k$ for every $k$ and $x \in P$, choose $x_k \in P_k$ with $x_k \to x$. Then $Ax_k \in P_k$, and Hausdorff convergence gives $Ax \in P$. Finally, by convexity of the norm,

$$\max_{z\in P_k}|z| = \max_j |z_{k,j}|, \qquad \max_{z\in P}|z| = \max_j |z_j|,$$

which proves convergence of the maxima. □

**Lemma A.5** (Strict area monotonicity)**.** *If $K \subseteq L$ are compact convex subsets of $\mathbb{R}^2$, both with nonempty interior, and $K \ne L$, then*

$$\operatorname{area}(K) < \operatorname{area}(L).$$

*Proof.* Choose $y \in L \setminus K$. By lemma A.2, there is a linear functional $\ell$ with

$$\ell(y) > a := \max_{x\in K}\ell(x).$$

Choose a closed disk $D \subset \operatorname{int} K$. The convex hull $C = \operatorname{conv}(D \cup \{y\})$ satisfies $C \subseteq L$. Because $D \subset \operatorname{int} K$, compactness gives $\max_D \ell < a < \ell(y)$. Choose $t \in (0,1)$ sufficiently close to 1 that

$$(1-t)\min_D \ell + t\ell(y) > a.$$

Then

$$D_t = (1-t)D + ty$$

is a disk of positive area contained in $C$, and every $x \in D_t$ satisfies $\ell(x) > a$. Hence $D_t \subseteq L\setminus K$, so $L \setminus K$ has positive area. This proves the strict inequality. □

**Lemma A.6** (Lattice parallelogram count)**.** *Let $u, v \in \mathbb{Z}^2$ be linearly independent and put $D = |\det(u, v)|$. The half-open parallelogram*

$$\mathcal{P} = \{\alpha u + \beta v : 0 \leq \alpha, \beta < 1\}$$

*contains exactly $D$ integer points, one in each coset of $\mathbb{Z}^2/(\mathbb{Z}u + \mathbb{Z}v)$.*

*Proof.* Let $z \in \mathbb{Z}^2$. Since $u$ and $v$ are linearly independent over $\mathbb{R}$, there are unique $a, b \in \mathbb{R}$ such that $z = au + bv$. Then the integer point $z - \lfloor a \rfloor u - \lfloor b \rfloor v \in \mathcal{P}$ represents the coset of $z$. Two integer points of $\mathcal{P}$ cannot represent the same coset: their difference is $mu + nv$ with integers $m, n$, while both coefficient differences lie in $(-1, 1)$, forcing $m = n = 0$. Thus the integer points of $\mathcal{P}$ are in bijection with the quotient group.

It remains to compute the order of that quotient. Integer elementary row and column operations, with determinant $\pm 1$, do not change either the index of the generated sublattice or the absolute determinant. Applying the Euclidean algorithm to the $2 \times 2$ matrix with columns $u, v$ reduces it to Smith normal form

$$\begin{pmatrix} d_1 & 0 \\ 0 & d_2 \end{pmatrix}, \qquad d_1, d_2 > 0.$$

For this diagonal lattice the quotient has exactly $d_1 d_2$ elements, and $d_1 d_2$ is the absolute determinant. Reversing the unimodular operations gives the result for $u, v$. □

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# Part II: Invariant Polygons and the Boundary of Stochastic Spectra

**Interface with Part I.** Part II applies the structural results proved in Part I, in particular theorem 1.4. The integer closing exponent $e = s - dq$ follows the convention stated there: it may be negative, and in that case the polynomial equation is the primary formulation.

## II.1 Introduction and proof architecture

For $n \geq 1$, define

$$\Theta_n = \bigcup \left\{ \operatorname{spec} A : A \in \mathbb{R}^{n\times n}_{\geq 0},\ A\mathbf{1} = \mathbf{1} \right\}. \tag{II.1.1}$$

Throughout, we identify $\mathbb{C}$ with the oriented plane $\mathbb{R}^2$ and use

$$\langle z, w\rangle = \operatorname{Re}(\overline{z}w), \qquad \det(z, w) = \operatorname{Im}(\overline{z}w).$$

Positive cyclic order means counterclockwise order; oriented boundary intervals inherit that convention.

The problem is to determine the possible location of one eigenvalue, not an entire spectrum. The answer is a compact radial region: its unit-circle points are roots of unity of order at most $n$, and its noncircular boundary arcs are indexed by consecutive fractions in the Farey sequence of order $n$.

The stochastic eigenvalue-region problem was developed by Dmitriev and Dynkin [1] and solved in full by Karpelevič [4]. Ito's formulation of the Karpelevič theorem [2] expresses the boundary through one-parameter polynomial families indexed by Farey neighbors. Later matrix realizations and structural treatments include [3, 5, 6]. The present argument derives that boundary directly from invariant polygons.

The proof passes from stochastic spectra to critical planar dynamics and then to a sharp one-dimensional inequality. Its logical spine is

$$\begin{aligned}
&\text{stochastic eigenpair} \iff \text{invariant polygon} \implies \text{new order-}N\text{ radial boundary point}\\
&\implies N\text{-criticality} \implies \underset{N\geq 4}{\text{critical-polygon finite product equation}}\\
&\implies \text{varying-parameter product equation and factor-argument bounds}\\
&\implies \text{strict log-sine convexity} \implies \text{radial comparison on each Farey interval}\\
&\implies \text{explicit attainment and Farey nesting} \implies \text{boundary in Ito's formulation.}
\end{aligned} \tag{II.1.2}$$

The invariant-polytope criterion identifies a radial boundary point satisfying (II.4.3) with an intrinsic $N$-critical elliptic contraction. For $N \geq 4$, the finite-product theorem in Part I converts that criticality into consecutive Farey fractions, a finite product equation, an equation for chosen real arguments, and factor-argument bounds needed here. Orders at most three are handled directly and do not pass through the projective cyclic-successor argument. The remaining steps are the local scalar comparison, explicit realization, and global boundary ordering.

**Dependency order.** Under (II.4.3), the upper bound uses, in order, the invariant-polytope criterion theorem II.4.2, the implication to criticality proposition II.4.7, the Part I finite-product theorem II.5.1, and the varying-parameter inequality theorem II.6.1. The reverse inclusion is independent of the Part I input: it comes from the sparse stochastic realization theorem II.7.3. Candidate nesting theorem II.8.4 is used afterward to pass to all orders. Thus neither realization nor nesting is used to establish criticality or the finite product equation. The direct arguments for orders at most three precede this chain.

The Part I assertion is existential: it selects one of the two adapted complex orientations and supplies the corresponding consecutive Farey fractions and finite product equation. No uniqueness is required for the invariant polygon, its contact data, or the stochastic matrix realizing the selected point.

**Background scope.** No prior Karpelevič boundary reduction or parametrization is used. We use elementary finite-dimensional convexity and compactness for the invariant-polytope criterion and radial structure. The intrinsic critical-polygon geometry, including the boundary conclusion for every invariant polygon within the vertex bound and contact reduction, is proved in Part I; for $N \geq 4$ its return towers and projective cyclic-successor argument culminate in theorem 1.4. Part II uses only the complex finite product equation, equation for chosen real arguments, and factor-argument bounds through theorem II.5.1; every other problem-specific implication used in Part II is proved below. The cases $n \leq 3$ are treated directly.

## II.2 Farey data, candidate arcs, and numerical evaluation

### II.2.1 Intervals between consecutive Farey fractions

Let $F_n$ denote the Farey sequence of order $n$ on $[0, 1]$. We use

$$F_n^+ = F_n \cap \left[0, \frac{1}{2}\right]$$

for the fractions corresponding, under $t \mapsto \exp(2\pi it)$, to points on the closed upper semicircle. For $x \in [0, 1/2]$, the ray at argument $2\pi x$ is $\{\rho \exp(2\pi ix) : \rho \geq 0\}$.

The following is the same criterion proved in lemma 8.1; we repeat its short proof so that the Part II construction remains self-contained.

**Lemma II.2.1** (Farey adjacency criterion)**.** *Two reduced fractions $a_0/b_0 < a_1/b_1$ belonging to $F_n$ are consecutive in $F_n$ if and only if*

$$b_0 a_1 - a_0 b_1 = 1, \qquad b_0 + b_1 > n. \tag{II.2.1}$$

*Proof.* Assume first that $b_0 a_1 - a_0 b_1 = 1$. If a reduced fraction $h/k$ lies strictly between $a_0/b_0$ and $a_1/b_1$, define

$$m = a_1 k - b_1 h, \qquad \ell = b_0 h - a_0 k.$$

Cross multiplication shows that $m$ and $\ell$ are positive integers. The determinant-one relation gives the exact decomposition

$$(k, h) = m(b_0, a_0) + \ell(b_1, a_1).$$

Looking at the first coordinate yields $k = mb_0 + \ell b_1 \geq b_0 + b_1$. Hence, if $b_0 + b_1 > n$, no fraction of denominator at most $n$ can lie between the two endpoints, so they are consecutive in $F_n$.

Conversely, suppose the two fractions are consecutive in $F_n$ and put $D = b_0 a_1 - a_0 b_1$. The determinant is a positive integer. If $D > 1$, the half-open parallelogram spanned by the primitive integer vectors $u = (b_0, a_0)$ and $v = (b_1, a_1)$ contains a nonzero integer point

$$w = \mu u + \nu v, \qquad 0 \leq \mu, \nu < 1.$$

Indeed, lemma A.6 says that the half-open fundamental parallelogram contains exactly $D = |\det(u, v)|$ integer points. When $D > 1$, one of them is different from the origin. If $\mu = 0$, then $w = \nu v$ is a nonzero integer point in the relative interior of $[0, v]$, which is impossible because $v$ is primitive. Likewise, $\nu = 0$ would give a nonzero integer point in the relative interior of $[0, u]$. Thus $0 < \mu, \nu < 1$. Both $w$ and $u + v - w$ are positive combinations of $u, v$, so their slopes lie strictly between $a_0/b_0$ and $a_1/b_1$. Their first coordinates are positive integers $K$ and $b_0 + b_1 - K$; one is at most $(b_0 + b_1)/2 \le n$ because $b_0, b_1 \le n$. After reducing that intermediate slope, its denominator can only decrease. This produces an element of $F_n$ strictly between the alleged consecutive fractions, a contradiction. Therefore $D = 1$. Finally, if $b_0 + b_1 \le n$, the reduced mediant $(a_0 + a_1)/(b_0 + b_1)$ has denominator at most $n$ and lies strictly between the endpoints. Thus consecutiveness also forces $b_0 + b_1 > n$. □

Let $f < g$ be consecutive elements of $F_n^+$. Label the two endpoints, independently of their left-right order, as

$$\frac{p}{q}, \qquad \frac{r}{s}, \qquad q < s, \tag{II.2.2}$$

so that $|rq - ps| = 1$. The denominators are unequal: if $q = s$, this determinant equation forces $q = s = 1$, but $0/1$ is the only element of $F_n^+$ with denominator 1. Thus the strict labeling $q < s$ is always available. Put

$$d = \left\lfloor \frac{n}{q} \right\rfloor, \qquad e = s - dq. \tag{II.2.3}$$

The integer $d$ is the number of complete $q$-returns allowed by the maximum of $n$ vertices.

The Part I product equation (1.6) allows the coefficients $\beta_1, \ldots, \beta_d$ to vary, with $\alpha_j = 1 - \beta_j$. At the algebraic level, the common-coefficient specialization

$$\beta_1 = \cdots = \beta_d = \beta, \qquad \alpha_1 = \cdots = \alpha_d = \alpha$$

gives the Ito equation below in the corresponding selected orientation. This is only a candidate specialization at this stage: Topic X proves the sharp equality statement, including the reflection back to the original orientation, and Topic XI proves stochastic realization.

**Definition II.2.2** (Ito polynomial family)**.** For $0 \le \alpha \le 1$, put $\beta = 1 - \alpha$. The Ito equation associated with the interval $[f, g]$ is

$$\lambda^s(\lambda^q - \beta)^d = \alpha^d \lambda^{dq}. \tag{II.2.4}$$

For $\lambda \neq 0$, equivalently, after cancelling the common power of $\lambda$ and clearing a possible negative exponent,

$$\begin{cases} \lambda^e(\lambda^q - \beta)^d = \alpha^d, & e \ge 0, \\ (\lambda^q - \beta)^d = \alpha^d \lambda^{-e}, & e < 0. \end{cases} \tag{II.2.5}$$

We define the candidate set on each Farey interval by its radial graph, rather than by an unproved choice of an $\alpha$-parametrized algebraic root branch. The next proposition constructs exactly one point of the algebraic family at every argument in the open interval.

### II.2.2 The scalar radius equation

Let $x \in (f, g)$ and write $\lambda = \rho \exp(2\pi i x)$. Define

$$A = 2\pi\,|qx - p|, \qquad B = \frac{2\pi}{d}\,|sx - r|. \tag{II.2.6}$$

The range used below is

$$A > 0, \qquad B > 0, \qquad A + B < \pi. \tag{II.2.7}$$

**Proposition II.2.3** (A unique modulus at each prescribed argument)**.** *There is a unique* $\rho^{(n)}_{f,g}(x) \in (0,1)$ *satisfying*

$$\rho^{s/d}\sin A + \rho^q \sin B = \sin(A+B). \tag{II.2.8}$$

*For this radius,*

$$\alpha = \frac{\rho^{s/d}\sin A}{\sin(A+B)}, \qquad \beta = \frac{\rho^q \sin B}{\sin(A+B)} \tag{II.2.9}$$

*are positive and sum to one, and* $\lambda = \rho\exp(2\pi i x)$ *satisfies* (II.2.4)*. The functions* $\rho^{(n)}_{f,g}, \alpha, \beta$*, and* $x \mapsto \rho^{(n)}_{f,g}(x)\exp(2\pi i x)$ *are* $C^\infty$ *on the open interval. Both sides of* (II.2.10)*, the vector equation constructed below, depend smoothly on* $x$ *there.*

*Proof.* First verify (II.2.7). Put

$$t = \frac{|x - p/q|}{|r/s - p/q|} = s|qx - p| \in (0,1).$$

The determinant-one relation gives

$$|qx - p| = \frac{t}{s}, \qquad |sx - r| = \frac{1-t}{q},$$

regardless of which labeled endpoint is on the left. Hence

$$\frac{A+B}{2\pi} = \frac{t}{s} + \frac{1-t}{dq}.$$

For $n \geq 3$ and $q < s$, one has $s \geq 3$: otherwise $q = 1$, $s = 2$, and the only possible endpoints would be $0/1$ and $1/2$, with $1/3 \in F_n$ strictly between them. One also has $dq \geq 2$, because either $q \geq 2$ or $q = 1$ and $d = n \geq 3$. Thus $A, B > 0$, while the displayed strict convex combination is smaller than $1/2$: if $dq > 2$, both numbers $1/s$ and $1/(dq)$ are smaller than $1/2$, and if $dq = 2$, the number $1/2$ has the strict weight $1 - t < 1$ while $1/s < 1/2$. This proves (II.2.7).

The left side of (II.2.8) is strictly increasing in $\rho$. At $\rho = 0$ it is zero, while at $\rho = 1$ it exceeds the right side because

$$\sin A + \sin B - \sin(A+B) = 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{A+B}{2} > 0.$$

This proves existence and uniqueness. On $(f,g)$ the signs inside the absolute values defining $A$ and $B$ are fixed, so, denoting the left side minus the right side of (II.2.8) by $F_x(\rho)$, this function is $C^\infty$ in $(x,\rho)$ and has

$$\frac{\partial F_x}{\partial \rho} = \frac{s}{d}\rho^{s/d-1}\sin A + q\rho^{q-1}\sin B > 0.$$

The implicit-function theorem therefore makes $\rho^{(n)}_{f,g}$ a $C^\infty$ function of $x$ on the open interval; the explicit formulas in (II.2.9) and the radial parametrization give the same conclusion for $\alpha$, $\beta$, and $x \mapsto \rho^{(n)}_{f,g}(x)\exp(2\pi i x)$.

Combining (II.2.9) with (II.2.8) gives $\alpha + \beta = 1$. All factors in the two numerators and in the common denominator are positive, so $\alpha, \beta > 0$.

Set

$$z = (\overline{\lambda})^{-1} = \frac{1}{\overline{\lambda}} = \rho^{-1}\exp(2\pi i x), \qquad w = w(x) := \rho^{-s/d}\exp\left(\frac{2\pi i(sx - r)}{d}\right).$$

Then $w^d = z^s$. Equivalently, with

$$\omega = \exp(-2\pi i r/d),$$

the notation $w = \omega z^{s/d}$ denotes this explicit continuous choice along the parametrized arc; it does not assert a global branch of the multivalued fractional power. Put

$$\sigma = \operatorname{sgn}(qx - p).$$

The two quantities $qx - p$ and $sx - r$ have opposite signs. Hence

$$z^q = \rho^{-q} \exp(i\sigma A), \qquad w = \rho^{-s/d} \exp(-i\sigma B).$$

Moreover (II.2.9) gives

$$\beta\rho^{-q} = \frac{\sin B}{\sin(A+B)}, \qquad \alpha\rho^{-s/d} = \frac{\sin A}{\sin(A+B)}.$$

The cancellation in the vector equation is completely explicit. The imaginary part of the proposed right side is

$$\sigma\frac{\sin B \sin A - \sin A \sin B}{\sin(A+B)} = 0,$$

and its real part is

$$\frac{\sin B \cos A + \sin A \cos B}{\sin(A+B)} = 1.$$

Thus resolving the two vectors into real and imaginary parts gives

$$1 = \beta z^q + \alpha w. \tag{II.2.10}$$

Raising to the $d$th power after moving the first term gives

$$(1 - \beta\overline{\lambda}^{-q})^d = \alpha^d \overline{\lambda}^{-s},$$

and therefore

$$(\overline{\lambda}^q - \beta)^d = \alpha^d \overline{\lambda}^{dq-s}.$$

Conjugating gives

$$(\lambda^q - \beta)^d = \alpha^d \lambda^{dq-s}.$$

Since $\lambda \neq 0$, multiplication by $\lambda^s$ yields

$$\lambda^s(\lambda^q - \beta)^d = \alpha^d \lambda^{dq},$$

which is (II.2.4). The established smoothness of $\rho$, together with the explicit formulas above, proves that $\alpha, \beta, z, w$ are $C^\infty$ functions of $x$ on the open interval; consequently both sides of (II.2.10) depend smoothly on $x$ there. $\square$

**Proposition II.2.4** (Endpoint limits, including the exceptional order-three interval)**.** *On every open Farey interval, $x \mapsto \rho^{(n)}_{f,g}(x)$ is continuous. If $n \geq 4$, it extends continuously to the closed interval with value 1 at both endpoints, and the corresponding points tend to the two endpoint roots of unity. The same conclusion holds for $n = 3$ except on the exceptional interval $[1/3, 1/2]$. On that interval,*

$$\lim_{x \downarrow 1/3} \rho^{(3)}_{1/3,1/2}(x) = 1, \qquad \lim_{x \uparrow 1/2} \rho^{(3)}_{1/3,1/2}(x) = \frac{1}{2}, \tag{II.2.11}$$

*so the nonreal radial graph tends to $-1/2$, not to the endpoint root $-1$.*

*Proof.* Continuity on the open interval was proved in proposition II.2.3. It remains to determine the endpoint limits. Let $x_k$ tend to an endpoint and pass to a subsequence for which $\rho(x_k) \to \rho_* \in [0,1]$.

At the endpoint $p/q$, where $A \to 0$, determinant one gives

$$B_0 = \frac{2\pi}{dq}.$$

Thus $0 < B_0 \leq \pi$. Equality can occur only when $dq = 2$; under $n \geq 3$ and $q < s$, this is exactly the exceptional order-three interval. Indeed, $q = 1$ would give $d = n \geq 3$, so $q = 2$ and $d = 1$; then $d = \lfloor n/2 \rfloor = 1$ and $n \geq 3$ force $n = 3$. The endpoint with smaller denominator is $p/q = 1/2$, and its consecutive neighbor in $F_3^+$ is $r/s = 1/3$. Outside that single case, $B_0 \in (0,\pi)$ and the limiting equation

$$\rho_*^q \sin B_0 = \sin B_0$$

forces $\rho_* = 1$.

At the other endpoint $r/s$, where $B \to 0$, one has

$$A_0 = \frac{2\pi}{s} \in (0,\pi),$$

because $s \geq 3$. Hence $\rho_*^{s/d} \sin A_0 = \sin A_0$ and again $\rho_* = 1$.

It remains to evaluate the exceptional limit. Put $x = 1/2 - \varepsilon$ in the interval $[1/3, 1/2]$. Then

$$A = 4\pi\varepsilon, \qquad B = \pi - 6\pi\varepsilon, \qquad A + B = \pi - 2\pi\varepsilon,$$

and (II.2.8) is

$$\rho^3 \sin(4\pi\varepsilon) + \rho^2 \sin(6\pi\varepsilon) = \sin(2\pi\varepsilon).$$

After division by $2\pi\varepsilon$ and passage to the limit, every subsequential limit satisfies

$$2\rho_*^3 + 3\rho_*^2 = 1, \qquad (2\rho_* - 1)(\rho_* + 1)^2 = 0.$$

The unique solution in $[0,1]$ is $\rho_* = 1/2$. Since every subsequence has the same limit, all asserted endpoint limits follow. In the exceptional asymptotics, (II.2.9) also gives $\alpha \to (1/2)^3 \cdot 4/2 = 1/4$, so the nonreal branch meets the selected real branch at $\alpha = 1/4$. At every nonexceptional endpoint, $\rho(x) \to 1$ and $\lambda(x) = \rho(x)\exp(2\pi i x)$, so $\lambda(x)$ tends to $\exp(2\pi i p/q)$ or $\exp(2\pi i r/s)$, as appropriate. □

**Definition II.2.5** (Compact candidate arc on a Farey interval)**.** Define the parametrization

$$\gamma_{f,g}^{(n)}(x) = \rho_{f,g}^{(n)}(x)\exp(2\pi i x), \qquad f < x < g. \tag{II.2.12}$$

For every interval other than the exceptional order-three interval, define

$$\Gamma_{f,g}^{(n)} = \mathrm{cl}_{\mathbb{C}}\big(\gamma_{f,g}^{(n)}((f,g))\big).$$

By proposition II.2.4, the parametrization extends continuously to $[f,g]$ using the stated endpoint limits. Its values have positive modulus, and distinct parameters in $[f,g] \subseteq [0,1/2]$ have distinct arguments, so this extension is injective. Its image is the closure displayed above. A continuous injection from a compact interval into $\mathbb{C}$ is a homeomorphism onto its image; hence $\Gamma_{f,g}^{(n)}$ is an arc. For the exceptional interval, define

$$\Gamma_{1/3,1/2}^{(3)} = \mathrm{cl}_{\mathbb{C}}\big(\gamma_{1/3,1/2}^{(3)}((1/3,1/2))\big) \cup [-1,-1/2]. \tag{II.2.13}$$

Every point in the parametrized image satisfies the Ito equation by proposition II.2.3. For the exceptional interval, that equation reduces to

$$\lambda^3 - (1-\alpha)\lambda - \alpha = (\lambda - 1)(\lambda^2 + \lambda + \alpha) = 0.$$

For $1/4 \le \alpha \le 1$, the root of the quadratic factor in the closed upper half-plane is

$$\lambda(\alpha) = -\frac{1}{2} + \frac{i}{2}\sqrt{4\alpha - 1}.$$

For interior $x$, the point has positive imaginary part, so the real quadratic factor has negative discriminant and $\alpha > 1/4$; also $\alpha < 1$ because $\beta > 0$. The displayed quadratic formula therefore places the image in the open vertical segment between $-1/2$ and $\exp(2\pi i/3)$. The image is connected because $(1/3, 1/2)$ is connected and the parametrization is continuous by proposition II.2.3; hence its closure is connected as well. That closure contains both endpoint limits $-1/2$ and $\exp(2\pi i/3)$ by proposition II.2.4. A connected subset of this vertical line containing both endpoints contains the segment between them, while the preceding containment excludes every other point. Therefore

$$\mathrm{cl}_{\mathbb{C}}(\gamma^{(3)}_{1/3,1/2}((1/3,1/2))) = \left\{ -\frac{1}{2} + iy : 0 \le y \le \frac{\sqrt{3}}{2} \right\}.$$

For $0 \le \alpha \le 1/4$, the two real roots are

$$\lambda_{\pm}(\alpha) = \frac{-1 \pm \sqrt{1-4\alpha}}{2}.$$

The selected root is

$$\lambda_{-}(\alpha) = \frac{-1 - \sqrt{1-4\alpha}}{2} \in [-1, -1/2],$$

which continues from $-1$ at $\alpha = 0$ and meets the nonreal branch at $-1/2$ when $\alpha = 1/4$. The other root lies in $[-1/2, 0]$ and is not part of this candidate arc. Thus the exceptional candidate arc is exactly the vertical segment displayed above together with the horizontal segment $[-1, -1/2]$. These two closed segments meet only at $-1/2$. Traversing first from $-1$ to $-1/2$ and then from $-1/2$ to $\exp(2\pi i/3)$ gives a continuous injective piecewise-linear parametrization by a closed interval. Hence the exceptional candidate set is also an arc.

**Algorithm II.2.6** (Certified numerical evaluation of a candidate arc)**.** Fix $n \ge 3$, an argument parameter $x \in [0, 1/2]$ represented exactly, and a radial error target $\varepsilon > 0$. Here exact representation means that $x$ supports exact comparison with every rational Farey endpoint and arbitrarily narrow rational enclosures. All evaluations of real powers, trigonometric functions, $F_x$, and $F'_x$ below use outward-rounded interval enclosures whose widths can be reduced arbitrarily by increasing the working precision. The algorithm has two possible outputs: a point approximation $(\widehat{\rho}, x)$ with a certified radial error, or, only when $n = 3$ and $x = 1/2$, the exact interval $\{(\rho, 1/2) : 1/2 \le \rho \le 1\}$ representing $[-1, -1/2]$.

1. Generate $F_n^+$ in increasing order and determine whether $x$ is an endpoint or belongs to a unique open interval $(f, g)$ between consecutive fractions, using the stipulated exact comparisons.
2. If $x$ is an endpoint, return the exact polar point $(1, x)$. The sole exception is $n = 3$ and $x = 1/2$, where the return value is the exact interval

$$\{(\rho, 1/2) : 1/2 \le \rho \le 1\},$$

which represents $[-1, -1/2]$.

3. If $x \in (f, g)$, label the endpoint with smaller denominator $p/q$ and the other $r/s$, compute $d = \lfloor n/q \rfloor$, and compute $A, B$ from (II.2.6). Define

$$F_x(\rho) = \rho^{s/d} \sin A + \rho^q \sin B - \sin(A + B).$$

By proposition II.2.3, $F_x(0) < 0 < F_x(1)$ and $F_x$ is strictly increasing on $[0, 1]$.

4. Apply bisection to the initial bracket $[0, 1]$. Stop when the bracket has length at most $2\varepsilon$, and let $\widehat{\rho}$ be its midpoint. Otherwise let $m$ be the midpoint. Evaluate $F_x(m)$ with the stipulated outward-rounded enclosure. If an exact zero is certified, set $\widehat{\rho} = m$ and proceed to Step 5. If a strict sign is certified, retain the half-bracket containing the zero. There is also a terminating enclosure test when an outward-rounded evaluation contains zero without certifying equality. Since the bracket length is greater than $2\varepsilon$, the interval $[m - \varepsilon, m + \varepsilon]$ lies inside the current bracket and has positive left endpoint. Obtain a certified lower bound $c > 0$ for $F_x'$ on this interval. If the outward-rounded enclosure for $F_x(m)$ is contained in $[-c\varepsilon, c\varepsilon]$, set $\widehat{\rho} = m$ and proceed to Step 5; strict monotonicity and the mean-value theorem give

$$|m - \rho_{f,g}^{(n)}(x)| \leq \varepsilon.$$

Otherwise increase the working precision until either this enclosure test or a strict sign is certified. Thus an exact midpoint root cannot cause the procedure to wait forever. On termination, write the returned value as $\widehat{\rho}$. Then

$$|\widehat{\rho} - \rho_{f,g}^{(n)}(x)| \leq \varepsilon.$$

A Newton trial may replace the midpoint only when it lies strictly inside the current bracket and is subjected to the same certified sign and enclosure tests; otherwise retain the bisection midpoint.

5. Return the certified polar point approximation $(\widehat{\rho}, x)$, which symbolically represents

$$\widehat{\lambda} = \widehat{\rho} \exp(2\pi i x)$$

Values of $\alpha, \beta$ computed from (II.2.9) are auxiliary numerical approximations unless interval enclosures are also propagated through those formulas. Because $x$ is exact,

$$|\widehat{\lambda} - \gamma_{f,g}^{(n)}(x)| = |\widehat{\rho} - \rho_{f,g}^{(n)}(x)| \leq \varepsilon.$$

If numerical Cartesian coordinates are requested instead of this symbolic polar output, assign a separate certified angular budget: if $|\widehat{\theta} - 2\pi x| \leq \eta$, then

$$\left| \widehat{\rho} e^{i\widehat{\theta}} - \gamma_{f,g}^{(n)}(x) \right| \leq \varepsilon + \eta,$$

because $\widehat{\rho} \leq 1$ and $|e^{iu} - e^{iv}| \leq |u - v|$. Any final outward rounding of sine and cosine must likewise be included in the requested Cartesian tolerance. No angular budget is needed for the exact-angle polar output. No separate error bound for the reported approximations to $\alpha, \beta$ is claimed here.

At this stage only membership in the Ito polynomial family has been proved. Topic X establishes the upper comparison, Topic XI constructs stochastic matrices realizing the candidate points, Topic XII proves nesting with the matrix order, and Topic XIII identifies the resulting arcs with the boundary of $\Theta_n$.

## II.3 Main theorem

**Theorem II.3.1** (Karpelevič theorem in Ito's formulation)**.** *Let $n \geq 3$. For every consecutive pair $f < g$ in $F_n^+$, let $\Gamma_{f,g}^{(n)}$ be the candidate arc of definition II.2.5. Then these arcs are the Karpelevič boundary arcs, and their ordered concatenation is the upper boundary of $\Theta_n$, from $1$ to $-1$. The lower boundary is its complex conjugate, and*

$$\Theta_n = \{t\zeta : 0 \leq t \leq 1,\ \zeta \in \partial\Theta_n\}\,. \tag{II.3.1}$$

*The only points of $\Theta_n$ on the unit circle are roots of unity of order at most $n$.*

*For $n = 3$, the terminal candidate arc contains the boundary segment $[-1, -1/2]$ before leaving the real axis. For $n = 1$, one has $\Theta_1 = \{1\}$; for $n = 2$, one has $\Theta_2 = [-1, 1]$.*

The proof occupies appendix II.4–appendix II.9. The principal upper-bound input is theorem 1.4 from Part I, restated under condition (II.4.3) in appendix II.5. Everything after it is one-dimensional convexity, explicit realization, and Farey nesting.

## II.4 Invariant polygons and elementary structure

**Proposition II.4.1** (Compactness, conjugation, and disk bound)**.** *For every $n$, the set $\Theta_n$ is compact, invariant under complex conjugation, and contained in the closed unit disk.*

*Proof.* Let $\mathcal{S}_n$ be the set of real row-stochastic $n \times n$ matrices. Every entry lies in $[0, 1]$, and the row-sum equations define a closed subset of $[0, 1]^{n^2}$; hence $\mathcal{S}_n$ is compact. Write

$$\overline{\mathbb{D}} = \{z \in \mathbb{C} : |z| \leq 1\}.$$

Every eigenvalue lies in $\overline{\mathbb{D}}$: if $Av = \lambda v$, choose an index with $|v_i| = \|v\|_\infty > 0$; then

$$|\lambda|\|v\|_\infty = |(Av)_i| \leq \sum_{j=1}^{n} a_{ij}|v_j| \leq \|v\|_\infty \sum_{j=1}^{n} a_{ij} = \|v\|_\infty.$$

Consequently the eigenpair set

$$\mathcal{E}_n = \{(A, \lambda) \in \mathcal{S}_n \times \overline{\mathbb{D}} : \det(\lambda I - A) = 0\}$$

is closed in a compact space and is therefore compact. Its projection to the $\lambda$-coordinate is exactly $\Theta_n$, so $\Theta_n$ is compact. Real characteristic polynomials give conjugation symmetry. □

**Theorem II.4.2** (Invariant-polytope criterion)**.** *Let $n \geq 2$. For $\lambda \in \mathbb{C}$, the following are equivalent.*

(i) $\lambda \in \Theta_n$.

(ii) *There is a non-singleton convex polytope $P \subseteq \mathbb{C} \simeq \mathbb{R}^2$ with at most $n$ vertices such that*

$$\lambda P \subseteq P. \tag{II.4.1}$$

*Proof.* If $Av = \lambda v$, put $P = \operatorname{conv}\{v_1, \ldots, v_n\}$. Every coordinate satisfies

$$\lambda v_i = \sum_j a_{ij} v_j \in P,$$

so (II.4.1) holds. If the hull is a singleton, all coordinates of the nonzero eigenvector are equal to the same nonzero number, so $\lambda = 1$; in that case any fixed segment may be used.

Conversely, let $x_1, \dots, x_m$ be the vertices of an invariant polytope, $m \le n$. Choose barycentric coordinates

$$\lambda x_i = \sum_{j=1}^{m} a_{ij} x_j, \qquad a_{ij} \ge 0, \qquad \sum_j a_{ij} = 1.$$

Then $A = (a_{ij})$ is row-stochastic and $Ax = \lambda x$. If $m < n$, set

$$\widetilde{A} = A \oplus I_{n-m}, \qquad \widetilde{x} = (x_1, \dots, x_m, 0, \dots, 0)^T.$$

Then $\widetilde{A}$ is row-stochastic and $\widetilde{A}\widetilde{x} = \lambda \widetilde{x}$. □

The same block-diagonal construction gives the natural nesting

$$\Theta_m \subseteq \Theta_n \qquad (1 \le m \le n).$$

**Corollary II.4.3** (Star-shapedness with respect to the origin)**.** *Let $n \ge 2$. If $\lambda \in \Theta_n$, then $t\lambda \in \Theta_n$ for every $0 \le t \le 1$.*

*Proof.* If $|\lambda| < 1$, take an invariant polytope $P$. Iteration gives $\lambda^k P \subseteq P$ and $\lambda^k x \to 0$, hence $0 \in P$; convexity then gives $t\lambda P \subseteq P$.

Suppose $|\lambda| = 1$, and take an invariant polytope $P$ with at most $n$ vertices. If $P$ has nonempty interior, equality of areas in $\lambda P \subseteq P$ gives $\lambda P = P$ by lemma A.5. Thus multiplication by $\lambda$ permutes the vertices, and the orbit of a nonzero vertex gives $\lambda^k = 1$ for some $k \le n$. If $P$ is a segment, equality of lengths again gives $\lambda P = P$; preservation of its line direction forces $\lambda = \pm 1$, whose orders are at most $n$. Hence in all cases $\lambda$ is a root of unity of order $k \le n$.

If $\lambda = 1$, the segment $[0,1]$ satisfies $t[0,1] \subseteq [0,1]$. Otherwise the regular orbit polygon

$$Q = \operatorname{conv}\{1, \lambda, \dots, \lambda^{k-1}\}$$

contains 0 because $1 + \lambda + \cdots + \lambda^{k-1} = 0$, so 0 is the equal-weight average of its vertices. It satisfies $\lambda Q = Q$, and therefore $t\lambda Q = tQ \subseteq Q$. The invariant-polytope criterion completes the proof. □

**Proposition II.4.4** (Unit-circle points)**.** *Let $n \ge 2$. If $\lambda \in \Theta_n$ and $|\lambda| = 1$, then $\lambda$ is a root of unity of order at most $n$. Conversely every such root belongs to $\Theta_n$.*

*Proof.* Take an invariant polytope $P$ with at most $n$ vertices. If $P$ has nonempty interior, then $\lambda P \subseteq P$ and equality of areas forces $\lambda P = P$ by lemma A.5. Multiplication by $\lambda$ permutes the vertices, so a nonzero vertex has an orbit of length at most $n$ and $\lambda^k = 1$ for some $k \le n$. If $P$ is a segment, then $\lambda P$ and $P$ have the same positive length; inclusion therefore gives $\lambda P = P$. Thus multiplication by $\lambda$ preserves the direction of the affine line containing $P$, so $\lambda = \pm 1$.

Conversely, if $\lambda$ has order $k \le n$, the permutation matrix of a $k$-cycle is row-stochastic and has $\lambda$ as an eigenvalue. If $k < n$, take its direct sum with $I_{n-k}$. □

**Lemma II.4.5** (Interior origin for a nonreal contraction)**.** *Let $P$ be a non-singleton compact convex polytope and let $\lambda = \rho e^{i\theta}$ with $0 < \rho < 1$ and $\theta \not\equiv 0, \pi \pmod{2\pi}$. If $\lambda P \subseteq P$, then $P$ has nonempty interior and $0 \in \operatorname{int} P$.*

*Proof.* Iteration gives $0 \in P$. If $P$ were a segment, its affine line would pass through 0 and would have to contain the segment of positive length $\lambda P$; hence multiplication by $e^{i\theta}$ would preserve a real line, forcing $\theta \equiv 0$ or $\pi$, a contradiction. Thus $P$ has nonempty interior; equivalently, it is two-dimensional.

Suppose $0 \in \partial P$. The supporting-line theorem gives a nonzero real linear functional $\ell$ such that $\ell \geq 0$ on $P$ and $\ell(0) = 0$. Because $P$ has nonempty interior, it is not contained in $\ker \ell$; hence we may choose $z \in P$ with $\ell(z) > 0$. For every $k \geq 0$, invariance gives

$$\lambda^k z = \rho^k e^{ik\theta} z \in P, \qquad 0 \leq \ell(\lambda^k z) = \rho^k \ell(e^{ik\theta} z).$$

Because $\rho^k > 0$, it follows that $\ell(e^{ik\theta} z) \geq 0$ for every $k \geq 0$. If $e^{i\theta}$ has finite order $m$, then $m \geq 3$ and $\sum_{k=0}^{m-1} e^{ik\theta} z = 0$; summing the displayed nonnegative quantities forces them all to vanish, contrary to $\ell(z) > 0$. If $e^{i\theta}$ has infinite order, its powers are dense in the unit circle, so some $e^{ik\theta} z$ lies in the open half-plane where $\ell < 0$, again a contradiction. Hence 0 is interior. □

For $n \geq 2$ and $0 < \theta < \pi$, define the radial function

$$R_n(\theta) = \max\left\{\rho \in [0,1] : \rho e^{i\theta} \in \Theta_n\right\}. \tag{II.4.2}$$

Indeed, $1 \in \Theta_n$, so corollary II.4.3 gives $0 \in \Theta_n$; hence the set in (II.4.2) is nonempty. Compactness of $\Theta_n$ gives attainment of the maximum, while star-shapedness identifies the entire ray intersection as

$$\Theta_n \cap \{\rho e^{i\theta} : \rho \geq 0\} = \{\rho e^{i\theta} : 0 \leq \rho \leq R_n(\theta)\}.$$

On an open Farey interval it is strictly less than one by proposition II.4.4.

**Definition II.4.6** (Polygonal complexity and $N$-critical maps)**.** For a compact convex set $P$, let $\operatorname{Ext} P$ denote its set of extreme points. For a real-linear map $A : \mathbb{R}^2 \to \mathbb{R}^2$, let

$$\nu_{\text{poly}}(A) = \min\left\{|\operatorname{Ext} P| : \begin{array}{c} P \text{ is a compact convex polygon with nonempty interior,} \\ AP \subseteq P \end{array}\right\}.$$

If the displayed family is nonempty, its vertex counts are positive integers, so the minimum exists by well-ordering; if the family is empty, set $\nu_{\text{poly}}(A) = \infty$. For $N \geq 3$, an elliptic contraction $T$ is *$N$-critical* when

$$\nu_{\text{poly}}(T) = N, \qquad \nu_{\text{poly}}(tT) > N \quad (t > 1).$$

Here elliptic means $(\operatorname{tr} T)^2 < 4 \det T$. The spectral radius is

$$\operatorname{spr}(T) = \max\{|\mu| : \mu \text{ is an eigenvalue of } T\},$$

and the manuscript term *elliptic contraction* means an elliptic map with $\operatorname{spr}(T) \in (0,1)$. This is a spectral, hence asymptotic, contraction condition; it does not assert an operator-norm bound in an arbitrary chosen norm.

For later use, write $T_\lambda(z) = \lambda z$. For every $n \geq 2$ and every nonreal $\lambda$ with $0 < |\lambda| < 1$, the invariant-polytope criterion and lemma II.4.5 give the exact equivalence

$$\lambda \in \Theta_n \iff \nu_{\text{poly}}(T_\lambda) \leq n.$$

For $N \geq 4$ and $0 < \theta < \pi$, consider a radial boundary point that belongs to $\Theta_N$ but not to $\Theta_{N-1}$, namely a point satisfying

$$\lambda = R_N(\theta) e^{i\theta} \in \Theta_N \setminus \Theta_{N-1}, \qquad 0 < |\lambda| < 1. \tag{II.4.3}$$

The modulus condition excludes the origin and the unit circle. Moreover, nesting gives $R_{N-1}(\theta) \leq R_N(\theta)$, and attainment of the $\Theta_{N-1}$ radial endpoint together with $\lambda \notin \Theta_{N-1}$ makes the inequality strict:

$$R_{N-1}(\theta) < R_N(\theta).$$

The restriction $N \geq 4$ is deliberate: this bridge is used to invoke the Part I critical-polygon theorem, whose nontrivial range is $N \geq 4$, while orders at most three are handled directly.

**Proposition II.4.7** (A radial boundary point in $\Theta_N \setminus \Theta_{N-1}$ yields an $N$-critical map)**.** *If* (II.4.3) *holds, then multiplication $T_\lambda : \mathbb{C} \to \mathbb{C}$, regarded as a real-linear map, is an $N$-critical elliptic contraction.*

*Proof.* In the real basis $(1, i)$,

$$[T_\lambda] = \begin{pmatrix} \operatorname{Re}\lambda & -\operatorname{Im}\lambda \\ \operatorname{Im}\lambda & \operatorname{Re}\lambda \end{pmatrix}.$$

Since $0 < \theta < \pi$, the multiplier is nonreal, and therefore

$$(\operatorname{tr} T_\lambda)^2 - 4\det T_\lambda = -4(\operatorname{Im}\lambda)^2 < 0.$$

Its spectral radius is $|\lambda| \in (0, 1)$, so it is an elliptic contraction.

By the displayed equivalence, membership $\lambda \in \Theta_N$ gives $\nu_{\mathrm{poly}}(T_\lambda) \le N$. If this complexity were at most $N-1$, the same equivalence would give $\lambda \in \Theta_{N-1}$, contrary to (II.4.3). Consequently

$$\nu_{\mathrm{poly}}(T_\lambda) = N.$$

Finally, let $t > 1$. If $t|\lambda| \le 1$, then $t\lambda$ lies farther out on the same ray than the maximal point $R_N(\theta)e^{i\theta}$, so $t\lambda \notin \Theta_N$. If $t|\lambda| > 1$, the closed-unit-disk bound in proposition II.4.1 gives the same conclusion. If $\nu_{\mathrm{poly}}(tT_\lambda) \le N$, a polygon with nonempty interior realizing that inequality would put $t\lambda$ in $\Theta_N$ by theorem II.4.2, a contradiction. Hence $\nu_{\mathrm{poly}}(tT_\lambda) > N$ for all $t > 1$, which is precisely $N$-criticality. ☐

## II.5 Critical-polygon input from Part I

For a nonzero complex number $z$ that is not positive real, let

$$\arg_+(z) \in (0, 2\pi)$$

denote its unique argument in that interval. The following theorem records exactly the conclusions from Part I used below.

**Theorem II.5.1** (Finite product theorem from Part I)**.** *Let $N \ge 4$ and suppose* (II.4.3) *holds. Retain $\lambda = |\lambda|e^{i\theta}$ and $x = \theta/(2\pi)$ for the original upper-half-plane multiplier. There is a multiplier*

$$\mu \in \{\lambda, \overline{\lambda}\}, \qquad \vartheta = \arg_+(\mu), \qquad y = \frac{\vartheta}{2\pi},$$

*and there are consecutive reduced fractions in $F_N$,*

$$\frac{p}{q} < y < \frac{r}{s}, \qquad rq - ps = 1, \qquad q < s.$$

*Thus $y = x$ when $\mu = \lambda$ and $y = 1 - x$ when $\mu = \overline{\lambda}$. Put*

$$d = \lfloor N/q \rfloor, \qquad e = s - dq.$$

*There are parameters*

$$0 \le \beta_j < 1, \qquad \alpha_j = 1 - \beta_j > 0 \qquad (1 \le j \le d) \tag{II.5.1}$$

*such that*

$$\mu^s \prod_{j=1}^{d} (\mu^q - \beta_j) = \mu^{dq} \prod_{j=1}^{d} \alpha_j. \tag{II.5.2}$$

*Equivalently, since $\mu \neq 0$,*

$$\mu^e \prod_{j=1}^{d} (\mu^q - \beta_j) = \prod_{j=1}^{d} \alpha_j. \tag{II.5.3}$$

*Any factor with $\beta_j = 0$ introduced only to extend the finite list of recurrence relations to its closing index is not claimed to be an additional relative-interior contact.*

*Set*

$$A = q\vartheta - 2\pi p, \qquad M = \arg_+(\mu^q - 1).$$

*For the unique arguments $u_j \in (0, \pi)$ determined by*

$$e^{iu_j} = \frac{\mu^q - \beta_j}{|\mu^q - \beta_j|},$$

*one has*

$$0 < A < M < \pi, \qquad u_j \in [A, M) \quad (1 \leq j \leq d), \tag{II.5.4}$$

*and the exact equation for the chosen real arguments*

$$e\vartheta + \sum_{j=1}^{d} u_j = 2\pi(r - dp). \tag{II.5.5}$$

*Proof.* By proposition II.4.7, $T_\lambda$ is an $N$-critical elliptic contraction. Apply the Part I theorem on consecutive Farey fractions and a finite product equation, theorem 1.4, including its factor-argument bounds. The determinant equation for the consecutive Farey endpoints follows from lemma II.2.1. The two complex structures in which $T_\lambda$ is multiplication by a complex scalar give precisely the choice between $\lambda$ and $\overline{\lambda}$. □

**Lemma II.5.2** (Effect of complex conjugation on the Farey interval)**.** *In theorem II.5.1, suppose $\mu = \overline{\lambda}$, so that $y = 1 - x$. Then*

$$\frac{s - r}{s} < x < \frac{q - p}{q}$$

*is the reflected order-$N$ Farey interval. With the denominator-based labeling used in (II.2.6),*

$$\begin{aligned} 2\pi\, |qx - (q - p)| &= 2\pi(qy - p) = A, \\ \frac{2\pi}{d}\, |sx - (s - r)| &= \frac{2\pi}{d}(r - sy) = \frac{2\pi r - s\vartheta}{d}. \end{aligned}$$

*Thus reflection preserves $q, s, d, e$, the modulus $|\lambda| = |\mu|$, and the radial equation (II.2.8). Conjugating the constant-parameter Ito equation for $\mu$ gives the Ito equation (II.2.4) for $\lambda$ in this reflected Farey interval.*

*Proof.* The map $t \mapsto 1 - t$ reverses the endpoint order and preserves denominators, giving the displayed interval. Since $x = 1 - y$,

$$qx - (q - p) = p - qy < 0, \qquad sx - (s - r) = r - sy > 0,$$

which proves the two equalities. All remaining assertions follow directly from the definitions and complex conjugation. □

## II.6 Log-sine convexity and the sharp radial inequality

The finite product equation supplied by Part I gives two scalar constraints. Taking moduli turns the factors into a sum of values $F(u_j)$ of one function, while the equation for the chosen real arguments fixes their arithmetic mean at $A + B$. On the interval given by the factor-argument bounds above, $F'' > 0$, so equality in Jensen's inequality is possible only when all factor arguments, and hence all parameters, agree. The theorem below gives the radial upper bound $\rho \le \rho_*$. The later sparse stochastic realization supplies the reverse inequality at a radial boundary point satisfying (II.4.3).

Retain the multiplier selected in theorem II.5.1 and write

$$\mu = \rho e^{i\vartheta}, \qquad y = \frac{\vartheta}{2\pi}.$$

We work in its open Farey interval

$$\frac{p}{q} < y < \frac{r}{s}, \qquad rq - ps = 1, \qquad q < s. \tag{II.6.1}$$

Put

$$A = q\vartheta - 2\pi p, \qquad B = \frac{2\pi r - s\vartheta}{d}. \tag{II.6.2}$$

These are the non-absolute versions of (II.2.6). The same determinant-one computation as in (II.2.7) gives

$$A > 0, \qquad B > 0, \qquad A + B < \pi. \tag{II.6.3}$$

### II.6.1 The strictly convex log-sine function

Fix $\mu = \rho e^{i\vartheta}$ with $0 < \rho < 1$ and let

$$M = \mathrm{Arg}(\mu^q - 1) \in (A, \pi)$$

be the unique argument in $(0, \pi)$. Since $\mu^q = \rho^q e^{iA}$ lies in the open upper half-plane, so does $\mu^q - \beta$ for every $0 \le \beta < 1$. Let

$$u(\beta) = \mathrm{Arg}(\mu^q - \beta) \in [A, M)$$

be its unique argument in $(0, \pi)$, and define

$$F(u) = \log \sin M - \log \sin(M - u) \qquad (A \le u < M).$$

The triangle with vertices $\beta$, 1, and $\mu^q$ gives

$$F(u(\beta)) = \log \frac{|\mu^q - \beta|}{1 - \beta}, \tag{II.6.4}$$

and hence

$$F''(u) = \csc^2(M - u) > 0. \tag{II.6.5}$$

At $\beta = 0$, the same triangle gives

$$\rho^q = \frac{\sin M}{\sin(M - A)}. \tag{II.6.6}$$

**Theorem II.6.1** (Sharp inequality for varying parameters)**.** *Let $N \geq 4$, let $\mu = \rho e^{i\vartheta}$ with $0 < \rho < 1$, and put $y = \vartheta/(2\pi)$. Suppose $p/q < y < r/s$ are consecutive reduced fractions in $F_N$, with*

$$rq - ps = 1, \qquad q < s,$$

*and put*

$$d = \lfloor N/q \rfloor, \qquad e = s - dq, \qquad A = q\vartheta - 2\pi p, \qquad B = \frac{2\pi r - s\vartheta}{d}.$$

*Let $M = \operatorname{Arg}(\mu^q - 1) \in (A, \pi)$. Suppose*

$$0 \leq \beta_j < 1, \qquad \alpha_j = 1 - \beta_j > 0 \qquad (1 \leq j \leq d),$$

*and*

$$\mu^e \prod_{j=1}^{d} (\mu^q - \beta_j) = \prod_{j=1}^{d} \alpha_j.$$

*For each $j$, let $u_j \in (0, \pi)$ be determined by*

$$e^{iu_j} = \frac{\mu^q - \beta_j}{|\mu^q - \beta_j|},$$

*and suppose*

$$u_j \in [A, M), \qquad e\vartheta + \sum_{j=1}^{d} u_j = 2\pi(r - dp).$$

*Then*

$$\rho^{s/d} \sin A + \rho^q \sin B \leq \sin(A + B). \tag{II.6.7}$$

*Let $\rho_* \in (0, 1)$ be the unique solution of*

$$t^{s/d} \sin A + t^q \sin B = \sin(A + B).$$

*Then $\rho \leq \rho_*$. Equality in either inequality holds if and only if*

$$\beta_1 = \cdots = \beta_d. \tag{II.6.8}$$

*Proof.* The equation for the chosen real arguments gives

$$\sum_{j=1}^{d} u_j = 2\pi(r - dp) - (s - dq)\vartheta = d(A + B).$$

Because every $u_j$ belongs to $[A, M)$, their mean $A + B$ belongs to that interval. Since $B > 0$, in fact

$$A < A + B < M, \qquad 0 < M - A - B < M - A < \pi.$$

Thus $\sin A$, $\sin B$, $\sin(A + B)$, $\sin M$, $\sin(M - A)$, and $\sin(M - A - B)$ are all positive.

Taking moduli in (II.5.3) gives

$$\sum_{j=1}^{d} F(u_j) = (dq - s) \log \rho.$$

Because every $u_j$ belongs to the interval $[A, M)$, their average $A + B$ belongs to the same interval and $F$ is strictly convex on the entire convex hull of the arguments. Jensen's inequality for the strictly convex function $F$ gives

$$dF(A + B) \leq (dq - s) \log \rho. \tag{II.6.9}$$

Equality holds exactly when all $u_j$ coincide. The argument map $u(\beta) = \operatorname{Arg}(\mu^q - \beta)$ is strictly increasing because

$$\frac{du}{d\beta} = \frac{\operatorname{Im}(\mu^q)}{|\mu^q - \beta|^2} = \frac{\rho^q \sin A}{|\mu^q - \beta|^2} > 0.$$

Therefore equality is equivalent to all $\beta_j$ coinciding.

Exponentiating (II.6.9) and using (II.6.6) yields

$$\frac{\sin(M - A - B)}{\sin M} \geq \rho^{s/d-q}.$$

Using (II.6.6) and the identity

$$\sin(M - A)\sin(A + B) - \sin M \sin B = \sin A \sin(M - A - B)$$

gives

$$\frac{\sin(M - A - B)}{\sin M} = \frac{\rho^{-q}\sin(A + B) - \sin B}{\sin A}. \tag{II.6.10}$$

Multiplication by $\rho^q \sin A$ gives (II.6.7).

Finally, the function

$$H(t) = t^{s/d} \sin A + t^q \sin B$$

is strictly increasing for $t > 0$. Moreover $H(0) = 0 < \sin(A + B)$, while

$$H(1) - \sin(A + B) = 4 \sin\frac{A}{2} \sin\frac{B}{2} \sin\frac{A + B}{2} > 0.$$

It therefore has the stated unique equality radius $\rho_* \in (0, 1)$. The inequality $H(\rho) \leq H(\rho_*)$ gives $\rho \leq \rho_*$, and equality holds exactly in the equality case already identified above. □

**Corollary II.6.2** (Constant parameters at a new order-$N$ boundary point)**.** *Under* (II.4.3) *in an open Farey interval, the parameters in the finite product equation selected by theorem II.5.1 satisfy*

$$\beta_1 = \cdots = \beta_d = \beta, \qquad \alpha = 1 - \beta,$$

*and the original point lies on* (II.2.4)*. Its modulus is the unique solution of* (II.2.8)*.*

*Proof.* Let $\mu$, $\vartheta$, $y$, and the consecutive Farey fractions be supplied by theorem II.5.1. That theorem supplies the finite-product equation, chosen real arguments, and factor-argument bounds required by theorem II.6.1. Applying the latter theorem to $\mu$ gives

$$|\mu| \leq \rho_*(\mu),$$

where $\rho_*(\mu)$ is the unique equality radius for the selected Farey interval. If $\mu = \lambda$, this is the original interval and ray. If $\mu = \overline{\lambda}$, the two displayed identities in lemma II.5.2 show that reflection preserves $q, s, d$, the two positive angles in the scalar equation, and the modulus $|\mu| = |\lambda|$. Hence in either case $\rho_*(\mu)$ is the unique radius $\rho_*$ for the original ray supplied by proposition II.2.3, and therefore, with $\rho = |\lambda|$,

$$\rho \leq \rho_*.$$

By corollary II.7.4, the point of modulus $\rho_*$ on that ray belongs to $\Theta_N$. Since $\rho = R_N(\theta)$ is the maximal attained radius on the ray, $\rho_* \leq \rho$. Combining the two inequalities gives $\rho = \rho_*$. Equality therefore holds in theorem II.6.1, whose equality statement yields $\beta_1 = \cdots = \beta_d$. For $\mu = \lambda$ the resulting constant-parameter equation is (II.2.4); for $\mu = \overline{\lambda}$, conjugation and lemma II.5.2 give the same equation for $\lambda$ in the reflected Farey interval. □

## II.7 Explicit stochastic realization of the candidate curve

We now prove the inner inclusion directly, without importing an Ito realization theorem.

For a weighted directed graph, an edge $u \to v$ of weight $w(u, v)$ is encoded in the adjacency matrix by

$$A_{uv} = w(u, v), \tag{II.7.1}$$

with $A_{uv} = 0$ when the arrow is absent. Thus the sum of the weights of edges leaving $u$ is the row sum of $A$ at $u$. In the construction below, a displayed edge of weight zero may be retained for bookkeeping; omitting it does not change the adjacency matrix.

**Lemma II.7.1** (Directed-cycle expansion of the characteristic polynomial)**.** *Let $\Gamma$ be a weighted directed graph on $N$ vertices, with adjacency matrix $A$ and edge-weight function $w$. Then*

$$\det(tI_N - A) = \sum_{\mathcal{C}} (-1)^{|\mathcal{C}|} t^{N-|V(\mathcal{C})|} \prod_{e \in E(\mathcal{C})} w(e),$$

*where the sum ranges over all collections $\mathcal{C}$ of pairwise vertex-disjoint directed cycles in $\Gamma$, including the empty collection. Here $V(\mathcal{C})$ and $E(\mathcal{C})$ are the unions of the vertices and edges of the selected cycles. Loops are allowed, and the empty product is one.*

*Proof.* Expand the determinant by Leibniz and then expand every selected diagonal factor $t - A_{uu}$. Each resulting monomial is obtained by selecting a collection of pairwise vertex-disjoint permutation cycles, with every remaining fixed vertex contributing $t$. A directed cycle $u_1 \to u_2 \to \cdots \to u_\ell \to u_1$ has permutation sign $(-1)^{\ell-1}$, while selecting the $-A$ entry at its $\ell$ vertices contributes $(-1)^\ell$. Its net sign is therefore $-1$, independently of $\ell$. For $\ell = 1$, this says precisely that selecting the loop chooses $-A_{uu}$ rather than $t$ from $t - A_{uu}$. Thus a collection of $|\mathcal{C}|$ cycles has sign $(-1)^{|\mathcal{C}|}$, contributes the product of its edge weights, and leaves $N - |V(\mathcal{C})|$ diagonal factors $t$. Summing over all collections, including the empty one, gives the formula. □

**Lemma II.7.2** (Directed cycles in the realization digraph)**.** *Let $d, q \in \mathbb{N}$. For each $j \in \mathbb{Z}/d\mathbb{Z}$, let*

$$B_j = \{v_{j,0}, \ldots, v_{j,q-1}\}$$

*be a set of vertices, with the sets $B_j$ pairwise disjoint, and put in the directed path*

$$v_{j,0} \longrightarrow v_{j,1} \longrightarrow \cdots \longrightarrow v_{j,q-1}.$$

*From each terminal vertex $v_{j,q-1}$ add a return edge to $v_{j,0}$ and an inter-block edge to a specified vertex $v_{j+1,a_{j+1}}$ of the next block, where $0 \le a_{j+1} < q$. At most one inter-block edge may instead be replaced by a directed path through new vertices, each of which has its unique outgoing edge to the next vertex of that path. When $d = 1$ and $a_0 = 0$, require this inter-block connection to be subdivided, so that it is distinct from the return edge.*

*Then every directed simple cycle is either the within-block $q$-cycle*

$$C_j : v_{j,0} \to \cdots \to v_{j,q-1} \to v_{j,0}$$

*for some $j$, or the unique directed cycle $C_*$ containing all $d$ inter-block connections. The cycle $C_*$ meets every $C_j$ at the terminal vertex of block $j$. Consequently the collections of pairwise vertex-disjoint directed cycles are exactly*

(i) *an arbitrary subset of $\{C_0, \ldots, C_{d-1}\}$, or*

(ii) *the singleton collection $\{C_*\}$.*

*Proof.* A directed cycle using no inter-block edge is confined to one block. The directed path forces it to the terminal vertex, and the only edge that both remains in that block and closes a cycle is the return edge. Hence it is $C_j$.

Now let a simple cycle use an inter-block connection into block $j$. From its entry vertex $v_{j,a_j}$, the directed suffix is forced until the terminal vertex. If the cycle then used the return edge, the path from $v_{j,0}$ would reach the already visited entry vertex before the cycle could return through its preceding inter-block connection, contradicting simplicity. Thus it must use the inter-block connection at every terminal it reaches. These connections advance cyclically through all $d$ blocks, so they determine the unique cycle $C_*$. Any subdivision vertices lie on that cycle and create no additional choice.

The cycle $C_*$ contains the terminal vertex of every block, which also belongs to $C_j$. Thus $C_*$ is not vertex-disjoint from any $C_j$, whereas the cycles $C_j$ in distinct blocks are pairwise vertex-disjoint. This is the asserted classification. □

**Theorem II.7.3** (Sparse stochastic realization of the reduced Ito polynomial)**.** *Let $f < g$ be consecutive elements of $F_n^+$, and let $p/q, r/s$ be their denominator-ordered labels from* (II.2.2)*, so $q < s$. Put $d = \lfloor n/q \rfloor$. For every $\alpha \in [0, 1]$ and $\beta = 1 - \alpha$, there is a row-stochastic matrix $A$ of order*

$$N_0 = \max\{dq, s\} \le n \tag{II.7.2}$$

*whose characteristic polynomial is, when $s \le dq$,*

$$\det(tI_{N_0} - A) = (t^q - \beta)^d - \alpha^d t^{dq-s}, \tag{II.7.3}$$

*and, when $s > dq$,*

$$\det(tI_{N_0} - A) = t^{s-dq}(t^q - \beta)^d - \alpha^d. \tag{II.7.4}$$

*Thus the spectrum of $A$ contains every root of the reduced Ito polynomial* (II.2.5)*, and hence every nonzero root of* (II.2.4)*. If $N_0 < n$, then*

$$\widetilde{A} = A \oplus I_{n-N_0}$$

*is row-stochastic of order $n$ and*

$$\det(tI_n - \widetilde{A}) = (t-1)^{n-N_0} \det(tI_{N_0} - A),$$

*so every eigenvalue of $A$ remains an eigenvalue of $\widetilde{A}$.*

*Proof.* Take $d$ blocks

$$B_j = \{v_{j,0}, \dots, v_{j,q-1}\}.$$

Inside each block put edges of weight 1, $v_{j,k} \to v_{j,k+1}$ for $k < q - 1$. At the terminal vertex $v_{j,q-1}$ put the within-block return edge $v_{j,q-1} \to v_{j,0}$ of weight $\beta$ and an inter-block edge of weight $\alpha$ entering block $j + 1$, with block indices read cyclically modulo $d$.

Suppose first that $s \le dq$. Farey adjacency gives $q + s > n \ge dq$, hence $s > (d-1)q$. Put

$$\ell_1 = s - (d-1)q, \qquad \ell_2 = \dots = \ell_d = q.$$

The inter-block edge entering block $j$ is, by convention, the edge whose tail is the terminal of block $j-1$. Route it to $v_{j,q-\ell_j}$. The local cycles have length $q$ and weight $\beta$. The unique cycle containing all $d$ inter-block edges uses all $d$ inter-block edges, has weight $\alpha^d$, and visits exactly $\ell_j$ vertices in block $j$, hence has length $s$. By lemma II.7.2, the only pairwise vertex-disjoint directed-cycle collections are arbitrary subsets of the within-block cycles or the singleton cycle containing all inter-block edges. Therefore the within-block subsets contribute

$$\sum_{J \subseteq \{1,\dots,d\}} (-1)^{|J|} \beta^{|J|} t^{dq-q|J|} = (t^q - \beta)^d,$$

and that cycle contributes $-\alpha^d t^{dq-s}$. By lemma II.7.1, this proves (II.7.3).

If $s > dq$, route every inter-block edge into the first vertex of the next block. Set $K = s - dq > 0$. On one such edge $u \to v$, introduce exactly the $K$ new vertices $w_1, \dots, w_K$ and replace that edge by the path

$$u \xrightarrow{\alpha} w_1 \xrightarrow{1} \cdots \xrightarrow{1} w_K \xrightarrow{1} v.$$

The cycle containing all inter-block edges has length $s$ and weight $\alpha^d$. The within-block cycles remain the $d$ disjoint $q$-cycles of weight $\beta$, and every collection of within-block cycles leaves all intermediate vertices unused. Using lemmas II.7.1 and II.7.2, those subsets contribute

$$t^{s-dq} \sum_{J \subseteq \{1,\dots,d\}} (-1)^{|J|} \beta^{|J|} t^{dq-q|J|} = t^{s-dq}(t^q - \beta)^d,$$

while the singleton cycle containing all inter-block edges contributes $-\alpha^d$. This proves (II.7.4).

Every nonterminal block vertex and every subdivision vertex has one outgoing edge of weight 1, while every block terminal has total outgoing weight $\alpha + \beta = 1$. All entries are nonnegative, so $A$ is row-stochastic. This remains literal at $\alpha = 0$ or $\beta = 0$: retain the corresponding zero-weight displayed edge for the cycle calculation, or omit it from the graph without changing the matrix or the formula. The two characteristic-polynomial identities are exactly the reduced forms in (II.2.5). The direct-sum assertion follows because an absorbing state contributes a factor $t - 1$ to the characteristic polynomial. □

**Corollary II.7.4** (Stochastic realization of the candidate point)**.** *Let $f < g$ be consecutive elements of $F_n^+$. For every $x \in (f, g)$, the point*

$$z = \rho_{f,g}^{(n)}(x) e^{2\pi i x}$$

*belongs to $\Theta_n$.*

*Proof.* By proposition II.2.3 and (II.2.9), there are $\alpha, \beta \in (0, 1)$ with $\alpha + \beta = 1$ such that the nonzero number $z$ satisfies the Ito equation (II.2.4). Equivalently, $z$ is a root of the corresponding reduced Ito polynomial (II.2.5). By theorem II.7.3, it is an eigenvalue of a row-stochastic matrix of order $N_0 \le n$; the direct-sum padding in that theorem gives a row-stochastic matrix of order exactly $n$ with the same eigenvalue. Hence $z \in \Theta_n$ by the definition of $\Theta_n$. □

## II.8 Farey refinement and nesting in the order

The finite-product theorem from Part I applies only under (II.4.3). To close the induction, we prove directly that the candidate radial regions are nested in $n$.

### II.8.1 A log-line lemma

Let $L$ be a line not through the origin. On any interval of angles for which $L$ has a positive radial intersection, write the logarithm of that radius as $\ell(\phi)$. For suitable constants $c, \phi_0$,

$$\ell(\phi) = c - \log\cos(\phi - \phi_0), \qquad \ell''(\phi) = \sec^2(\phi - \phi_0) > 0. \tag{II.8.1}$$

We shall use the following signed form of the chord calculation. Identify complex numbers with vectors in $\mathbb{R}^2$. For

$$X = Pe^{iA}, \qquad Y = Qe^{-iC}, \qquad P, Q > 0, \qquad A, C > 0, \qquad A + C < \pi,$$

define

$$\Delta_{X,Y}(r) := \det(Y - X, r - X) = r(P \sin A + Q \sin C) - PQ \sin(A + C), \qquad r > 0. \quad \text{(II.8.2)}$$

Thus the positive-real intercept of the line through $X$ and $Y$ is

$$I(P, Q) = \frac{PQ \sin(A + C)}{P \sin A + Q \sin C}, \qquad \text{(II.8.3)}$$

and

$$I(P, Q) > 1 \quad \Longleftrightarrow \quad \Delta_{X,Y}(1) < 0. \qquad \text{(II.8.4)}$$

The endpoint monotonicities also have fixed signs:

$$\frac{\partial I}{\partial P} = \frac{Q^2 \sin(A + C) \sin C}{(P \sin A + Q \sin C)^2} > 0, \qquad \frac{\partial I}{\partial Q} = \frac{P^2 \sin(A + C) \sin A}{(P \sin A + Q \sin C)^2} > 0.$$

All chord comparisons below will be reduced to (II.8.2); no diagrammatic side convention is used.

**Lemma II.8.1** (Mediant chord expansion in both denominator orientations). *Let $n \geq 4$, let $a/b < c/d$ be consecutive in $F_{n-1}$, and suppose the mediant*

$$\frac{a + c}{b + d}$$

*is newly inserted in $F_n$, so $n = b + d$. Fix an interior ray of one of the two new subcells and let $R > 1$ be the reciprocal of the old order-$(n - 1)$ candidate radius on that ray. At this same $R$, the positive-real intercept of the order-$n$ reciprocal chord is strictly larger than $1$, the intercept of the old reciprocal chord. At the common Farey endpoints the two candidate outer radii are both $1$ by definition. This holds whether $b < d$ or $d < b$.*

*Proof.* We first treat $b < d$. Put

$$m = \left\lfloor \frac{n - 1}{b} \right\rfloor, \qquad U = R^b e^{iA}, \qquad V = R^{d/m} e^{-iB},$$

where

$$A = 2\pi(bx - a), \qquad B = \frac{2\pi}{m}(c - dx).$$

At the old candidate radius, (II.2.10) gives

$$1 = \beta U + \alpha V, \qquad \alpha, \beta \in (0, 1).$$

Consequently $1 \in \operatorname{relint}[U, V]$. Let $L$ be the line through $U, V$, and let $\ell$ be its log-radial function on the angular interval $[-B, A]$. Then

$$\ell(0) = 0, \qquad \ell(A) = b \log R, \qquad \ell(-B) = \frac{d}{m} \log R,$$

and (II.8.1) says that $\ell'$ is strictly increasing.

Consider first the left subcell

$$\frac{a}{b} < x < \frac{a + c}{b + d},$$

equivalently $B > A/m$. Suppose initially that $b > 1$. Since $\gcd(b, d) = 1$, one has $b \nmid d$, and hence

$$\left\lfloor \frac{n}{b} \right\rfloor = \left\lfloor \frac{n - 1}{b} \right\rfloor = m.$$

The new reciprocal chord has endpoints

$$U, \qquad W = VU^{1/m}, \qquad \arg W = -B + \frac{A}{m}.$$

Put $\phi = -B + A/m \in (-B, 0)$. Strict monotonicity of $\ell'$ gives

$$\begin{aligned} \ell(\phi) - \ell(-B) &= \int_0^{A/m} \ell'(-B+t)\, dt \\ &< \int_0^{A/m} \ell'(mt)\, dt = \frac{1}{m}\ell(A). \end{aligned} \tag{II.8.5}$$

Therefore

$$\log |W| = \ell(-B) + \frac{1}{m}\ell(A) > \ell(\phi).$$

Let

$$V_* = \exp(\ell(\phi))e^{i\phi} \in [\,V, 1\,], \qquad W = \gamma V_*, \qquad \gamma > 1.$$

Since $U, V_*, 1$ are collinear,

$$\det(V_* - U, 1 - U) = 0.$$

The signed determinant of the new chord at 1 is therefore

$$\begin{aligned} \Delta_{U,W}(1) &= \det(W - U, 1 - U) \\ &= (\gamma - 1)\det(U, 1) \\ &= -(\gamma - 1)R^b \sin A < 0. \end{aligned} \tag{II.8.6}$$

By (II.8.4), its positive-real intercept is strictly larger than 1.

The right subcell satisfies

$$\frac{a+c}{b+d} < x < \frac{c}{d}, \qquad \text{equivalently} \qquad A > mB.$$

This calculation is valid also when $b = 1$. Since $b < d$, the new multiplicity at the endpoint of denominator $d$ is one. With the positive-angle endpoint written first, the new rooted endpoints are

$$X = UV^m, \qquad Y = V^m;$$

indeed their arguments are $A - mB > 0$ and $-mB < 0$. The new line is $V^m L$, because $1, U \in L$. Moreover

$$\ell(mB) + m\ell(-B) = m \int_0^B (\ell'(mt) - \ell'(-t))\, dt > 0. \tag{II.8.7}$$

Put

$$Z = \exp(\ell(mB))e^{imB} \in [\,1, U\,], \qquad \eta = V^{-m}.$$

Equation (II.8.7) says that $\eta = \gamma Z$ for some $0 < \gamma < 1$. Multiplication by $V^m$ multiplies real determinants by the positive number $|V|^{2m}$, so

$$\begin{aligned} \Delta_{X,Y}(1) &= \det\big(V^m(1 - U), 1 - V^m U\big) \\ &= |V|^{2m}\det(1 - U, \eta - U) \\ &= |V|^{2m}(\gamma - 1)\det(1 - U, U) \\ &= |V|^{2m}(\gamma - 1)R^b \sin A < 0. \end{aligned} \tag{II.8.8}$$

Here the third equality uses the collinearity of $1, U, Z$. Again (II.8.4) gives an intercept strictly larger than 1.

It remains to handle the left subcell when $b = 1$. Farey consecutiveness then gives the old cell

$$\frac{0}{1} < \frac{1}{d},$$

and the new left cell has right endpoint $1/(d+1)$. Set

$$\phi_{\text{old}} = \frac{2\pi}{d}, \qquad \phi_{\text{new}} = \frac{2\pi}{d+1}.$$

For $0 < \theta < \phi_{\text{new}}$, the radial intersection of the chord from 1 to $e^{i\phi}$ is

$$r_\phi(\theta) = \frac{\sin\phi}{\sin\theta + \sin(\phi - \theta)} = \frac{\cos(\phi/2)}{\cos(\theta - \phi/2)}.$$

Its logarithmic derivative has the explicit sign

$$\begin{aligned}\frac{\partial}{\partial\phi}\log r_\phi(\theta) &= -\frac{1}{2}\left(\tan\frac{\phi}{2} + \tan\left(\theta - \frac{\phi}{2}\right)\right)\\ &= -\frac{\sin\theta}{2\cos(\phi/2)\cos(\theta - \phi/2)} < 0.\end{aligned} \tag{II.8.9}$$

Hence

$$r_{\phi_{\text{new}}}(\theta) > r_{\phi_{\text{old}}}(\theta).$$

At $R = r_{\phi_{\text{old}}}(\theta)^{-1}$, the intercept of the new reciprocal chord is, by (II.8.3),

$$I_{\text{new}} = \frac{R\sin\phi_{\text{new}}}{\sin\theta + \sin(\phi_{\text{new}} - \theta)} = \frac{r_{\phi_{\text{new}}}(\theta)}{r_{\phi_{\text{old}}}(\theta)} > 1.$$

Finally suppose $d < b$. Complex conjugation sends $[a/b, c/d]$ to the oppositely oriented interval $[-c/d, -a/b]$. Translating both endpoint fractions by the same integer does not change their roots, reciprocal radii, scalar equations, or positive-real chord intercepts. The reflected interval has smaller denominator $d$ at its left endpoint, so the case already proved applies and conjugation gives the two required inequalities in the original interval. Equality $b = d$ would force $b = d = 1$ from $bc - ad = 1$, and then $b + d = 2$, contrary to $n \geq 4$.

At every inherited Farey endpoint the two candidate radii are both 1 by definition. □

**Lemma II.8.2** (Multiplicity padding)**.** *Let $n \geq 4$. Suppose a Farey interval is unchanged from order $n-1$ to order $n$, but*

$$\left\lfloor\frac{n}{q}\right\rfloor = \left\lfloor\frac{n-1}{q}\right\rfloor + 1 = M.$$

*Let $\lambda = \rho e^{2\pi i x}$ be the order-$(n-1)$ constant-parameter candidate on an open ray of this cell. Orient the cell—conjugating $\lambda$ and reflecting its Farey endpoints when necessary—to obtain $\mu = \rho e^{i\vartheta}$ and $y = \vartheta/(2\pi)$ such that*

$$\frac{p}{q} < y < \frac{r}{s}, \qquad rq - ps = 1, \qquad q < s. \tag{II.8.10}$$

*Then the order-$(n-1)$ constant-parameter candidate is a varying-parameter order-$n$ product obtained by adjoining one factor with $\beta_M = 0$, $\alpha_M = 1$. Its chosen factor arguments satisfy the factor-argument bounds and the order-$n$ equation for chosen real arguments. Consequently its radius does not exceed the order-$n$ constant-parameter radius.*

*Proof.* Put $m = M - 1$ and define

$$A = q\vartheta - 2\pi p, \qquad B_m = \frac{2\pi r - s\vartheta}{m}. \tag{II.8.11}$$

If the smaller-denominator endpoint is already on the left, take $\mu = \lambda$ and $y = \vartheta/(2\pi)$. Otherwise replace $\lambda$ by $\overline{\lambda}$ and the cell by its reflected interval; denominators, moduli, and the scalar candidate equation are unchanged. Thus (II.8.10) holds, and the determinant-one calculation used in (II.6.3) gives

$$A > 0, \qquad B_m > 0, \qquad A + B_m < \pi.$$

The order-$(n-1)$ candidate has the constant parameter list $\beta_1 = \cdots = \beta_m = \beta$ and $\alpha_1 = \cdots = \alpha_m = \alpha = 1 - \beta$, where the scalar formulas at multiplicity $m$ are

$$\beta = \frac{\rho^q \sin B_m}{\sin(A + B_m)}, \qquad \alpha = \frac{\rho^{s/m} \sin A}{\sin(A + B_m)}. \tag{II.8.12}$$

Since $\mu^q = \rho^q e^{iA}$, direct resolution into real and imaginary parts gives

$$\mu^q - \beta = \frac{\rho^q \sin A}{\sin(A + B_m)} e^{i(A+B_m)}. \tag{II.8.13}$$

Hence every old factor has the chosen argument

$$u_1 = \cdots = u_m = A + B_m.$$

In particular, the equation for the old chosen real arguments is not an additional assumption:

$$\begin{aligned}(s - mq)\vartheta + \sum_{j=1}^{m} u_j &= (s - mq)\vartheta + m(A + B_m) \\ &= 2\pi(r - mp).\end{aligned} \tag{II.8.14}$$

The old constant-parameter Ito equation is

$$\mu^{s-mq}(\mu^q - \beta)^m = \alpha^m. \tag{II.8.15}$$

Now set

$$\beta_M = 0, \qquad \alpha_M = 1, \qquad u_M = A.$$

Because $M = m + 1$, multiplying (II.8.15) by $\mu^{-q}(\mu^q - 0) = 1$ gives the order-$n$ varying-parameter product

$$\mu^{s-Mq}(\mu^q - \beta)^m(\mu^q - 0) = \alpha^m. \tag{II.8.16}$$

Likewise, (II.8.14) and $A = q\vartheta - 2\pi p$ give

$$\begin{aligned}(s - Mq)\vartheta + \sum_{j=1}^{M} u_j &= 2\pi(r - mp) - q\vartheta + A \\ &= 2\pi(r - Mp).\end{aligned} \tag{II.8.17}$$

Finally, $0 < \beta < 1$ on an open ray. The argument map $t \mapsto \operatorname{Arg}(\mu^q - t)$ is strictly increasing on $[0, 1)$, so with $M_* = \operatorname{Arg}(\mu^q - 1)$ one has

$$u_1 = \cdots = u_m = A + B_m \in (A, M_*), \qquad u_M = A.$$

Thus all factors in (II.8.16) lie in the interval $[A, M_*)$ required by theorem II.6.1, and (II.8.17) is exactly its equation for the chosen real arguments at multiplicity $M$. Put

$$B_M = \frac{2\pi r - s\vartheta}{M} = \frac{m}{M} B_m.$$

Applying theorem II.6.1 to (II.8.16)–(II.8.17) gives the explicit order-$n$ scalar sign

$$\rho^{s/M} \sin A + \rho^q \sin B_M - \sin(A + B_M) < 0. \tag{II.8.18}$$

The inequality is strict: the $M$ factor parameters consist of $m$ copies of $\beta \in (0,1)$ and one copy of 0, so they are not all equal, and the equality condition in theorem II.6.1 cannot hold. The left side of (II.8.18) is strictly increasing as a function of $\rho > 0$ and vanishes at the order-$n$ constant-parameter radius. Thus the old radius is strictly smaller than the new one on every open ray. Conjugating back preserves the modulus. □

For $n \geq 3$, define the candidate outer radius $K_n$ as follows. For $x = \theta/(2\pi)$ in an open Farey interval and $0 < \theta < \pi$, use the unique scalar radius of proposition II.2.3; at a Farey endpoint in $[0, \pi)$ set $K_n = 1$; and set

$$K_n(\pi) = 1. \tag{II.8.19}$$

For $n \geq 4$ this is the continuous closed-cell scalar graph. For $n = 3$ it has the necessary jump $\lim_{\theta \uparrow \pi} K_3(\theta) = 1/2$ but $K_3(\pi) = 1$, representing the outer endpoint $-1$ of the terminal radial segment.

**Lemma II.8.3** (Exhaustive order-refinement split)**.** *Let $n \geq 4$ and let $\theta/(2\pi) \in [0, 1/2]$. Comparing $F^+_{n-1}$ with $F^+_n$, exactly one of the following alternatives applies.*

(i) *The ray is an inherited Farey endpoint; both candidate outer radii are defined to be* 1.
(ii) *The ray is a newly inserted endpoint of denominator $n$; the order $n$ candidate outer radius is* 1*, while the order $n-1$ ray is interior to one old cell.*
(iii) *The ray is interior to an old cell which is not split at this order. Then the endpoint pair is unchanged, and for the smaller denominator $q$ the multiplicity changes from $\lfloor (n-1)/q \rfloor$ to $\lfloor n/q \rfloor$, hence either stays the same or increases by exactly one.*
(iv) *The ray is interior to one of the two cells obtained by inserting the mediant $(a+c)/(b+d)$ between consecutive old endpoints $a/b < c/d$; in this case $b + d = n$.*

*There are no further cases.*

*Proof.* The only fractions which can appear in $F_n$ but not in $F_{n-1}$ have reduced denominator exactly $n$. If such a fraction $h/n$ is inserted between consecutive old endpoints $a/b < c/d$, the Farey adjacency criterion lemma II.2.1 gives $bc - ad = 1$ and $b + d \geq n$. Since $h/n$ lies between them, the decomposition argument in the proof of lemma II.2.1 writes

$$(n, h) = m(b, a) + \ell(d, c)$$

with positive integers $m, \ell$. Thus $n = mb + \ell d \geq b + d$. Hence $b + d = n$, $m = \ell = 1$, and $h/n = (a + c)/(b + d)$. This proves that every new interior split is exactly the mediant case and that no other old cell is split.

If the ray itself is a new fraction, it is case (ii). If it is an old endpoint, it is case (i). Otherwise it is interior to an old cell. When that old cell is split, case (iv) applies; when it is not split, the endpoint pair is unchanged. Increasing the order from $n - 1$ to $n$ can change $\lfloor n/q \rfloor$ by at most one, so the unchanged-cell alternatives in case (iii) are exhaustive. The upper-half restriction merely discards the reflected lower-half copies and does not introduce another case. □

**Theorem II.8.4** (Candidate nesting)**.** *With this definition,*

$$K_{n-1}(\theta) \le K_n(\theta) \tag{II.8.20}$$

*for every $0 \le \theta \le \pi$ and every $n \ge 4$; the lower-half-plane version follows by conjugation.*

*Proof.* Fix first an angle $\theta$ which is interior to a cell of both $F_{n-1}^{+}$ and $F_n^{+}$. Label the endpoints of its order-$n$ cell by $p/q, r/s$ with $q < s$, put

$$D = \left\lfloor \frac{n}{q} \right\rfloor,$$

and let $A, B$ be the positive quantities in (II.2.6). Define the order-$n$ scalar residual

$$\mathcal{F}_{n,\theta}(t) := t^{s/D} \sin A + t^q \sin B - \sin(A + B), \qquad t > 0. \tag{II.8.21}$$

Its derivative has the fixed sign

$$\mathcal{F}'_{n,\theta}(t) = \frac{s}{D} t^{s/D-1} \sin A + q t^{q-1} \sin B > 0. \tag{II.8.22}$$

By proposition II.2.3, $K_n(\theta)$ is the unique zero of $\mathcal{F}_{n,\theta}$.

Set

$$\rho_- = K_{n-1}(\theta), \qquad R = \rho_-^{-1}.$$

By lemma II.8.3, there are three interior-ray possibilities.

If the Farey interval and its multiplicity are unchanged, then its scalar equation is unchanged and

$$\mathcal{F}_{n,\theta}(\rho_-) = 0.$$

If the cell is unchanged but its multiplicity increases, then (II.8.18) gives

$$\mathcal{F}_{n,\theta}(\rho_-) < 0.$$

It remains to consider a cell created by inserting a new denominator-$n$ mediant. The two reciprocal-chord endpoints of the new scalar chord at $\rho_-$ are, after choosing their common orientation,

$$R^q e^{iA}, \qquad R^{s/D} e^{-iB}.$$

Its positive-real intercept is

$$I_R = \frac{R^{q+s/D} \sin(A + B)}{R^q \sin A + R^{s/D} \sin B}.$$

Direct subtraction, without a side convention, gives

$$I_R - 1 = \frac{R^{q+s/D}(-\mathcal{F}_{n,\theta}(\rho_-))}{R^q \sin A + R^{s/D} \sin B}. \tag{II.8.23}$$

By lemma II.8.1, $I_R > 1$; every other factor in (II.8.23) is positive. Hence

$$\mathcal{F}_{n,\theta}(\rho_-) < 0.$$

Thus in all three cases

$$\mathcal{F}_{n,\theta}(\rho_-) \le 0.$$

Together with (II.8.22) and $\mathcal{F}_{n,\theta}(K_n(\theta)) = 0$, this proves

$$K_{n-1}(\theta) = \rho_- \le K_n(\theta)$$

on every ray interior to both Farey decompositions. The inequality is strict whenever the cell is split or its multiplicity increases.

Now let $\theta/(2\pi)$ be a newly inserted fraction. It is interior to one order-$(n-1)$ cell, so

$$K_{n-1}(\theta) < 1$$

by proposition II.2.3, whereas $K_n(\theta) = 1$ by the endpoint definition. At every inherited Farey endpoint below $\pi$, both values are 1. At $\theta = 0$ the same is true, and at $\theta = \pi$ both sides are 1 by (II.8.19). The latter separate clause is essential for $n = 4$, because $\lim_{\theta\uparrow\pi} K_3(\theta) = 1/2$ although $K_3(\pi) = 1$. This proves (II.8.20) everywhere. Conjugation gives the lower-half-plane statement. ☐

## II.9 Completion of the proof

**Lemma II.9.1** (Compact star-shaped sets with continuous radial function)**.** *Let $S \subset \mathbb{C}$ be compact and star-shaped with respect to the origin, and suppose that*

$$r_S(\phi) := \max\{r \geq 0 : re^{i\phi} \in S\}$$

*is positive and continuous on the circle. Then*

$$S = \{t\, r_S(\phi)e^{i\phi} : 0 \leq t \leq 1,\ -\pi < \phi \leq \pi\}, \qquad \partial S = \{r_S(\phi)e^{i\phi} : -\pi < \phi \leq \pi\}. \tag{II.9.1}$$

*Proof.* Compactness makes each maximum attainable, and star-shapedness with respect to the origin then says that the intersection of $S$ with the ray of angle $\phi$ is exactly $\{re^{i\phi} : 0 \leq r \leq r_S(\phi)\}$. Taking the union over $\phi$ gives the first formula. A point $r_S(\phi)e^{i\phi}$ cannot be interior, since a sufficiently small outward radial displacement would contradict maximality.

Conversely, let $z = re^{i\phi}$ with $0 < r < r_S(\phi)$. Choose $\eta > 0$ with $r + 3\eta < r_S(\phi)$. By continuity, $r_S(\psi) > r + 2\eta$ for all $\psi$ in some neighborhood of $\phi$. A sufficiently small Euclidean ball about $z$ consists of points $w$ with $\arg w$ in that neighborhood and $|w| < r + \eta$; the ray description then puts the whole ball in $S$. Thus every nonzero point strictly below the radial graph is interior. Finally, positivity and compactness of the circle give $c := \min_\phi r_S(\phi) > 0$, so the disk $|z| < c$ lies in $S$ and 0 is interior. No further boundary points remain. ☐

### II.9.1 The base orders

**Proposition II.9.2** (Orders one, two, and three)**.** *One has*

$$\Theta_1 = \{1\}, \qquad \Theta_2 = [-1, 1],$$

*and, with $\omega = e^{2\pi i/3}$,*

$$\Theta_3 = \operatorname{conv}\{1, \omega, \overline{\omega}\} \cup [-1, -1/2]. \tag{II.9.2}$$

*Proof.* For order one the only matrix is $[1]$. For order two, every stochastic matrix has the form

$$\begin{pmatrix} a & 1-a \\ b & 1-b \end{pmatrix}, \qquad 0 \leq a, b \leq 1,$$

and its second eigenvalue is $a - b$; hence the first two assertions follow.

Let a $3 \times 3$ stochastic matrix have nonreal eigenvalues $\lambda = x + iy$ and $\overline{\lambda}$. Its spectrum is $1, \lambda, \overline{\lambda}$. Since the diagonal is nonnegative,

$$1 + 2x = \operatorname{tr} A \geq 0,$$

so $x \geq -1/2$. Also

$$(\operatorname{tr} A)^2 \leq 3 \sum_i a_{ii}^2 \leq 3 \operatorname{tr}(A^2),$$

because every term $a_{ij}a_{ji}$ in $\operatorname{tr}(A^2)$ is nonnegative. Substituting

$$\operatorname{tr} A = 1 + 2x, \qquad \operatorname{tr}(A^2) = 1 + 2(x^2 - y^2)$$

gives

$$3y^2 \leq (1 - x)^2.$$

Thus the nonreal region lies in the triangle in (II.9.2); every real stochastic eigenvalue lies in $[-1, 1]$.

The chord $[1, \omega]$ is realized by $(1 - \alpha)I + \alpha C_3$, $0 \leq \alpha \leq 1$. The terminal sparse realization of theorem II.7.3 has polynomial

$$\lambda^3 - (1 - \alpha)\lambda - \alpha = (\lambda - 1)(\lambda^2 + \lambda + \alpha). \tag{II.9.3}$$

For $0 \leq \alpha \leq 1/4$, the root $(-1 - \sqrt{1 - 4\alpha})/2$ runs from $-1$ to $-1/2$. For $1/4 < \alpha \leq 1$, the upper root is

$$\lambda = -\frac{1}{2} + \frac{i}{2}\sqrt{4\alpha - 1},$$

and runs from $-1/2$ to $\omega$. Conjugation supplies the lower edges, and star-shapedness with respect to the origin fills the triangle and the real segment. This proves (II.9.2).

It remains to identify the two nonreal boundary pieces with the canonical radius $K_3$ used by the induction. On the Farey interval $[0, 1/3]$, the denominator labeling of (II.2.2) is

$$(q, s) = (1, 3), \qquad d = \lfloor 3/1 \rfloor = 3.$$

For $0 < \theta = 2\pi x < 2\pi/3$, (II.2.6) gives

$$A = \theta, \qquad B = \frac{2\pi}{3} - \theta, \qquad A + B = \frac{2\pi}{3}.$$

Thus (II.2.8) is

$$\rho \sin\theta + \rho \sin\left(\frac{2\pi}{3} - \theta\right) = \sin\frac{2\pi}{3}. \tag{II.9.4}$$

With the coefficients from (II.2.9), this equation is exactly $\alpha + \beta = 1$, and direct real and imaginary parts give

$$\rho e^{i\theta} = \beta + \alpha\omega.$$

Indeed, the imaginary part is $\alpha \sin(2\pi/3) = \rho \sin\theta$, while the real part is

$$\frac{\rho \sin(2\pi/3 - \theta) + \rho \sin\theta \cos(2\pi/3)}{\sin(2\pi/3)} = \rho \cos\theta.$$

For $\lambda \neq 0$, the Ito equation (II.2.4) reduces here to $(\lambda - \beta)^3 = \alpha^3$, and its upper branch is precisely $\lambda = \beta + \alpha\omega$. Moreover

$$\frac{d}{d\alpha} \arg(1 + \alpha(\omega - 1)) = \frac{\Im\omega}{|1 + \alpha(\omega - 1)|^2} > 0.$$

Hence the chord meets every ray with angle in $(0, 2\pi/3)$ exactly once; the uniqueness in proposition II.2.3 identifies that modulus with $K_3(\theta)$.

On the terminal cell $[1/3, 1/2]$, denominator order gives

$$(q, s) = (2, 3), \qquad d = \lfloor 3/2 \rfloor = 1.$$

For $2\pi/3 < \theta = 2\pi x < \pi$ one has

$$A = 2(\pi - \theta), \qquad B = 3\theta - 2\pi, \qquad A + B = \theta,$$

and the scalar equation becomes

$$\rho^3 \sin(2(\pi - \theta)) + \rho^2 \sin(3\theta - 2\pi) = \sin\theta. \tag{II.9.5}$$

The nonzero Ito equation is now $\lambda(\lambda^2 - \beta) = \alpha$, equivalently the polynomial (II.9.3). From $\lambda^3 = \beta\lambda + \alpha$, with $\lambda = \rho e^{i\theta}$, imaginary and real parts give

$$\beta = \frac{\rho^2 \sin(3\theta - 2\pi)}{\sin\theta}, \qquad \alpha = \frac{\rho^3 \sin(2(\pi - \theta))}{\sin\theta}.$$

Consequently $\alpha + \beta = 1$ is exactly (II.9.5). Along the upper root $\lambda = -1/2 + iy$,

$$\frac{d}{dy} \arg\left(-\frac{1}{2} + iy\right) = \frac{-1/2}{1/4 + y^2} < 0.$$

As $y$ increases from 0 to $\sqrt{3}/2$, the argument therefore decreases strictly from $\pi$ to $2\pi/3$. This boundary piece meets every ray in $(2\pi/3, \pi)$ exactly once, and proposition II.2.3 again identifies its modulus with $K_3(\theta)$. The real range $0 \le \alpha \le 1/4$ supplies the additional segment $[-1, -1/2]$ on the endpoint ray; it is not an open-cell branch.

The trace bounds proved that no order-three point lies beyond these pieces, while the displayed matrices attain them. At the Farey endpoint rays $0, 2\pi/3, \pi$, the outer points are respectively $1, \omega, -1$, and the endpoint convention gives $K_3 = 1$. We have therefore proved the base identity required at order four:

$$R_3(\theta) = K_3(\theta) \qquad (0 \le \theta \le \pi), \tag{II.9.6}$$

where at 0 and $\pi$ the notation $R_3$ denotes the attained outer radial modulus. The jump from the terminal nonreal arc to $-1$ is exactly the exceptional order-three behavior recorded in proposition II.2.4. ☐

### II.9.2 Induction on the order

*Proof of theorem II.3.1.* The orders $1, 2, 3$ are proposition II.9.2. Assume now that $n \ge 4$ and that the theorem holds at order $n - 1$.

First note the natural nesting

$$\Theta_{n-1} \subseteq \Theta_n : \tag{II.9.7}$$

if $A$ is an $(n-1) \times (n-1)$ row-stochastic matrix, then $A \oplus [1]$ is an $n \times n$ row-stochastic matrix with every eigenvalue of $A$. Consequently

$$R_{n-1}(\theta) \le R_n(\theta) \qquad (0 < \theta < \pi).$$

Fix an angle $\theta$ lying in an open interval between consecutive terms of $F_n^+$ and put

$$\lambda_n = R_n(\theta) e^{i\theta}.$$

By corollary II.7.4,

$$K_n(\theta) e^{i\theta} \in \Theta_n, \qquad \text{hence} \qquad 0 < K_n(\theta) \le R_n(\theta).$$

Since $\theta/(2\pi)$ is not a Farey endpoint of order $n$, $e^{i\theta}$ is not a root of unity of order at most $n$. Thus proposition II.4.4 gives

$$0 < R_n(\theta) < 1.$$

Suppose first that $\lambda_n \notin \Theta_{n-1}$. Then $\lambda_n$ satisfies precisely the radial-boundary conditions (II.4.3). By corollary II.6.2, its modulus is the unique solution of the scalar radius equation on this ray. By definition that solution is $K_n(\theta)$, and hence

$$R_n(\theta) = K_n(\theta).$$

Suppose instead that $\lambda_n \in \Theta_{n-1}$. Its membership and the definition of $R_{n-1}$ give

$$R_n(\theta) \le R_{n-1}(\theta),$$

whereas (II.9.7) gives the reverse inequality. Thus

$$R_n(\theta) = R_{n-1}(\theta).$$

Because an open order-$n$ ray cannot be an inherited Farey endpoint, the induction hypothesis applies to this same ray and gives

$$R_{n-1}(\theta) = K_{n-1}(\theta).$$

Now theorem II.8.4 and attainment give the fully explicit chain

$$R_n(\theta) = K_{n-1}(\theta) \le K_n(\theta) \le R_n(\theta).$$

Hence equality holds throughout, and again

$$R_n(\theta) = K_n(\theta).$$

We have proved this identity on every such open interval. If $\theta/(2\pi) = p/q \in F_n^+$ is a Farey endpoint, then $q \le n$ and the cyclic permutation matrix of order $q$, padded to order $n$, realizes $e^{i\theta}$. The disk bound in proposition II.4.1 therefore shows that the outer radial modulus on that endpoint ray is $1 = K_n(\theta)$. Extend $R_n$ to $\theta = 0, \pi$ by these outer radial moduli. Then

$$R_n(\theta) = K_n(\theta) \qquad (0 \le \theta \le \pi). \tag{II.9.8}$$

For $n \ge 4$, proposition II.2.4 now joins the open-cell graphs continuously at their Farey endpoints. Thus

$$\zeta_n(\theta) = K_n(\theta)e^{i\theta}, \qquad 0 \le \theta \le \pi,$$

traces exactly the ordered concatenation of the candidate arcs $\Gamma^{(n)}_{f,g}$, from 1 to $-1$. Every point of this chain belongs to $\Theta_n$ by corollary II.7.4 and the preceding endpoint argument. Conversely, the equality $R_n = K_n$ says that no point of $\Theta_n$ lies farther out on any upper-half-plane ray. Conjugation gives the corresponding lower chain.

For completeness, define the full-circle function

$$\widehat{K}_n(\phi) = \begin{cases} K_n(\phi), & 0 \le \phi \le \pi, \\ K_n(-\phi), & -\pi \le \phi \le 0. \end{cases}$$

Conjugation symmetry and (II.9.8) show that this is the radial maximum of $\Theta_n$ on every ray. It is positive by attainment and continuous on the circle by proposition II.2.4; at the identified angles $-\pi$ and $\pi$ both values are 1. The set $\Theta_n$ is compact by proposition II.4.1 and star-shaped with respect to the origin by corollary II.4.3. Therefore lemma II.9.1, applied with $S = \Theta_n$ and $r_S = \widehat{K}_n$, gives

$$\Theta_n = \{t\widehat{K}_n(\phi)e^{i\phi} : 0 \le t \le 1, -\pi < \phi \le \pi\}, \qquad \partial\Theta_n = \{\widehat{K}_n(\phi)e^{i\phi} : -\pi < \phi \le \pi\}.$$

Thus the upper and lower boundary arc chains are exactly the topological boundary, not merely the outer points on individual rays, and the first formula is the claimed radial hull. The unit-circle assertion is proposition II.4.4. The discontinuous terminal graph and the segment $[-1, -1/2]$ occur only at order three and were handled directly in proposition II.9.2; the continuous-radius lemma is not invoked for that exceptional base case. □

## II.10 The complete order-seven example

The upper Farey sequence of order seven is

$$0, \frac{1}{7}, \frac{1}{6}, \frac{1}{5}, \frac{1}{4}, \frac{2}{7}, \frac{1}{3}, \frac{2}{5}, \frac{3}{7}, \frac{1}{2}. \tag{II.10.1}$$

For each cell, the table lists the denominator pair $(q, s)$ with $q < s$, the multiplicity $d = \lfloor 7/q \rfloor$, the exponent $e = s - dq$, and the reduced Ito polynomial. Throughout $\beta = 1 - \alpha$.

| cell | $(q,s)$ | $d$ | $e$ | reduced Ito polynomial |
|---|---|---|---|---|
| $0 \to 1/7$ | $(1,7)$ | 7 | 0 | $(z-\beta)^7 - \alpha^7$ |
| $1/7 \to 1/6$ | $(6,7)$ | 1 | 1 | $z(z^6 - \beta) - \alpha$ |
| $1/6 \to 1/5$ | $(5,6)$ | 1 | 1 | $z(z^5 - \beta) - \alpha$ |
| $1/5 \to 1/4$ | $(4,5)$ | 1 | 1 | $z(z^4 - \beta) - \alpha$ |
| $1/4 \to 2/7$ | $(4,7)$ | 1 | 3 | $z^3(z^4 - \beta) - \alpha$ |
| $2/7 \to 1/3$ | $(3,7)$ | 2 | 1 | $z(z^3 - \beta)^2 - \alpha^2$ |
| $1/3 \to 2/5$ | $(3,5)$ | 2 | $-1$ | $(z^3 - \beta)^2 - \alpha^2 z$ |
| $2/5 \to 3/7$ | $(5,7)$ | 1 | 2 | $z^2(z^5 - \beta) - \alpha$ |
| $3/7 \to 1/2$ | $(2,7)$ | 3 | 1 | $z(z^2 - \beta)^3 - \alpha^3$ |

### II.10.1 A worked ray: $x = 3/8$

The ray $x = 3/8$ lies in the cell $1/3 < x < 2/5$. Here

$$q = 3, \qquad s = 5, \qquad d = 2, \qquad e = -1,$$

and

$$A = 2\pi\left(3 \cdot \frac{3}{8} - 1\right) = \frac{\pi}{4}, \qquad B = \pi\left(2 - 5 \cdot \frac{3}{8}\right) = \frac{\pi}{8}.$$

The boundary radius is the unique root of

$$\rho^{5/2} \sin\frac{\pi}{4} + \rho^3 \sin\frac{\pi}{8} = \sin\frac{3\pi}{8}, \tag{II.10.2}$$

namely

$$\begin{aligned} &\rho = 0.940100221928822853\ldots, \\ &\alpha = 0.655850787368397414\ldots, \quad \beta = 0.344149212631602586\ldots. \end{aligned} \tag{II.10.3}$$

Thus

$$\lambda = \rho e^{3\pi i/4} = -0.664751241920849 + 0.664751241920849\, i$$

and

$$(\lambda^3 - \beta)^2 = \alpha^2 \lambda. \tag{II.10.4}$$

A six-state sparse stochastic realization, padded by one absorbing state, is

$$A_7 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ \beta & 0 & 0 & \alpha & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & \alpha & 0 & \beta & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \tag{II.10.5}$$

The two local 3-cycles have weight $\beta$, while the unique cross cycle has length five and weight $\alpha^2$. Hence

$$\det(tI - A_7) = (t-1)\big((t^3 - \beta)^2 - \alpha^2 t\big),$$

which verifies (II.10.4) directly.

### II.10.2 The order-seven region

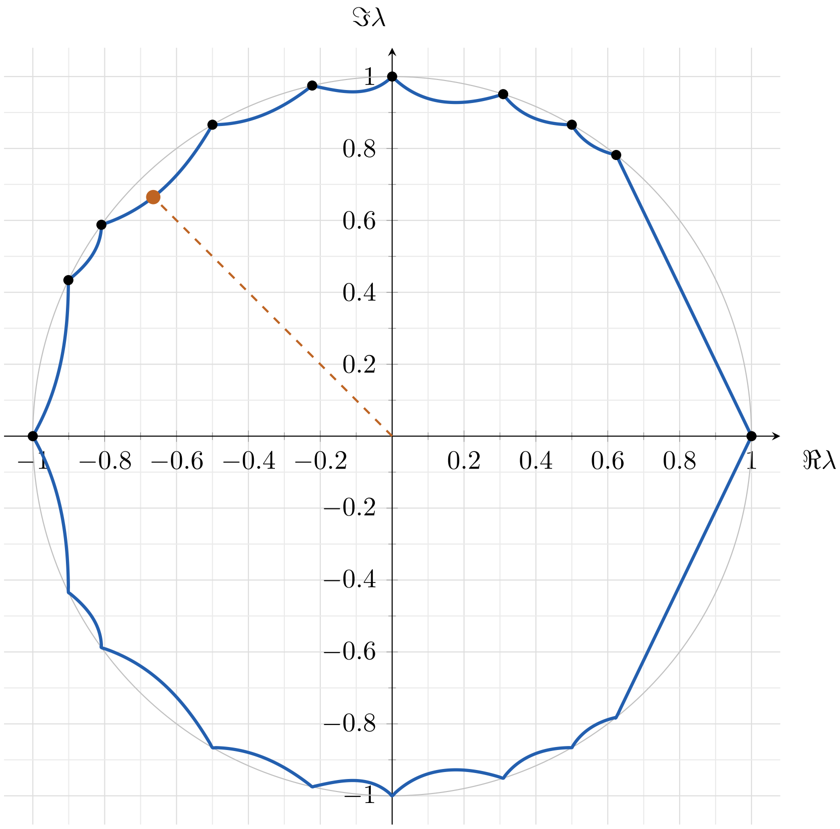


Figure 1: The boundary of $\Theta_7$. The dashed ray is $x = 3/8$, and the marked point is the solution of (II.10.2).

## II.11 Concluding structural remarks

The proof separates three statements that are often conflated.

First, the boundary equation is not a generic determinant identity for stochastic matrices. For $N \geq 4$, a point satisfying (II.4.3) first becomes an $N$-critical planar multiplier; only then does the critical-polygon finite-product theorem force the varying-parameter product equation and the equation for chosen real arguments. Orders at most three are settled by the direct small-order argument instead.

Second, the Part I finite-product theorem is existential. It selects one of the two adapted complex orientations and supplies the corresponding consecutive Farey fractions and finite product equation. The reflection formulas return these quantities to the original upper-half-plane point; the proof uses no uniqueness of either the invariant polygon or the stochastic matrix realizing the point.

Third, the scalar Farey geometry is downstream of the Part I input. Once the finite product equation, the equation for chosen real arguments, and the factor-argument bounds are available, the remainder is strict one-dimensional convexity: every nonconstant parameter list lies radially inside the constant-parameter case, and Farey refinement moves the constant-parameter envelope outward. The intrinsic contact and return theory that establishes this input is developed in Part I and culminates in theorem 1.4.

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